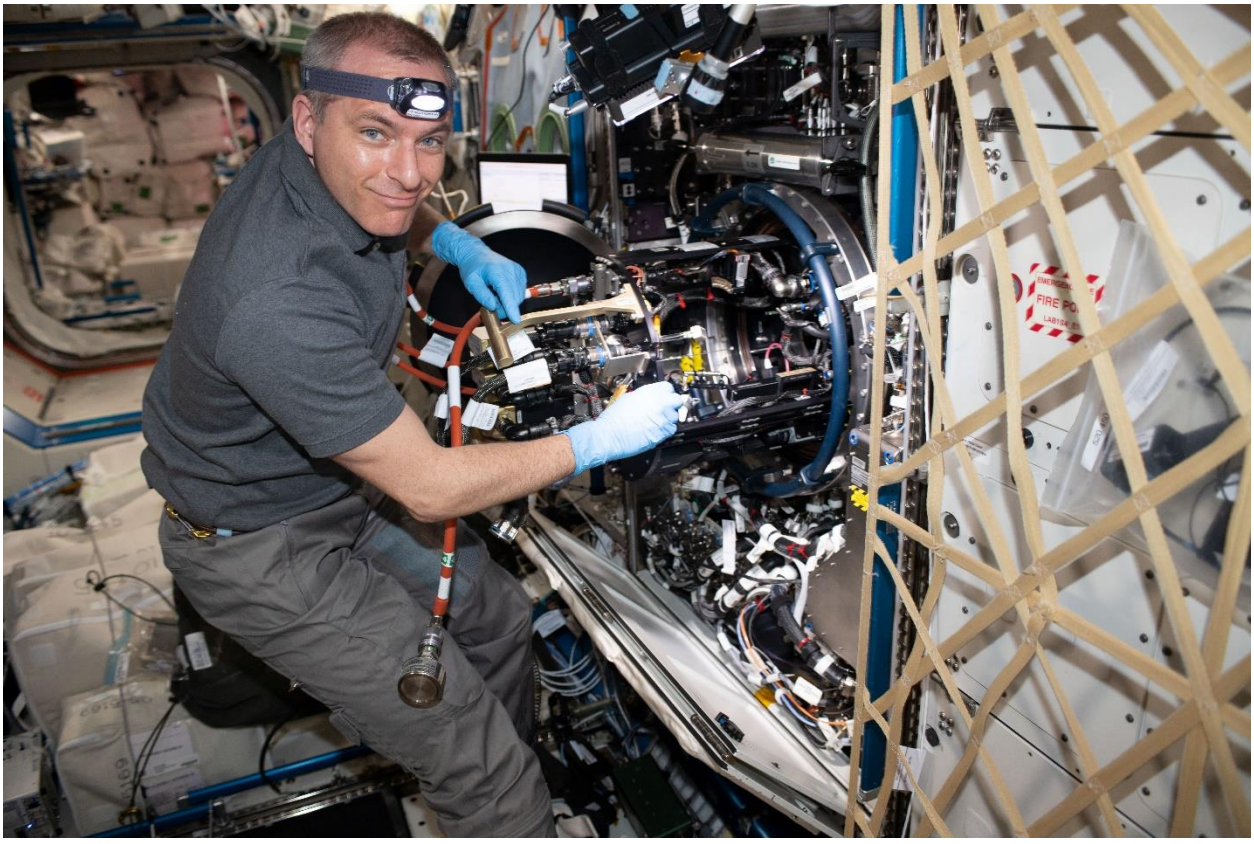
Advanced Combustion via Microgravity Experiments (ACME)

ACME Engineering Implementation

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Canadian astronaut David Saint-Jacques working to replace the igniter tip on the ACME chamber insert, which is partially withdrawn from the Combustion Integrated Rack (CIR), on 12 May 2019. [ISS059e060847]

Given that this document provides a limited overview of the ACME hardware and operations, readers searching for additional information are welcome to contact Dennis Stocker with questions.

Identification of commercial products is for clarification and does not indicate endorsement by NASA or the federal government.

Table of Contents

	Page
1. Introduction	**3**
2. Engineering Overview	**5**
A. General Description	5
B. Overview for E-FIELD Flames	13
3. Engineering Details (selected)	**19**
A. Chamber Insert Layout	20
B. Gases and Gas Delivery	22
C. Burners	25
1. Gas-Jet Burners	25
2. Coflow Burner	27
3. Spherical Burners	43
4. BRE Burners	45
D. Igniter Tips	50
E. Optical Layout	51
F. Radiometer Fields of View	57
G. Thermocouples	59
4. Engineering Documents	**60**
A. General Documents	60
B. Deviations and Waivers	69
5. Drawings	**75**
A. Top Level Assembly	75
B. Common Electrical	76
C. Quick Disconnects (QDs) and Hoses	77
D. Experiment Mounting Structure (EMS)	77
E. Gas Supply	78
F. Burners	79
G. Igniter	81
H. Analog Camera System	82
I. Diagnostic System	82
J. Measurement System	83
K. E-Field System	84
L. Cooling System	85
M. ACME Data Camera Package	86
N. ACME Avionics Package	88
O. Ground Support Equipment	90
P. Shipping Containers	92
Q. Trainer	92
6. Internet Links	**94**
7. Nomenclature	**94**

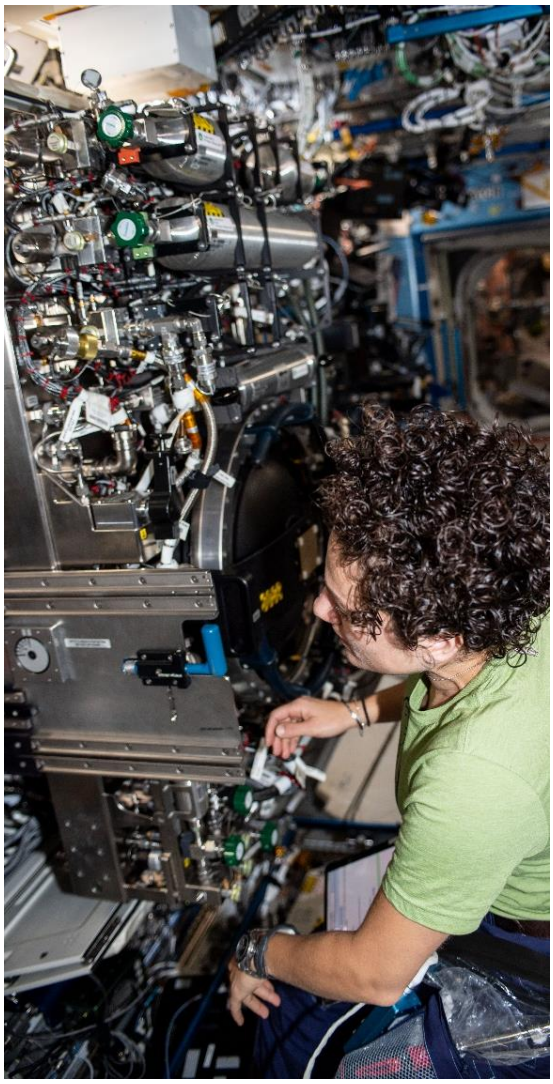


Figure 1. NASA astronaut Jessica Meir working on 24 Oct. 2019 to remove and replace a failed hard drive in one of the CIR's Image Processing and Storage Units (IPSUs). [Cropped detail from ISS061e013306]

PHOTOS - An extensive collection of photographs of the ACME hardware and crew activities can be found within the *Space Flames* Flickr site. Given that it includes photos from other projects and activities, it is best to start with a review of the available albums at www.flickr.com/photos/space-flames/albums. Updates on ACME and similar research, including photographs and videos, can also be seen at the *Space Flames* Facebook page, www.facebook.com/space.flames.

1. Introduction

The purpose of this report is to provide a description of the experimental hardware and operations of used in the Advanced Combustion via Microgravity Experiments (ACME) project that is sufficient for informed interpretation of the experimental results. In contrast, it is not the intent of this document to review the project management, engineering development, design process, etc. although such aspects may be briefly addressed.

The ACME suite of experiments are being carried out on the International Space Station (ISS) in the Combustion Integrated Rack (CIR). ACME features investigations with gaseous fuels, such as methane and ethylene. In addition to the fuel state, the current ACME experiments use laminar, non-premixed (i.e., diffusion) flames for their studies. To simplify analysis, these flames are furthermore either axisymmetric or spherically symmetric, depending upon the experiment.

Currently, ACME includes six independent experiments of which the first five were selected for testing in spaceflight by NASA while the sixth was selected by the Center for the Advancement of Science in Space (CASIS) and the National Science Foundation (NSF).

The CIR facility was launched in 2008 and with the Multi-user Droplet Combustion Apparatus (MDCA) was used for a series of liquid-fueled experiments from 2009 to 2017. The facility was reconfigured and set up for ACME testing by the ISS crew in September and October 2017, where over 25 astronauts from Canada, Germany, Italy, Japan, Russia and the United States have thus far supported the project's research. As of this writing, over one thousand ACME flames have been ignited on the ISS, where that began in Nov. 2017. It is expected that the planned ACME testing will be completed in 2021, when the CIR operations should subsequently shift to the Solid Fuel Ignition and Extinction (SoFIE) project.

The discussion of the CIR hardware herein is limited, but the research facility is described in various documents, such as those listed below. These are listed in reverse chronological order, where the oldest predate the facility's completion and have become outdated in some regards. References 1, 2 and 4 also discuss ACME, but were all prepared prior to the completion of ACME's spaceflight hardware.

1. Suleyman A. Gokoglu, Daniel L. Dietrich, Dennis P. Stocker, Paul V. Ferkul, Sandy L. Olson, Michael C. Hicks, Amelia Rai (editor), Neesha Hosein (editor), and Cory Duke (designer), "Combustion Science: A Researcher's Guide to International Space Station," NASA Publication NP-2015-10-034-JSC (2015).
2. J.M. Hickman, D.L. Dietrich, M.C. Hicks, V. Nayagam, V., and D. Stocker, "FLEX: A Decisive Step Forward in NASA's Combustion Research Program," 1st Annual International Space Station Research and Development Conference, Denver, Colorado 26-28 June (2012).
3. "Unlocking Mysteries in Microgravity: NASA Glenn Provides the Keys with the Fluids and Combustion Facility," NASA fact sheet FS-2011-06-035-GRC, PS-00538-0611 (2011).
4. T.F. O'Malley, W.A. Sheredy, and D.P. Stocker, "Combustion Research on the International Space Station," 59th International Astronautical Congress, Vol. 2, pp. 756-766 (2008).
5. Terence F. O'Malley, Karen J. Weiland, Craig A. Myhre, Frank G. Gati, Myron E. Hill, and John H. Snead, "The Fluids and Combustion Facility and First Payloads: The Multi-User Droplet Combustion Apparatus and the Light Microscopy Module: Enabling Research on the ISS," 57th International Astronautical Congress, IAC-06-A2.1.07, 2-6 Oct. 2006, Valencia, Spain (2006).
6. T. O'Malley and C. Myrhe, "The Fluids and Combustion Facility Combustion Integrated Rack and the Multi-user Droplet Combustion Apparatus: Microgravity Combustion Science Using Modular

Multi-User Hardware," 53rd International Astronautical Congress, The World Space Congress 2002, IAC-02-J.5.05, 10-19 Oct. 2002, Houston, Texas (2002).
7. Terence F. O'Malley and Karen J. Weiland, "The FCF Combustion Integrated Rack: Microgravity Combustion Science Onboard the International Space Station," NASA Technical Memorandum TM--2002-210981, AIAA-2001-4927, September 2002.
8. Robert L. Zurawski and Nora G. Bozzolo, "ISS Fluids and Combustion Facility: Combustion Integrated Rack: Payload Accommodations - Principal Investigator's Guide," NASA report, September 2000.

The CIR facility was preceded by the Combustion Module (CM) which flew on the space shuttle *Columbia*. In 1997, the combustion research facility was within Spacelab on the STS-83 and STS-94 (Microgravity Science Laboratory) missions. It was subsequently reconfigured and flew in 2003 within SPACEHAB, again in the shuttle's payload bay, in the STS-107 mission. The Combustion Module investigations included Laminar Soot Processes (LSP), Structure Of Flame Balls At Low Lewis-number (SOFBALL), and the Water-Mist Fire Suppression Experiment (Mist). Although all three experiments burned gaseous fuel, each had its own chamber insert, whereas ACME features a single modular chamber insert to accomplish multiple investigations. A significant difference between the CIR and its shuttle precursor is that the ISS tests are remotely commanded from the ground without crew involvement, which wasn't the case with the Combustion Module research. But in all cases the crew are required to configure the necessary hardware.

While extensive, this report is far from exhaustive but there are other reports which provide additional detail useful for interpreting ACME's experimental results. Some key examples include the following:

Camera characterization	
CFI Intensified Camera Package Characterization Report	MDC-XXX-1866-R1
Data Camera Characterization Report	ACME-RPT-010 Rev. A
Optical instrument characterization	
BRE Radiometer Characterization Report	ACME-RPT-012
Cool Flames Radiometer Characterization Report	CIR-RPT-4600
PMT Characterization Report	ACME-RPT-013
Radiometer Characterization Report	ACME-RPT-011
Requirement verification	
ERD Verification Report	ACME-VER-002 Rev. B
Inspection Verification Report	ACME-VER-001 Rev. A
Software and non-imaging data	
Software Interface Design Document	ACME-DOC-010 Rev. D

Abbreviations are addressed at the end of this report, but ERD stands for Engineering Requirements Document and many of the engineering requirements were directly derived from ACME's science requirements. Meanwhile, key hardware dimensions, for example of the burners, and other details are confirmed in the Inspection Verification Report. Also note that the Burning Rate Emulator (BRE) and Cool Flames radiometer reports are specific to certain ACME experiments, whereas the other reports are general to all of ACME.

ACME Project Scientist Dennis Stocker (216-433-2166, dennis.p.stocker@nasa.gov) may be contacted with questions about the ACME project.

2. Engineering Overview

A. General Description

This section provides a general description of the ACME engineering from the '3.0 Experimental' section of the following 2014 conference paper:

> *Dennis P. Stocker, Fumiaki Takahashi, J. Mark Hickman, and Andrew C. Suttles, "Gaseous Non-Premixed Flame Research Planned for the International Space Station," Spring Technical Meeting of the Central States Section of the Combustion Institute, Tulsa, OK, 16–18 March 2014.*

IMPORTANT: *The 2104 paper from which this section was extracted proceeded completion of the spaceflight hardware which was mostly launched in 2017. As a result, it does not accurately describe the final engineering in some regards. Therefore updates, clarifications, additional details, shortcomings, references, etc. have been included in boxed notes like this following the original description. As a minor note, the sizes of figures were adjusted in this report, but none were cropped.*

As previously discussed, the ACME experiments will be conducted within the Combustion Integrated Rack (CIR) using a chamber insert, depicted in Figure 2, which was designed to accommodate gas delivery to a variety of exchangeable burners. Besides the chamber insert, other ACME-unique hardware includes a high-definition color camera and an avionics package, which is for data acquisition and control.

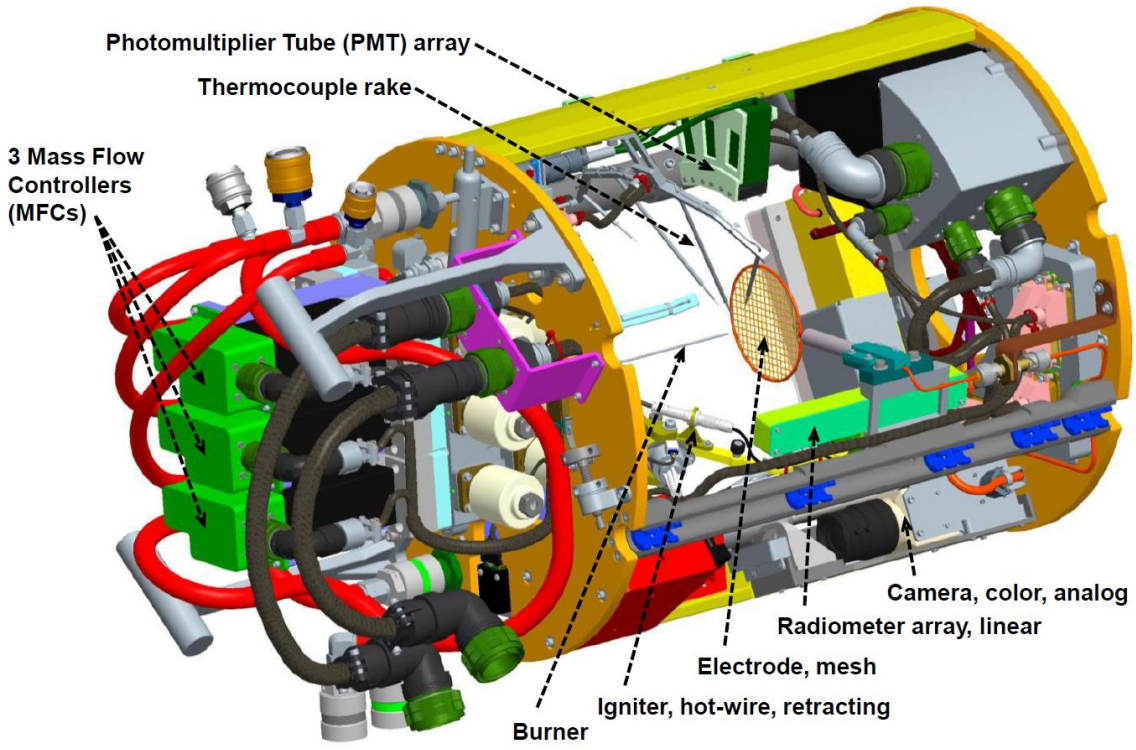


Figure 2: Design of the Advanced Combustion via Microgravity Experiments (ACME) chamber insert that will be used in the Combustion Integrated Rack (CIR) on ISS.

ACME's chamber insert, which is ~0.4 m in diameter, provides common mounting for a variety of burners on the chamber axis (but only one at a time) and includes a retracting resistive igniter. As depicted in Figure 3, there will initially be gas-jet burners (with inner diameters ranging from 0.4 to 3.2 mm), a small coflow burner (where the two concentric tubes have inner diameters of 2.1 and 25 mm), porous spherical burners (with diameters ranging from 6.35 to 12.7 mm), and two flat porous burners for the BRE experiment (with inner diameters of 25 and 50 mm). Most burners will be equipped with a surface thermocouple, and each BRE burner will have two heat flux transducers. The internal pressure of the porous burners will be measured to allow monitoring of the pressure differential across the outlet surface. The igniter will have a crew-exchangeable tip and arm (to accommodate the various burners) and its stepper-motor drive will allow for precise positioning.



Figure 3: ACME burner types, including from left to right, gas-jet burners, a co-flow burner, porous spherical burners, and flat porous burners for the BRE experiment.

Ultimately, the spherical burners were limited to a single diameter of 6.35 mm, and the fabricated gas-jet burners were limited to two sizes with nominal inner diameters of 1.3 and 2.1 mm. However, only the larger gas-jet burner was used during the spaceflight tests. The burners are addressed in detail in section 3.C.

Unfortunately, the spherical burners have not always yielded spherically symmetric flames, presumably as a result of (1) greater burner flow through the joint between the porous sphere and supply tube, and (2) heat loss to the support tube (encompassing the supply tube) causing local flame quenching.

Two types of igniter tips have been used during ACME testing. Most tests have been conducted using a loop igniter tip like those which had been successfully used in the droplet combustion experiments which preceded ACME research in the CIR. However, some tests were conducted with a coil igniter tip, where that included the second round testing for both the Burning Rate Emulator (BRE) and Structure and Response of Spherical Diffusion Flames (s-Flame) experiments, with the exception of the latter's first day testing. The CFI-G investigation will also use the coil igniter tip. For more information about the igniter tips, see section 3.D.

The chamber insert includes three mass flow controllers (visible on the left of Fig. 2) for the delivery of fuel, oxidizer or inert (e.g., helium or carbon dioxide), and nitrogen to the burner. The maximum fuel flow is 2 slpm (on a nitrogen basis), whereas much higher flows, e.g., 10 slpm (on a nitrogen basis), are possible for the oxidizer and nitrogen lines. The crew can exchange the mass flow controllers to provide appropriate ranges that maximize the flow accuracy. The gas delivery system enables on-orbit dilution of the fuel or oxidizer (but not both simultaneously), coflow,

inverse flames (e.g., oxidizer delivery to the center of a coflow burner), and premixed flames, although the latter are not planned for any of the current ACME experiments. Gases can also be delivered premixed to the space station where that is necessary in some circumstances, such as the blending of different fuel gases. CIR's standard bottle sizes are 1, 2.25, and 3.8 liters but ACME will sometimes use smaller bottles as a hazard control to limit the deficient reactant (e.g., fuel) that can be released into the combustion chamber.

Thus far, ACME tests were not conducted with carbon dioxide as a diluent. Furthermore, the fuel diluted with helium was premixed prior to launch rather than on orbit. Ultimately, oxidizer flows of up to 20 slpm (on a nitrogen basis) were possible, while the nitrogen flow was limited to a maximum of 5 slpm. ACME has not used gas bottles smaller than 1 liter as indicated above; the bottle fill pressure was instead limited as an additional hazard control.

ACME's primary flame diagnostics are a suite of imaging systems including five video cameras and two illumination systems, of which three cameras are part of the CIR facility and are already in use on orbit. All but one of the cameras is mounted outside of the chamber. Collectively, the imaging systems will enable color, CH*, and OH* imaging; pyrometry; measurement of the soot volume fraction; and operational support. However, it is unlikely that all functions would be exercised at any one time as will become apparent in the subsequent discussion.

Color imaging is provided by two cameras: a high-definition digital camera mounted outside of the chamber and a standard analog camera mounted on the chamber insert (with a turning mirror). Both cameras are part of the ACME hardware that is now in development. The analog camera signal will be downlinked in near real time to support the experiment operations. Light-emitting diodes within the chamber can be used to illuminate the igniter and burner to both support the operations and provide reference images. The 12-bit digital camera records at up to 30 full frames (1360x1024) per second, and is equipped with a motorized lens with independent control of the zoom, iris, and focus. The digital camera will be equipped with a filter assembly with four options, where selection of a clear "filter" allows for color imaging. Color imaging can also be carried out with a blue-green BG7 Schott glass filter, which balances the color planes to avoid saturation in the red while maintaining sensitivity in the blue and green. The natural flame color can be approximately reconstructed, but the primary purpose of the filter is pyrometry and measurement of the soot volume fraction, again via the techniques of [23-24]. Alternately, CH* imaging can be conducted with this camera because the filter assembly includes a 430-nm bandpass filter with a 10-nm full width at half-maximum transmission (FWHM). A 450-nm bandpass filter (again with a 10-nm FWHM) can be used with the CH* filter to effectively subtract interfering broadband soot emission. However, the filter assembly rotates slowly so such subtraction effectively requires a separate test with each filter and of course reproducible flame conditions.

The video from the analog operations (i.e., ops) camera, also referred to as the internal camera, is not archived in space. The video is instead recorded on the ground in the operations center. Its gain was sometimes adjusted for improved imaging of dim flames, e.g., approaching extinction.

The sensitivity of the color digital camera, often referenced as the ACME DataCam (i.e., Data Camera), to the dim microgravity flames was such that it was often used with long exposure times and thus low framing rates, for example on the order of 1 frame per second. The clear "filter" is

of fused silica. Unfortunately, the thickness of the blue-green BG7 Schott glass filter and the added antireflective (AR) coating caused too great a reduction in the red, so it does not balance the intensity of the three color planes as intended and thereby can limit the accuracy of temperature determination. Meanwhile, rather than the 450-nm bandpass filter described above, the high definition color camera was equipped with a bandpass filter for imaging formaldehyde emissions. For more information, see the Data Camera Characterization Report (ACME-RPT-010 Rev. A).

Imaging of the dim hydrogen flames in the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment was particularly challenging with these, if not all, cameras.

Pyrometry can alternately be accomplished with CIR's monochrome High-Bit Depth Multispectral (HiBMs) camera, equipped with a Liquid-Crystal Tunable Filter (LCTF), which allows for imaging at multiple spectra, but requires a quasi-steady flame for measurement at each desired band. This camera has a telecentric optical system and is recorded at 512x512 and 30 frames per second (fps). Although it has a 1024x1024 pixel array, binning has been necessary because of signal loss through the LCTF. Both soot pyrometry and Thin Filament Pyrometry (TFP) are possible, where ACME is equipped with a crew-replaceable array of 5 silicon carbide (SiC) fibers. The fibers are orthogonal to the flame axis and the array can be translated along the flame axis to the desired position downstream of the burner, e.g., based on the downlink of the analog video. During set up, the crew can manually orient the array so that it is orthogonal to either ACME's color digital camera or the HiBMs camera with the LCTF. An alternate fiber array including a platinum wire will be used for inflight characterization, which in combination with preflight calibration will enable determination of the soot volume fraction [24].

In contrast to the preceding paragraph, ACME did not use a HiBMs camera for pyrometry, but instead relied on use of ACME's color DataCam for such temperature measurements. For an example of how this can be done, see the techniques described in the following papers:

P.B. Kuhn, B. Ma, B.C. Connelly, M.D. Smooke, and M.B. Long, "Soot and Thin-filament Pyrometry Using a Color Digital Camera," Proceedings of the Combustion Institute, 33 (2011) 743-750.

B. Ma, M. B. Long, "Absolute light calibration using S-type thermocouples," Proceedings of the Combustion Institute, 34 (2) (2013) 3531-3539.

Thus far, TFP tests have been conducted with 14-micron fibers spaced apart by 5 mm for a total span of 20 mm (in the case of 5 intact ceramic fibers), neglecting optional translation of the array. Tests have been conducted using arrays with and without the platinum wire in the furthest downstream position. Tests have also been conducted with some of the fibers broken and subsequently removed, leaving only three fibers in the final testing for the BRE and Coflow Laminar Diffusion Flame (CLD Flame) experiments and at least the initial testing for the Cool Flames Investigation with Gases (CFI-G).

CIR has a second HiBMs camera, without a filter, that will be used in combination with CIR's illumination package for alternate measurements of the soot volume fraction. This will be accomplished via an extinction technique previously employed in microgravity [3,6,26], where the camera and the collimated light source (with its two fiber-coupled laser diodes) are positioned at opposite windows. The images are recorded at 1024x1024 and 30 fps. Thus far, the primary purpose of this combined system has been the measurement of the droplet diameter for determination of the droplet burning rate.

A HiBMs camera without the Liquid Crystal Tunable Filter (LCTF) has been used throughout much, but not all, of ACME's operations. In the first round of the CLD Flame experiment, it was used without a filter for flame imaging, but this proved to be largely unsuccessful because the camera is very insensitive to blue. Afterward, it was equipped with a red filter corresponding to the CIR illumination package for the rest of the ACME testing. Unfortunately, that filter is sufficiently broad that significant incandescence can pass through the filter limiting its utility for determinations of soot volume fraction via light extinction, e.g., through the technique described in the following papers:

D.L. Urban, Z.G. Yuan, P.B. Sunderland, K.C., Lin, Z. Dai, and G.M. Faeth, "Smoke-Point Properties of Nonbuoyant Round Laminar Jet Diffusion Flames," Proceedings of the Twenty-Eighth Symposium (International) on Combustion 28 (2000) 1965-1972.

F.J. Diez, C. Aalburg, P.B. Sunderland, D.L. Urban, Z.-G. Yuan, G.M. Faeth, "Soot properties of laminar jet diffusion flames in microgravity," Combustion and Flame, 156 (2009) 1514-1524.

P.S. Greenberg and J.C. Ku, "Soot Volume Fraction Imaging," Applied Optics, 36 (22) (1997) 5514-5522.

Nonetheless, soot volume fraction can also be determined from the color DataCam images.

During ACME operations, the HiBMs camera is typically archived at 15 fps because of data limits for transfer and downlink operations. And because operations are constrained by the data limits, image data from this camera are sometimes deleted without downlinking to enable additional testing. And in some testing, HiBMs images aren't even acquired given their lesser value.

Unfortunately, the HiBMs backlight illumination is significantly obstructed by the bracket for the electrode mesh further limiting its utility for the E-FIELD Flames experiment. This was a design trade-off necessitated primarily by ACME's large number of cameras and other optical instruments.

CIR's Low-Light-Level Ultraviolet (LLL-UV) package is a monochrome camera with an intensifier (for 280-700 nm) and a 310-nm bandpass filter (with a 10-nm FWHM) for OH* imaging. Again as a result of the signal loss through the filter, binning is employed such that the OH* images are recorded at 512x512 and 30 fps.

During ACME operations, the LLL-UV camera has only been used for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment.

While not discussed in the preceding section, another intensified camera was used throughout ACME testing, although its image data was sometimes deleted or not even acquired to allow the performance of additional tests. This was the Cool Flames Investigation (CFI) camera which was generally configured with its CH filter, where an exception is some s-Flame testing with particularly dim flames. Due to design trade-offs, its Field of View (FOV) was partially obstructed by the arms of the Thin Filament Pyrometry (TFP) fiber array when it was moved inwards to the flame. More information about this camera can be found in its characterization report, CIR-RPT-4600.*

The ACME chamber insert will also be equipped with non-imaging optical diagnostics including both thermopile radiometers and Photomultiplier Tubes (PMTs). There is a crew-exchangeable

linear array of radiometers of which there are two versions supporting the initial set of experiments. The array has a fixed position which is parallel to the chamber axis and centered (along the chamber length) relative to the chamber windows and thus most of the other instrumentation. One array is equipped with a single wide-view radiometer with two narrow-view radiometers that are downstream and two more that are upstream, where all views are orthogonal to the chamber axis, as depicted in Figure 4. The narrow-view radiometers are intended to support studies of gas-jet flames, where ACME can accommodate shorter burners, making the "lower" detectors useful. The other array, also seen in Fig. 4, provides for three wide-view measurements of the flame, where one view is masked to nominally block radiant emission from the burner. It was specifically designed for the BRE experiment but will be a resource available to other ACME experiments. Given that the radiometer array can be exchanged by the crew, alternate arrays could be developed.

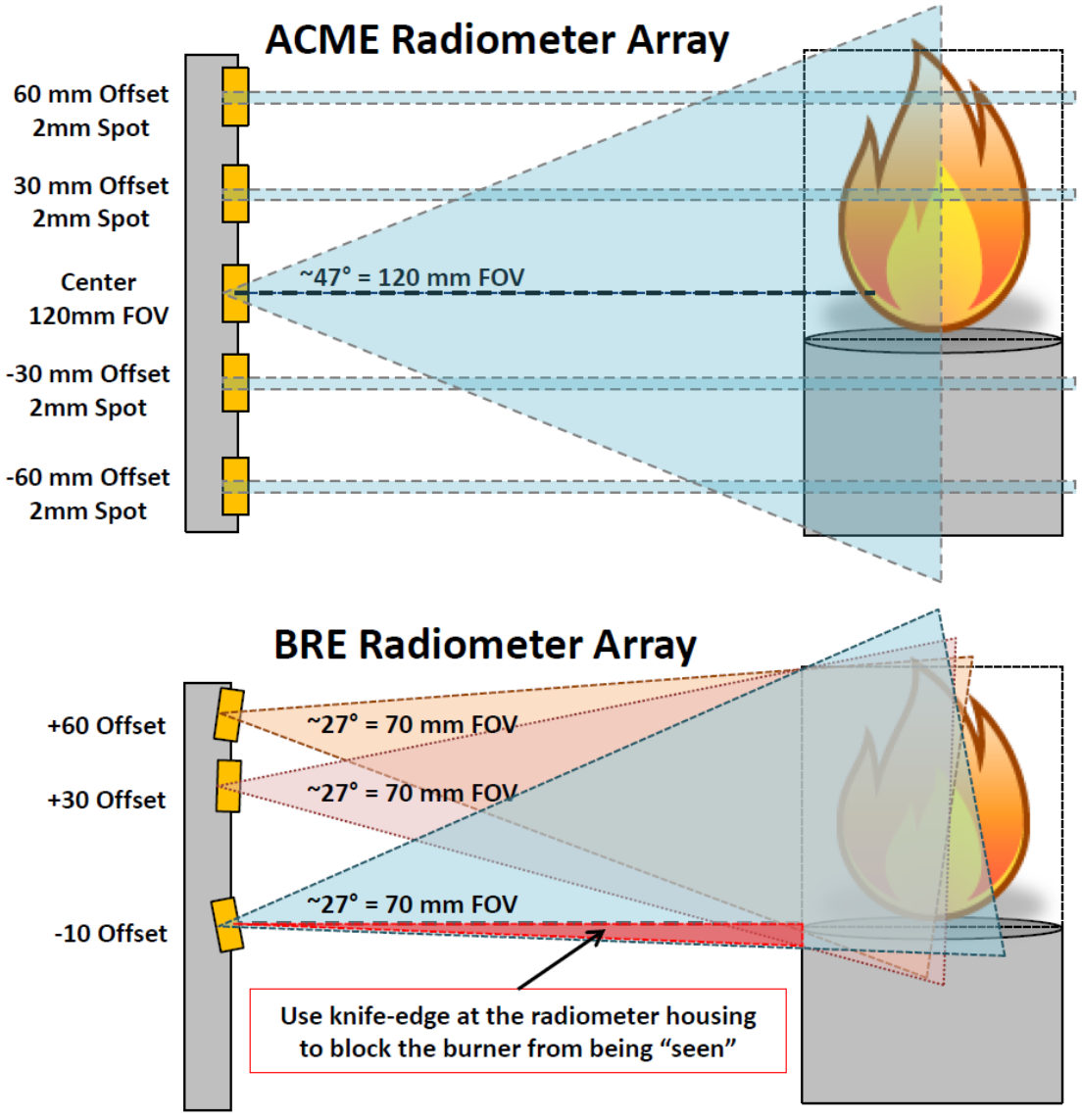


Figure 4: Crew-exchangeable radiometer arrays on the ACME chamber insert: (top) standard (ACME) array of 5 radiometers, and (bottom) array with three radiometers, designed for the BRE experiment, where the lowest detector is masked to avoid view of the burner.

A third radiometer array was developed for the CFI-G experiment, where it is important to understand that only a single radiometer array can be used at a time.

The narrow-view radiometers in the standard array are only fully sensitive in small spots, but they are each sensitive in a much greater Field of View (FOV) than represented in the preceding figure. They do not have lenses; their FOV is a result of an aperture where most of their FOV is within the penumbra which is only visible to a portion of the sensor. Figures representing the measured FOVs of the as-built standard and BRE radiometer arrays are found in section 3.F of this document.

Unfortunately, the radiometers in the BRE array typically saturated in the first round of BRE testing, so the standard array was used in its second round. However, those radiometers similarly saturated in many of round 2 tests. In this regard, each of ACME's BRE and standard radiometers have a fixed gain. Meanwhile, each sensor in the radiometer array developed for the CFI-G investigation has two fixed gains, i.e., a high gain and low gain.

For more information about the standard radiometer array used for most ACME testing, see the Radiometer Characterization Report (ACME-RPT-011). Meanwhile, you can learn about the other radiometer arrays from the BRE Radiometer Characterization Report (ACME-RPT-012) and the Cool Flames Radiometer Characterization Report (CIR-RPT-4600).

An additional wide-view radiometer, that is fixed and not crew exchangeable, is positioned downstream of the burner and flame at 30° from the chamber axis. This detector is primarily intended to support the spherical flame experiments and will have a quicker response than the more sensitive detectors that will used in the linear arrays. Like the radiometers on the two linear arrays, it will have a spectral range of nominally 0.2-11 micron and its field of view will be established by an appropriately selected aperture.

The FOV of the off-set radiometer is also shown in section 3.F. Learn more about this sensor from the Radiometer Characterization Report (ACME-RPT-011).

To detect the quenching of dim flames, the chamber insert will be equipped with a set of three PMTs. Each PMT will have a wide-angle view, a sight line that is orthogonal to the chamber axis, and will be centered (along the chamber length) relative to the chamber windows so that it is aligned with most of the other instrumentation. The PMTs' spectral range will be 230-700 nm, i.e., visible and near ultraviolet, but one will have a 310-nm bandpass filter and another will have a 430-nm bandpass filter (where both are with a 10-nm FWHM). The pair of detectors will enable measurement of OH* and CH* emission, respectively. The third PMT will be equipped with a neutral density filter to prevent saturation. Like many other components, the PMT array can be removed from the chamber insert by the crew.

Unlike the radiometers, each PMT had a readily adjustable gain. However, the PMT signals sometimes saturated when the gains were set too high. See the PMT Characterization Report (ACME-RPT-013) to learn more about the three Photomultiplier Tube (PMTs).

The chamber insert is also equipped with six far-field thermocouples to measure the spread of the thermal field in the spherical flame experiments. They are of limited utility for other experiments because they are positioned upstream of the burner tip. Therefore, they will

generally not be installed by the crew to reduce the chance of accidental damage. There will be two versions of a four-thermocouple rake to allow a choice of near and far measurements. Other arrays could be designed to support future experiments, but they cannot extend downstream of the burner tip without interfering with other (e.g., optical) measurements.

The two lone thermocouples, which are both at nominally 200 mm from the center of the spherical burner, have been installed throughout ACME testing. The tip of one was seen to be slightly bent early in testing, as discussed and depicted in section 3.G of this report.

Only the short (or far) version of the radiometer rake has been used thus far, where factors include the need for crew time to switch between rakes and the heat loss that could result from thermocouples in the long (close) rake. The short (far) rake's thermocouple tips should nominally be at 45, 60, 100, and 100 mm from the spherical burner center. For thermocouples at the same distance from the burner center, the angle between the rays from the burner center to the two thermocouples should be 90 degrees. The bead size of the thermocouples was not measured, but it is undoubtedly at least twice the size of the 0.2 mm diameter wire.

The two thermocouples closest to the burner outlets on each four-thermocouple rake are type B, while the others are type K.

Experiment-unique hardware is also mounted on the chamber insert. For example, to accommodate the E-FIELD Flames experiment, the insert is equipped with a high-voltage supply and a crew-removable electrode mesh which can be set between +10 and -10 kV. The circular mesh is orthogonal to the chamber axis and is positioned (during set up) at either ~35 or ~50 mm downstream of the insulated burner. The ion current between the burner and electrode mesh will be determined via measurement of the voltage drop across a known resistor and application of Ohm's Law.

Additional images of the chamber insert can be seen in subsequent sections of this report. The hardware specific to E-FIELD Flames is discussed further in section 2.B.

Additional supporting measurements will be made, where the chamber pressure is a simple example. The ACME experiments will use the CIR's gas chromatograph for pre- and post-test analysis of the chamber atmosphere, e.g., for oxygen, nitrogen, carbon dioxide, carbon monoxide, and fuel gases. Acceleration measurements will be provided by the Space Acceleration Measurement System (SAMS) [27], which is the primary instrument for such measurements on ISS and previously served that role for the space shuttle.

The CIR's Gas Chromatograph Instrumentation Package (GCIP) was not used for the CLD Flame or E-FIELD Flames investigations and thus far was only sometimes used during BRE, Flame Design, and s-Flame testing. Its use requires significant time and as such effectively reduces the number of tests that can be conducted. However, estimates of the chamber contents were computed assuming that the fuel in a normal flame test (for example) was either fully or not at all oxidized, e.g., for unsustained ignition in the latter case.

The CIR facility was equipped with a 3-axis SAMS sensor head and the acceleration measurements, along with relevant documentation, are accessible via the home page for the Principal Investigator

Microgravity Services (PIMS), https://gipoc.grc.nasa.gov/wp/pims/home/. To be clear, the acceleration measurements are not archived with the ACME experiment data in the Physical Sciences Informatics (PSI) website, www.nasa.gov/PSI.

All ACME experiments will be constant volume (i.e., rather than constant pressure) studies, where the chamber's free volume will be about 90 liters. Burner flows will generally be constrained to limit the pressure rise, and tests will typically be 0.5-2 minutes in duration.

The chamber's free volume is approximately 83 liters depending on the installed hardware.

The ACME experiments will be configured, but not conducted, by the ISS crew. Instead, the experiments will be commanded from the ground, specifically from the Telescience Support Center (TSC) at the NASA Glenn Research Center in Cleveland, Ohio. They can be conducted in a nominally automated mode using pre-programmed scripts. However, it will also be possible to effectively operate the experiments in a "manual" mode, where changes can be made in response to the analog video downlink and the numerical data available in the telemetry stream. For example, a flow rate could be adjusted to approach a stability or soot limit. Step and ramp changes can be made in the flow rate(s) or electric field potential. The TFP fiber array can be translated uniformly through a soot-free flame to get a continuum measurement of the temperature field. In summary, ACME's experiment control will allow for broad flexibility.

The TSC is now also referred to as the Glenn ISS Payload Operations Center (GIPOC). Thus far, the ACME tests have always been conducted in a nominally automated mode.

B. Overview for E-FIELD Flames

While focused on the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment, this section provides a good general description relevant to all ACME experiments. This text is from the 'Methods / Experimental' section of the following conference paper:

> *Y.-C. Chien, J. Tinajero, D. Stocker, U. Hegde, and D. Dunn-Rankin, "Electric Field Effects on Flames in Microgravity on the International Space Station," 2018 Spring Technical Meeting of the Central States Section of The Combustion Institute, Minneapolis, MN, 20-22 May 2018.*

See the paper itself for the references, etc. as they not included in this report. As a minor note, the sizes of some figures were adjusted in this report, but none were cropped. The figures have not been renumbered herein but are effectively 5-8 for this report (rather than 1-4).

In contrast to the preceding section, this paper was written after the beginning of E-FIELD Flames (and CLD Flame) operations on the ISS. However, additional details are again included within this section in boxed notes like this.

The E-FIELD Flames experiment is being conducted in 2018 within the ISS' Combustion Integrated Rack (CIR) [15], which was launched in 2008 and used from 2009 to 2017 for a series of droplet combustion experiments [16]. The CIR facility is similar in size to a vending machine as can be seen in Figure 1. It includes a 100-liter combustion chamber capable of operation at pressures of up to 300 kPa. The chamber features a ring of 8 sapphire windows and is equipped with a set of reconfigurable imaging systems. The CIR was designed with gas delivery, venting, thermal control, electrical power, data handling, image processing and storage, and vibration isolation to enable combustion research with gaseous, liquid, and solid fuels.

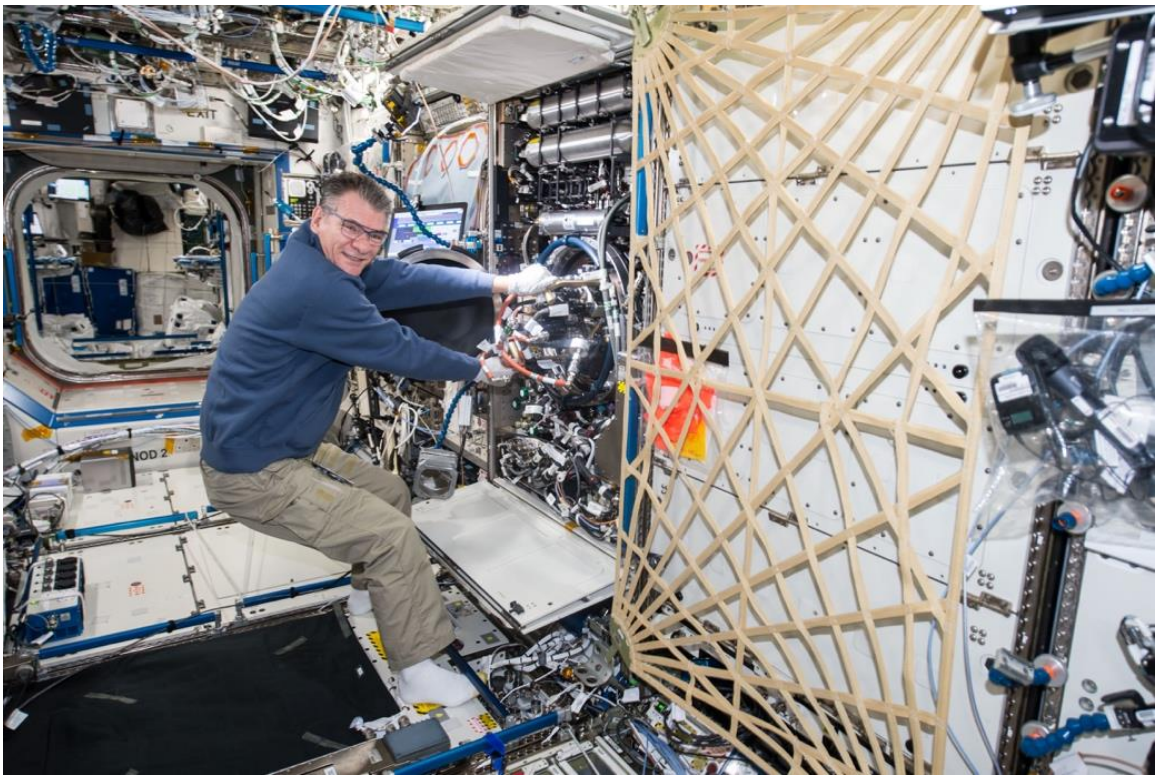


Figure 1: European Space Agency (ESA) astronaut Paolo Nespoli installing the chamber insert for the Advanced Combustion via Microgravity Experiments (ACME) in the Combustion Integrated Rack (CIR). This photograph [ISS053e098207] was taken on the International Space Station (ISS) on 12 Oct. 2017.

E-FIELD Flames is one of the six unique experiments featuring non-premixed laminar flames of gaseous fuels which currently make up the ACME project [17]. A modular set of hardware, including the critical chamber insert depicted in Figures 2 and 3, was developed to carry out the ACME investigations in the CIR facility. The 0.4-m diameter ACME chamber insert provides common mounting for a variety of burners on the chamber axis (but only one at a time) and a retracting resistively-heated igniter. The igniter has a crew-exchangeable tip and arm (to accommodate the various burners) and stepper-motor drive for radial positioning. The chamber insert includes three mass flow controllers (visible on the far left of Figures 2-3 behind an orange hose and black cables) for delivering fuel, oxidizer (or an inert), and nitrogen to the burner or chamber. The crew can exchange the mass flow controllers to provide appropriate accuracy for the desired flow ranges. ACME's gas delivery system enables on-orbit dilution of the gaseous fuel or oxidizer with nitrogen (available from the space station), but not both simultaneously. However, bottles of premixed fuel and oxidizer are also used, where the CIR's standard bottle sizes are 1.00, 2.25, and 3.80 liters. Fuels are stored in 1-liter bottles, while the 21/79 oxygen/nitrogen mixture (on a volume basis) used by E-FIELD Flames comes in 3.80-liter bottles. The ACME experiments are constant volume studies, where the chamber's free volume, with the insert installed, is about 90 liters. Burner flows are generally constrained to limit the pressure rise and vitiation of the atmosphere, where typical ACME tests are only 0.5-2 minutes in duration.

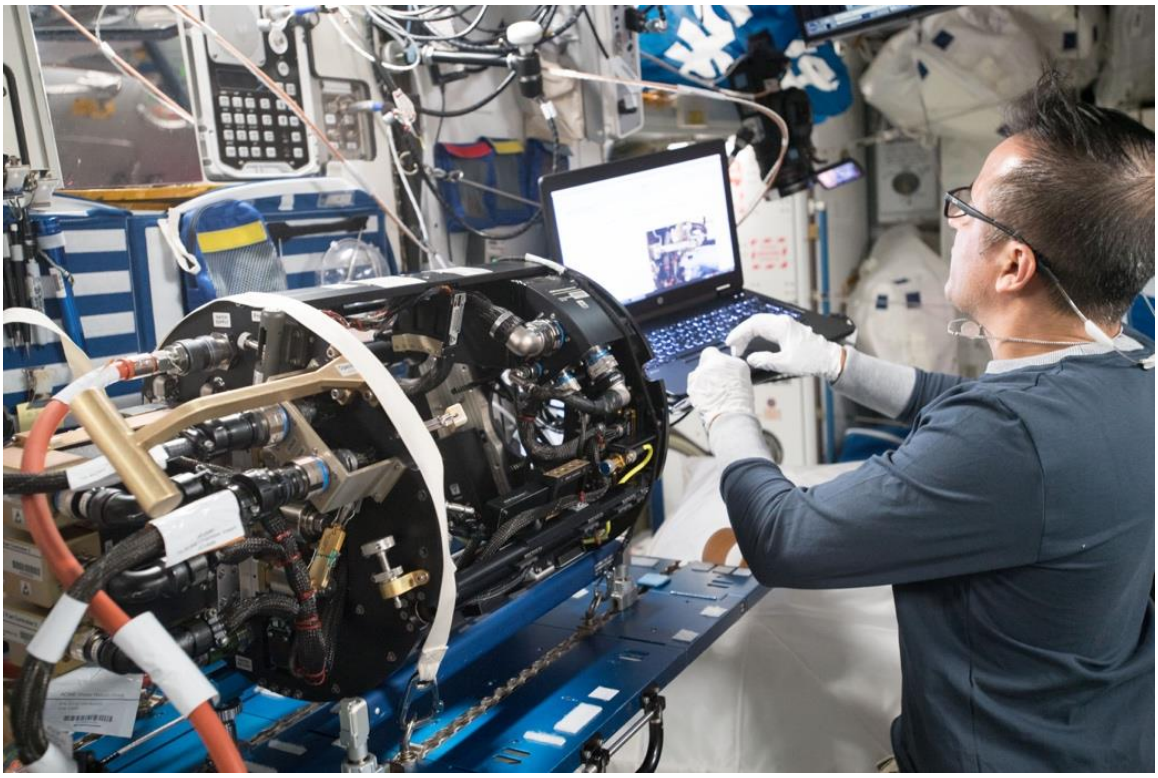


Figure 2: NASA astronaut Joe Acaba reviewing procedures to align the hot-wire igniter on the ACME chamber insert to his left, in a photograph [ISS054e014660] taken by NASA astronaut Mark Vande Hei on 5 Jan. 2018.

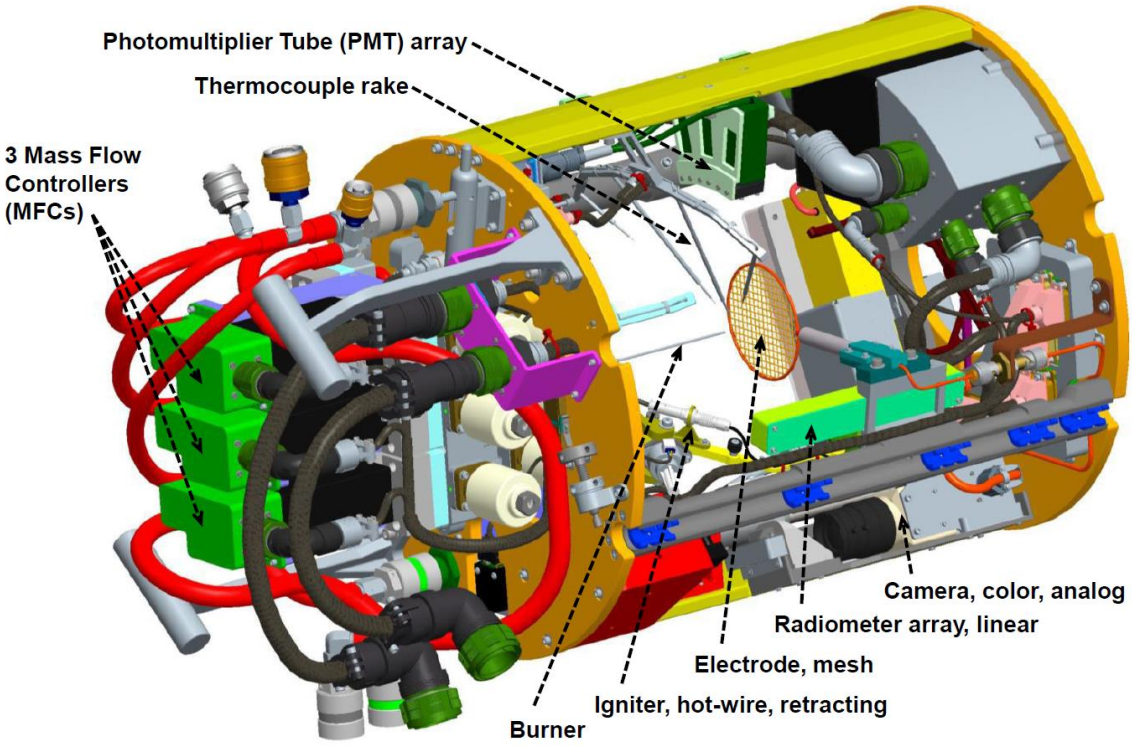


Figure 3: Solid model representation of the ACME chamber insert.

The E-FIELD Flames experiment is being conducted with two different burners: a simple gas-jet burner and a coflow burner, which can be seen in Figure 4(a). The results discussed in this paper are only from the first of the experiment's tests conducted with the coflow burner. The stainless steel fuel tube in the burner has inner and outer diameter of 2.2 and 2.4 mm, respectively. The coflow burner's steel outer tube has an inner diameter of 25 mm, where the annulus outlet is a 6.4-mm thick Hastelloy X honeycomb with 0.05-mm foil thickness and 0.81-mm cell size. The coflow burner's inner tube protrudes 3.1 mm beyond the surface of the honeycomb outlet. While the inner tube is being used exclusively for fuel flow in E-FIELD Flames, valves in the ACME chamber insert allow the fuel, oxidizer, and nitrogen to be directed to either tube, making this burner capable of inverse flame testing.

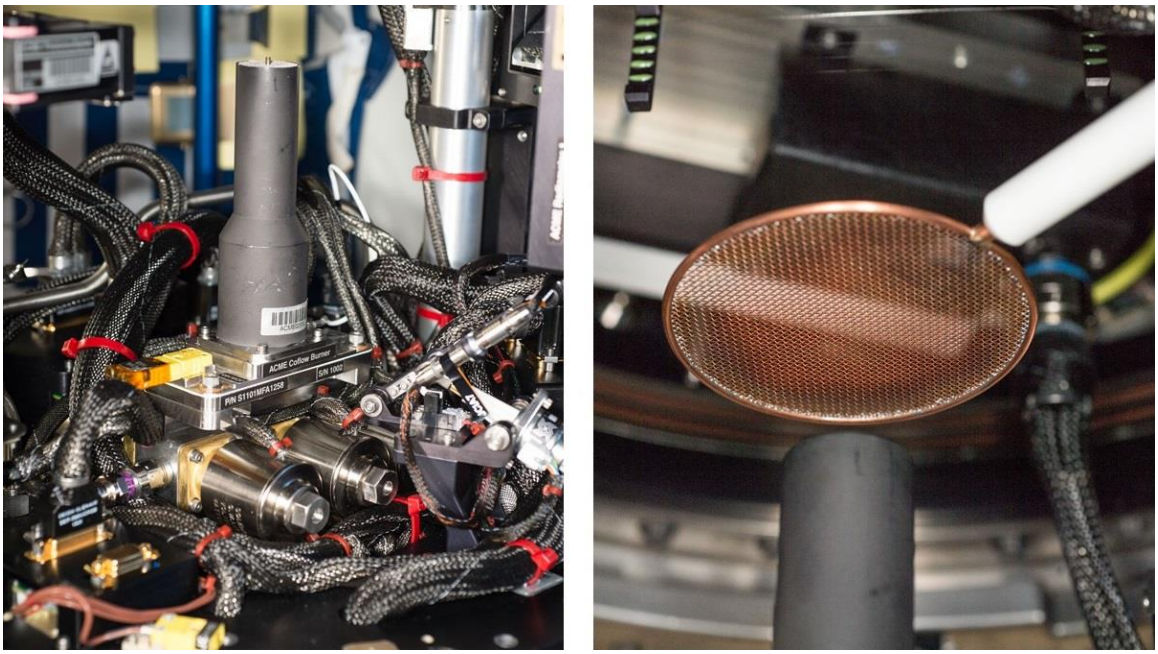


Figure 4: (a) Detail of the ACME chamber insert featuring the coflow burner (in the image center) used by the E-FIELD Flames experiment. This is from a photograph [ISS053e096074] taken by ESA astronaut Paolo Nespoli on 12 Oct. 2017 after he installed the burner on the chamber insert. (b) Detail of the ACME chamber insert where the copper electrode mesh, used by the E-FIELD Flames experiment, can be seen above the coflow burner. This is from a photograph [ISS055e001252] taken by NASA astronaut Scott Tingle on 8 March 2018 following his installation of the electrode mesh.

In this study, a high-voltage electric field is generated between the burner and a copper electrode mesh located above (i.e., downstream of) the burner outlet, as can be seen in Figure 4(b). The circular mesh is energized at up to 10,000 volts with either polarity, where the burner serves as the electrical return and the flow of flame-produced ions completes the circuit. The ion current between the burner and electrode mesh is measured from the voltage drop across a known resistor and application of Ohm's Law. With and without the copper ring, the electrode mesh is 8.9 and 8.4 cm in diameter, respectively. The square copper weave has 7.1 wires/cm, a wire diameter of 0.43 mm, and a 48% open area. The mesh permits gas to pass through while also acting as an electrical element. The electrode mesh is mounted 37 mm above the coflow burner tip and is centered on and orthogonal to the chamber axis. The ACME chamber insert can also

accommodate an alternate electrode position of 52 mm above the coflow burner tip, but the closer position is being used to produce the highest field strengths.

The slow response time of the high-voltage power supplies (recommended in the ACME Integrated Science Requirements Document, i.e., ISRD) did not meet the requirements necessary to achieve the requested time response performance, and as such inhibited determination of the flame's response time to rapid changes in the electric field strength.

The negative and positive high-voltage power supplies were to have been coupled, as described for example in Appendix C of the ISRD. However, switching between the two high-voltage power supplies led to excessive Electromagnetic Interference (EMI) in ground testing so the power supplies were completely decoupled, where only one was installed at a time. As such, testing was limited to a single polarity at a time in contrast to the experimental plans documented in the experiment's Science Requirements Document (SRD) and the ACME ISRD.

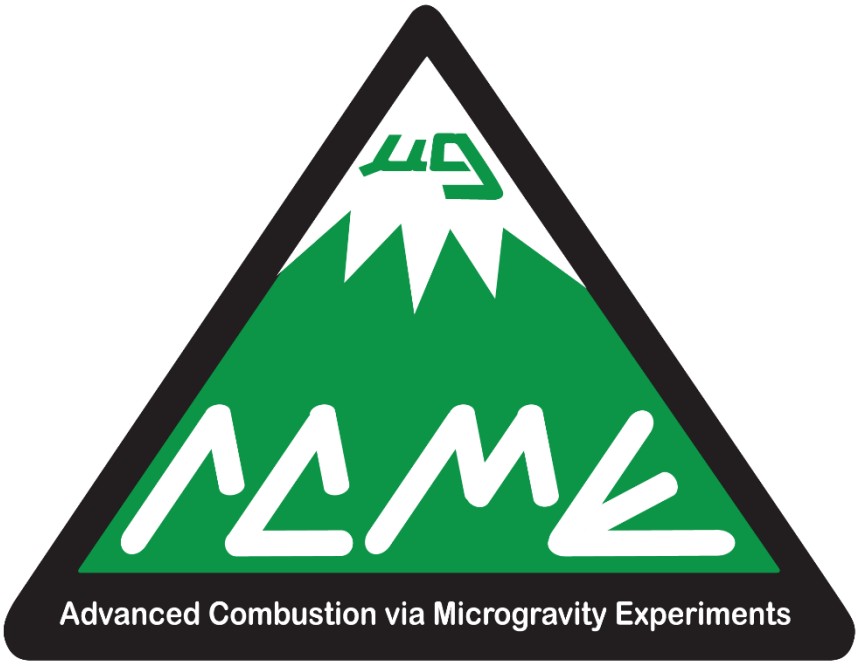
ACME's non-imaging measurements are recorded at 125 Hz. The measured test conditions include the chamber pressure and temperature, the three gas flow rates, and the electrical potential of the electrode mesh. Acceleration measurements are provided by the Space Acceleration Measurement System (SAMS) [18], which is the primary instrument for such measurements on ISS and previously served that role for the space shuttle.

The flame behavior is observed optically with a variety of instruments. For most ACME experiments, including E-FIELD Flames, the radiant emission is measured with 6 thermopile detectors with a spectral range of nominally 0.2-11 microns. Five radiometers are in a crew-exchangeable linear array that is mounted parallel to the chamber axis. The other detector is downstream and off axis from the burner. Chemiluminescent emission is measured with a set of three Photomultiplier Tube (PMT) modules. Their spectral range is 230-700 nm, i.e. in the visible and near ultraviolet, but two are equipped with narrow bandpass filters for measurement of OH* and CH* emissions, respectively. The third PMT is equipped with a neutral density filter to prevent saturation, although the gain values for all three PMTs are independently and readily adjustable. Like many of the components, the array of PMT modules can be removed from the chamber insert by the crew. In addition to their scientific value, PMT measurements are used operationally to detect ignition and automatically deactivate and retract the igniter. The radiometer or PMT measurements can similarly be used to detect extinction and automatically end the gas flow and test.

Imaging measurements of the flames complement the optical sensors described above. Color imaging is provided by an analog color camera within the chamber and a high-definition (1360 x 1024), 12-bit color camera mounted outside the chamber on CIR's optical bench. The latter is equipped with a motorized lens with independent control of the zoom, iris, and focus. It has a filter barrel with a fused silica window (for color imaging), a blue-green BG7 Schott glass filter, and bandpass filters for imaging CH* and HCHO (formaldehyde) chemiluminescence. The digital color camera was characterized prior to launch for color-ratio pyrometry and measurement of the soot volume fraction, via the techniques of [19, 20]. The camera makes soot pyrometry possible as well as Thin Filament Pyrometry (TFP), where the chamber insert is equipped with a translating array of silicon-carbide (SiC) fibers. A 14-micron fiber array can be seen at the top of Figure 4(b). However, the fibers interfere with the electric field preventing the use of Thin Filament Pyrometry in the E-FIELD Flames experiment.

Two intensified monochrome cameras mounted outside the combustion chamber are available for ACME use, although only one is being used by E-FIELD Flames. It is a high-definition (1390x1040), 12-bit camera equipped with a filter barrel similar to that used by the digital color camera discussed in the preceding paragraph, although it doesn't have a blue-green filter. The CH* filter has been used in the intensified photography of the E-FIELD Flames tests. While unused by E-FIELD Flames, the CIR is also equipped with a high-definition (1024x1024) camera with an intensifier extending into the ultraviolet which can be equipped with a bandpass filter and corresponding optics for OH* imaging.

Finally, a high-definition (1024x1024), 12-bit monochrome camera is mounted outside the chamber opposite a collimated light source with two fiber-coupled laser diodes emitting red light of 650-655 nm. This camera has a bandpass filter corresponding to the illumination source, allowing for determination of soot volume fraction through an extinction technique previously employed in microgravity research [21-23]. Unfortunately, because of trade-offs demanded by limited chamber volume and large number of instruments and components, the collimated light is obstructed substantially by the electrode mesh mount, limiting the utility of this camera for the E-FIELD Flames experiment.



3. Engineering Details (selected)

This section provides additional details about some of the key engineering elements, including the chamber insert (pp., 20-22), gases and gas delivery (pp. 22-25), burners (pp. 25-50), igniter tips (pp. 50-51), optical layout (pp. 51-56), radiometers (pp. 57-58), and thermocouples (pp. 59-60). Most of the engineering subsystems are depicted in Figure 9, which is color-coded for ACME and the Combustion Integrated Rack (CIR) facility in which the ACME research is being conducted. Thus far, ACME has not used the HiBMs-2 camera as shown in the chart, but the project instead has taken advantage of the Cool Flames Investigation (CFI) intensified camera, a CIR asset developed for the CFI experiment. While not shown in the figure, ACME has also sometimes used the CIR's Gas Chromatograph Instrumentation Package (GCIP) for the BRE, Flame Design, and s-Flame experiments. This is limited to pre- or post-test measurements of the chamber atmosphere for oxygen, nitrogen, carbon dioxide, carbon monoxide, and fuel gases.

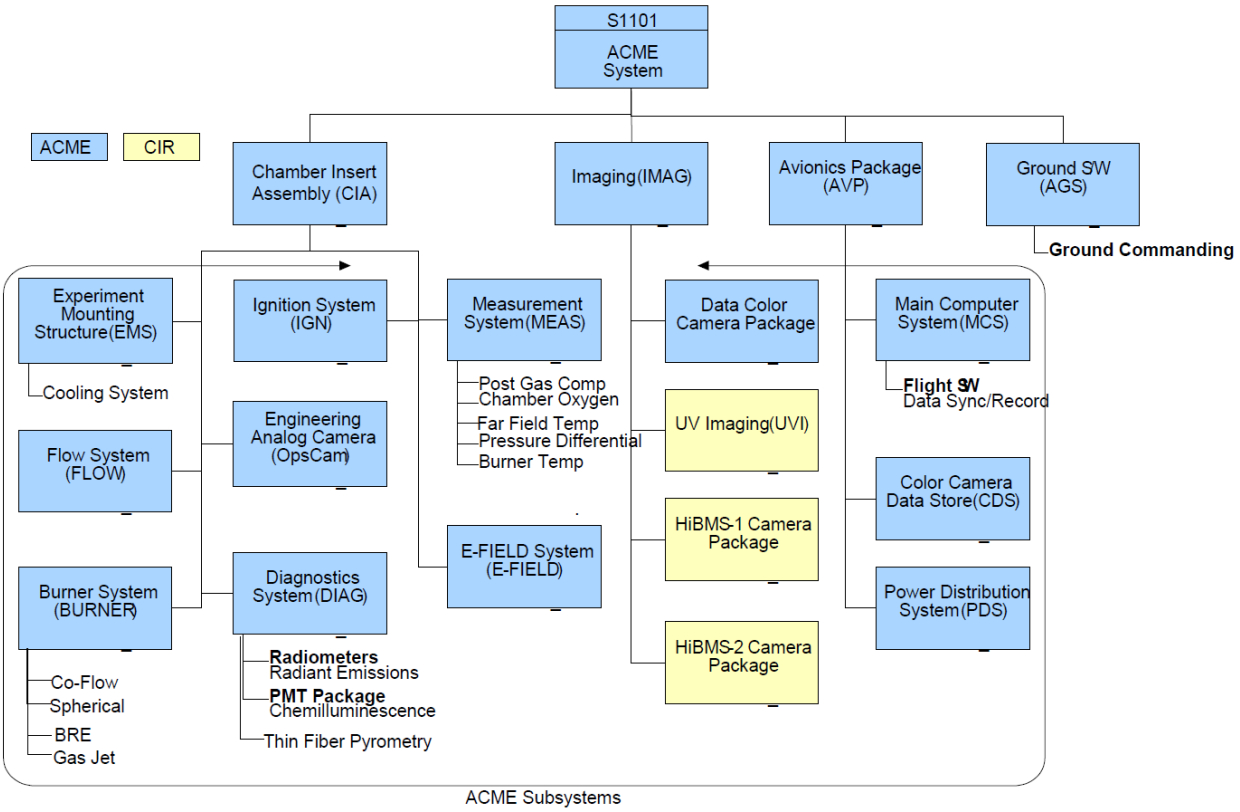


Figure 9. ACME subsystems as identified in ACME's Critical Design Review (CDR) held in Nov. 2013.

Several of the subsystems identified in the figure and preceding paragraph are addressed in specific reports:

ACME-RPT-010 Rev. A	Data Camera Characterization Report
ACME-RPT-011	Radiometer Characterization Report
ACME-RPT-012	BRE Radiometer Characterization Report
ACME-RPT-013	PMT Characterization Report
CIR-RPT-4600	Cool Flames Radiometer Characterization Report
MDC-XXX-1866-R1	CFI Intensified Camera Package Characterization Report

Meanwhile, a key document regarding the software and data files is:

ACME-DOC-010 Rev. D	ACME Software Interface Design Document

A. Chamber Insert Layout

This section provides the layout of the ACME chamber insert from different perspectives. While the layout is accurate in these solid model representations, some details are not.

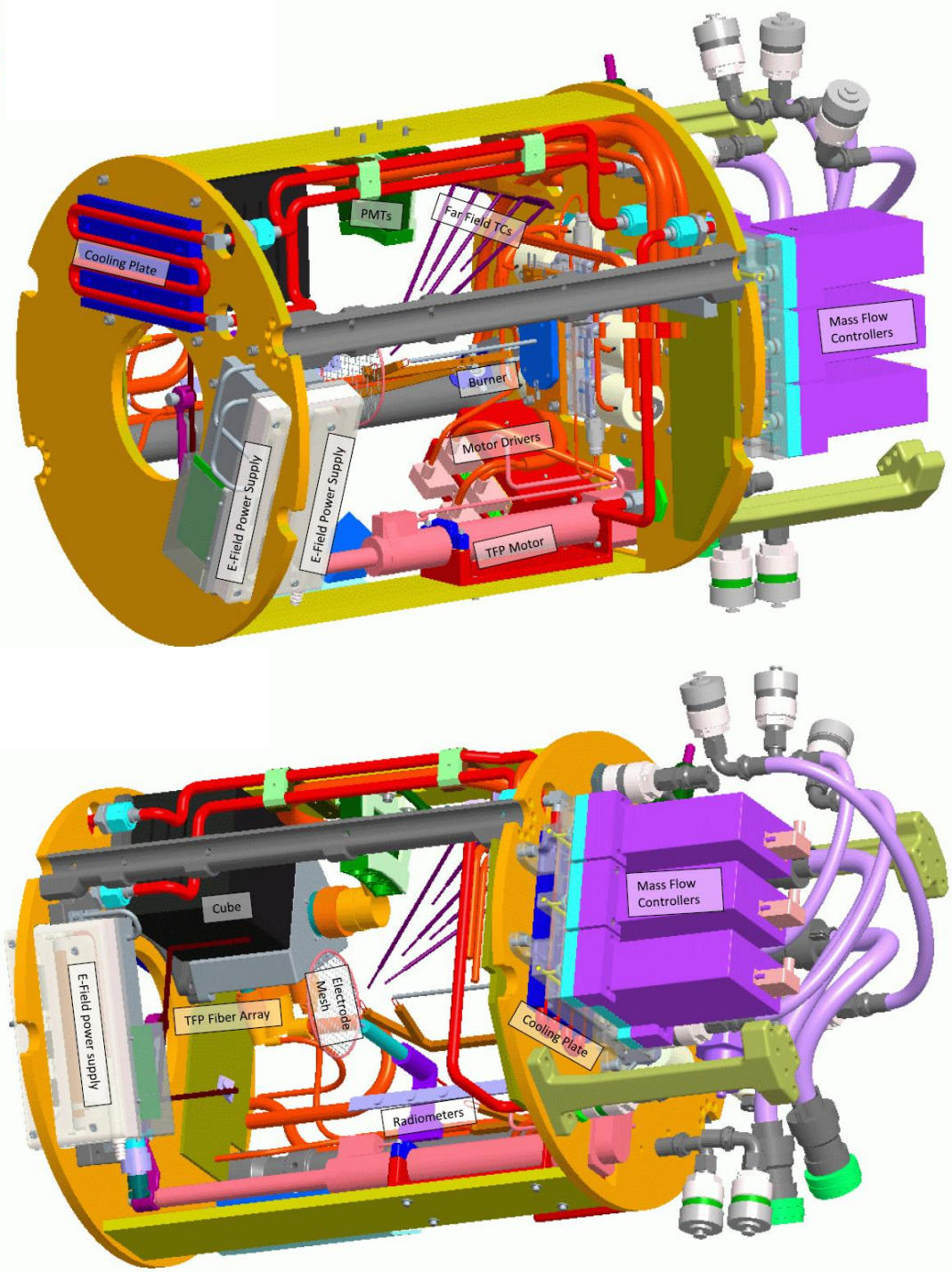


Figure 10. Solid model representation of the chamber insert from two external perspectives.

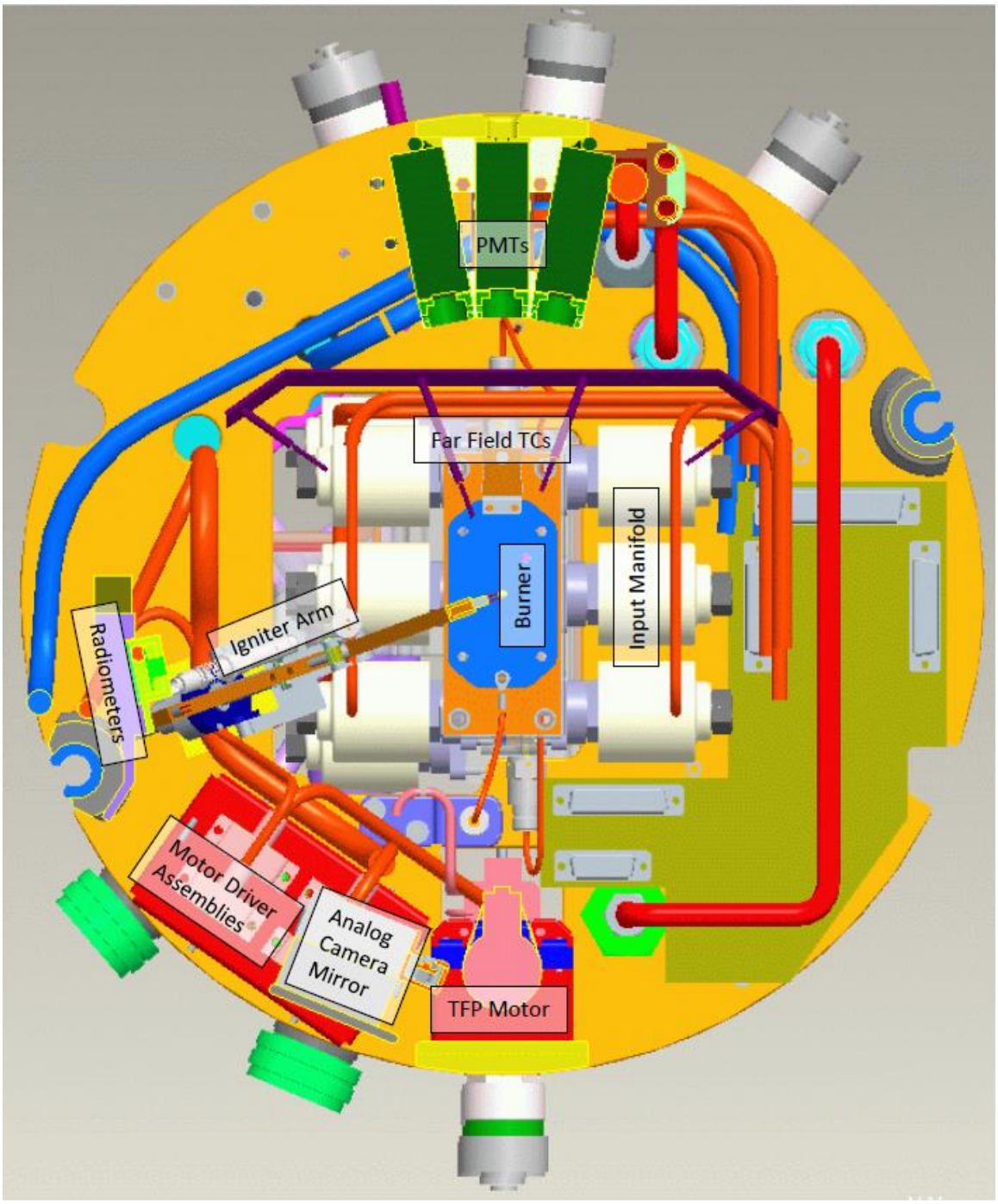


Figure 11. Solid model representation of the chamber insert's interior as seen looking toward the burner and the end plate on which it is mounted.

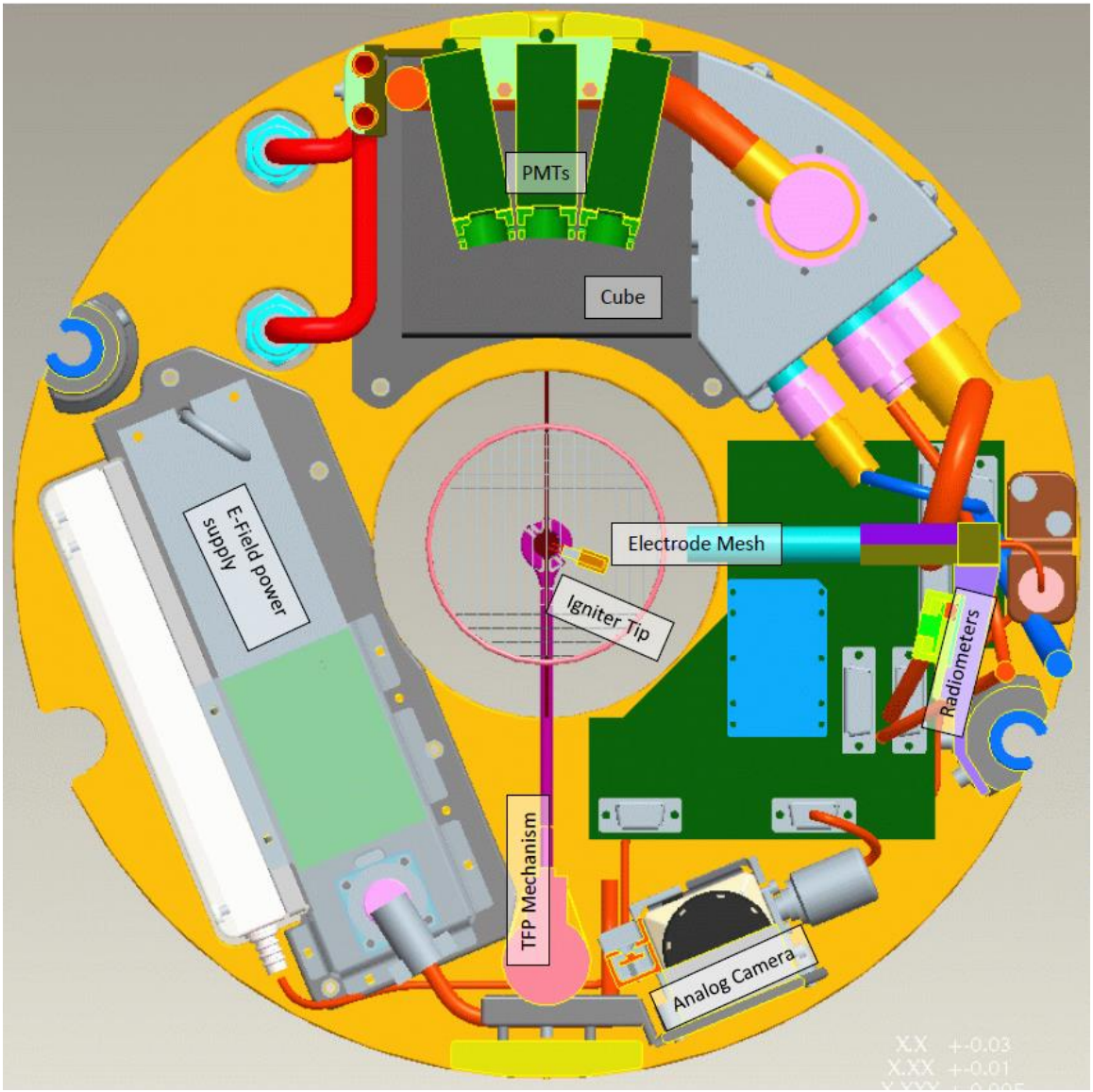


Figure 12. Solid model representation of the chamber insert's interior as seen looking away from the burner toward the opposite end plate. Note that the electrode mesh was only installed for the E-FIELD Flames experiment, where it obstructed use of the Thin Filament Pyrometry (TFP) fiber array. Furthermore, its mounting bracket interfered with illumination for possible soot volume fraction measurements via light extinction.

B. Gas Delivery

The tests are conducted using nitrogen available from the ISS and fuel, oxidizer, and inert gas bottles launched from Earth specifically for use with the CIR facility. This approach includes both the burner flow and the chamber atmosphere, where combustion tests were not conducted with the ISS cabin atmosphere, even including check-out or exploratory tests.

The pure and premixed fuel gases launched for flame tests are listed in Table 1. Planned tests for four of the six ACME experiments are limited to the use of methane and ethylene of varying dilutions. The two exceptions are the Structure and Response of Spherical Diffusion Flames (s-

Flame) experiment and Cool Flames Investigation with Gases (CFI-G). The s-Flame investigation included tests with hydrogen and hydrogen/methane mixtures while CFI-G features tests with ethane, propane, and n-butane.

Table 1. Fuel gases launched to support the ACME experiments, where mixture fractions on a volume basis. Note that calibration gases for the CIR's Gas Chromatograph Instrumentation Package (GCIP) are not included in this list.

Flammable gas(es)	**Fuels**
H_2 (hydrogen) *s-Flame only*	100% H_2 45% H_2, 55% He (helium)
H_2 and CH_4 (methane) *s-Flame only*	66.7% H_2, 33.3% CH_4 55.6% H_2, 44.4% CH_4 44.4% H_2, 55.6% CH_4 30% H_2, 15% CH_4, 55% He 25% H_2, 20% CH_4, 55% He 20% H_2, 25% CH_4, 55% He
CH_4	100% CH_4 45% CH_4, 55% N_2 (nitrogen) 40% CH_4, 60% N_2 30% CH_4, 70% N_2 45% CH_4, 55% He
C_2H_6 (ethane) *CFI-G only*	100% C_2H_6 30% C_2H_6, 70% N_2
C_2H_4 (ethylene)	100% C_2H_4 50% C_2H_4, 50% N_2 30% C_2H_4, 70% N_2 30% C_2H_4, 70% He 25% C_2H_4, 75% He 20% C_2H_4, 80% He
C_3H_8 (propane) *CFI-G only*	100% C_3H_8 30% C_3H_8, 70% N_2
C_4H_{10} (n-butane) *CFI-G only*	100% C_4H_{10} 50% C_4H_{10}, 50% N_2 30% C_4H_{10}, 70% N_2

The launched oxidizer bottles have included 21%, 30%, 40%, and 85% oxygen (by volume) with a balance of nitrogen and 30% oxygen with a balance of helium. It is important to understand that ACME testing is limited to 40% oxygen in the combustion chamber's atmosphere, where the 85% oxygen is for inverse flame testing, in which the oxidizer flows from the burner into the chamber filled with a fuel/inert mixture. Inverse flame tests are planned for both the CFI-G and Flame Design experiments, and no others, but have yet to be conducted as of this writing. Meanwhile, the only inert launched specifically to support ACME testing has been 100% helium where compressed nitrogen is available as a space station resource. Again, this discussion neglects gases for the GCIP, e.g., its carrier and calibration gases.

It is important to understand that the CIR and ACME hardware only allows for controlled nitrogen dilution of one gas at a time. For most experiments, this isn't a limitation since the chamber is

filled with its atmosphere before gas flows from the burner. But in tests using the coflow burner, either the inner or annular gas flows can be diluted, but not both at once in a controlled manner. This can be discerned from the schematic of the gas delivery found in Figure 13. As a result, one of the gases is premixed on the ground. For the CLD Flame and E-FIELD Flames investigation with the coflow burner, tests were conducted using premixed 21% O2 (79% N2) bottles while the ISS nitrogen was used to dilute the fuel.

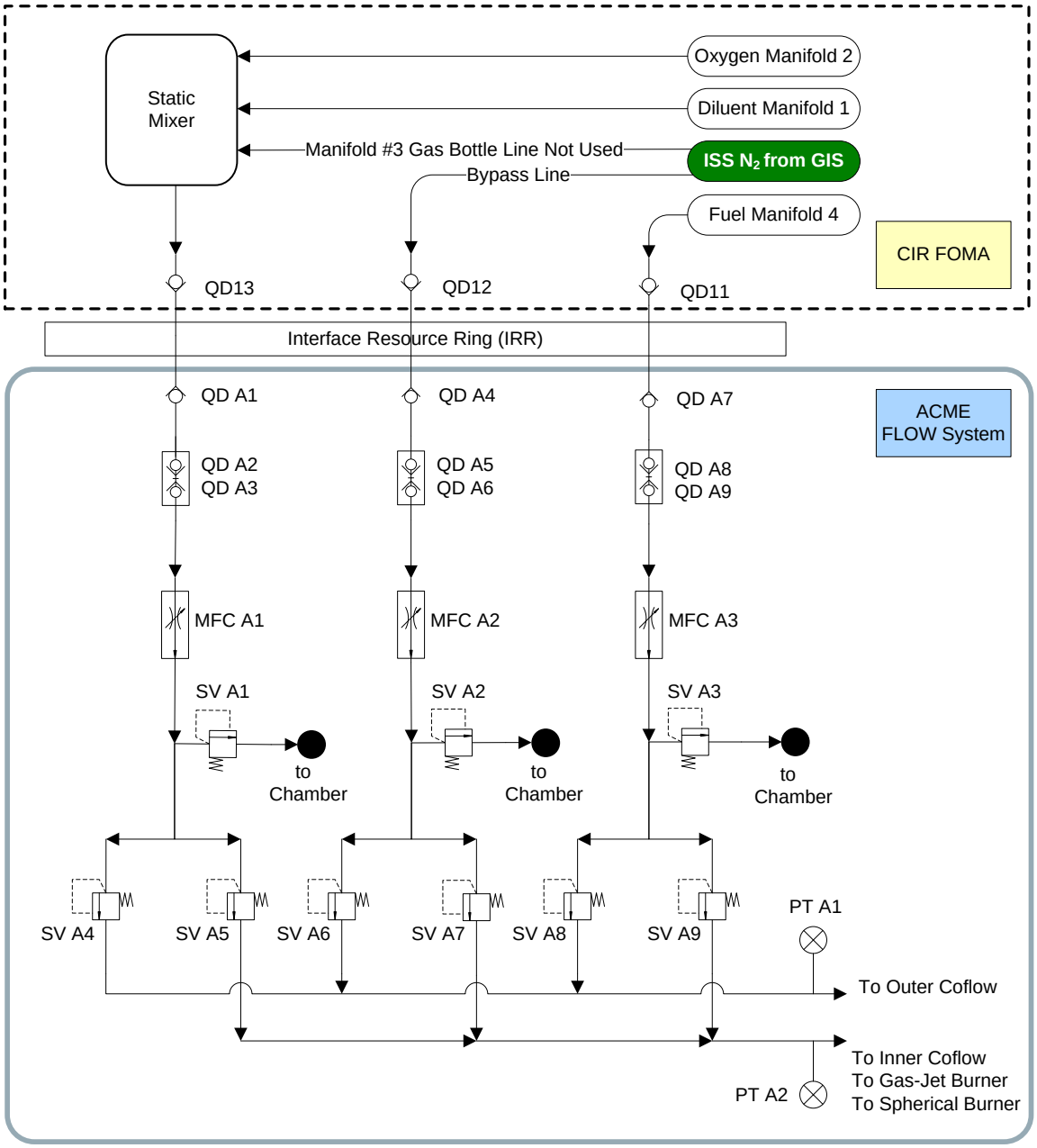


Figure 13. Schematic of gas delivery in ACME testing, where the FOMA is the CIR's Fuel and Oxidizer Management Assembly.

In ACME testing, the gas flow is controlled using the crew-exchangeable Mass Flow Controllers (MFCs) on the chamber insert, where there is one each used at a time for the fuel, oxidizer, and

nitrogen. These MFCs are available with various flow ranges as seen in Table 2, where it is advantageous in terms of flow control and uncertainty to use MFCs commensurate with the needed gas flow rates. However, the advantages of swapping MFCs must be balanced with need for crew time to swap MFCs so tests are not always conducted under optimal conditions. For example, control at small fractions of the total flow range is unlikely to be good. Note that the CIR facility is equipped with MFCs for large gas flow rates, but these were simply forced to be fully open when gas flow was desired. More specifically, the CIR MFCs are for 0-30 slpm (i.e., standard liters per minute) on a nitrogen basis, with the important exception of the fuel MFC which is 0-2 slpm on a nitrogen basis. Otherwise they are not used to control the gas flow rates in ACME tests.

Table 2. Mass Flow Controllers (MFCs) available for use on the ACME chamber insert

MFC	Gas type	# of MFCS	Maximum flow rates (on a nitrogen basis)
A1	Oxidizer (or inert)	2	0.5, 20 slpm
A2	Nitrogen	9	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 2, 5 slpm
A3	Fuel	6	0.05, 0.2, 0.1, 0.2, 0.5, 1, 2 slpm

C. Burners

As discussed in the Hardware Overview, there are four types of burners, where two types were fabricated in different sizes and multiple spaceflight copies were made and used of the spherical burner. Meanwhile there is only a single spaceflight copy of the coflow burner.

Table 3. Burner types and sizes used for the ACME experiments

Experiment	Burners used (or to be used)
BRE	BRE burners, 25-mm (rounds 1 and 2) and 50-mm (round 2 only)
CFI-G	Spherical burner(s)
CLD Flame	Coflow burner
E-FIELD Flames	Coflow burner (round 1), 2.1-mm gas-jet burner (round 2)
Flame Design	Spherical burners
s-Flame	Spherical burner

Additional details can be learned from the burner drawings listed in section 6.F.

1. Gas-Jet Burners

The fabricated gas-jet burners were limited to two sizes with nominal inner diameters of 1.3 and 2.1 mm, where two spaceflight copies of each were fabricated because of concerns that they might be damaged. However, only a single copy of the larger diameter burner has thus far been used during the spaceflight tests, where that use was limited to the second round of the E-FIELD Flames testing. This gas-jet burner was used to allow comparison with tests carried out with the coflow burner in E-FIELD Flames' first round. The same commercial stainless-steel tubing used for the coflow burner's inner tube was also used for the 2.1-mm gas-jet burner. However, it is important to note that the nominal inner diameter (ID) in both cases was 2.16 mm, i.e., approximately 2.2 mm rather than 2.1 mm. While potentially confusing, this met the tolerance requirement for the 2.1-mm gas-jet burner. Meanwhile, the nominal outer diameter (OD) was 2.41 mm, again for the 2.1-mm gas-jet burner and the inner tube of the coflow burner. It is also important to understand that the gas-jet burner is slightly taller than the coflow burner, where the

electrode mesh was nominally 35 and 37 mm downstream of the gas-jet and coflow burners, respectively (based on the outlet of the inner tube in the case of the coflow burner).

The gas-jet burner features an extremely fine, sheathed, type-K thermocouple mounted on the tube exterior with its tip 5 mm upstream of the burner outlet. In contrast, the coflow burner was not equipped with the thermocouple because of concerns that it might result in flow and thus flame non-uniformity.

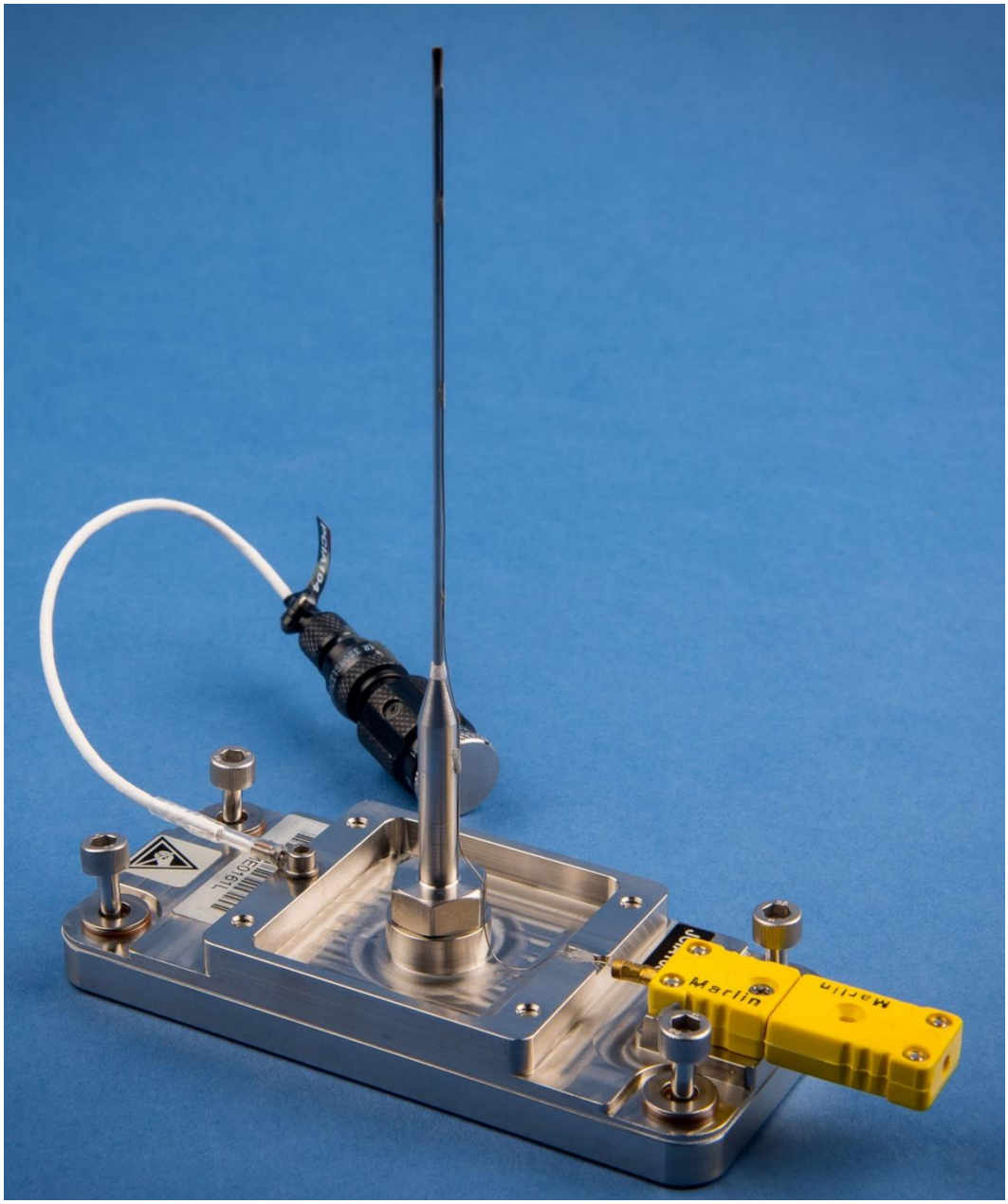


Figure 14. A gas-jet burner prior to launch. The white cable on the left was for grounding, while the yellow connector is for the burner's type-K thermocouple.

2. Coflow Burner

The coflow burner is effectively a 2.1-mm gas-jet burner within a large tube, where gas could also flow from the annulus between the two tubes. The stainless-steel fuel tube in the burner has inner and outer diameter of 2.16 and 2.41 mm, respectively. The coflow burner's steel outer tube has an inner diameter of 25 mm, where the annulus outlet is a 6.4-mm thick Hastelloy X honeycomb with 0.05-mm foil thickness and 0.81-mm cell size. The coflow burner's inner tube protrudes 3.1 mm beyond the surface of the honeycomb outlet.

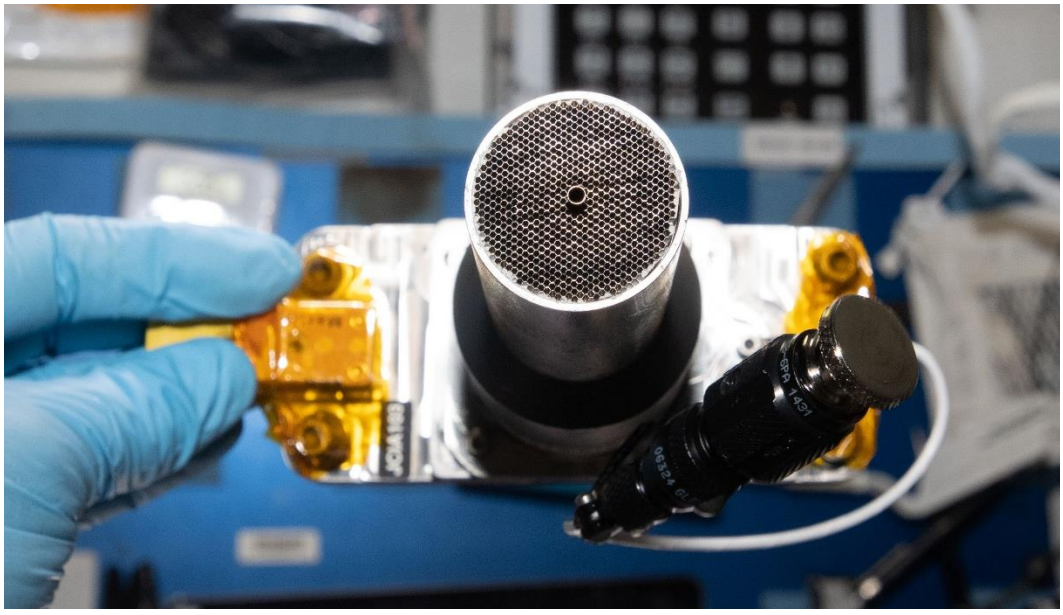


Figure 15. The coflow burner aboard the space station, where the inner tube extends downstream of the annular outlet featuring a fine honeycomb. The white cable with the black connector on the right is for grounding, while the yellow connector on the left was used to secure the cable for the burner thermocouple, even though the coflow burner isn't equipped with a thermocouple. [Cropped detail from ISS064e028847]

The coflow burner has thus far only been used for the CLD Flame and E-FIELD Flames experiments, where in both cases the inner tube was used exclusively for fuel flow. However, ACME's gas delivery manifold allows the fuel, oxidizer, and nitrogen to be directed to either tube, making the burner capable of inverse flame testing, as can be seen in section 3.B.

CLD Flame tests were always conducted with an oxidizer coflow, but some E-FIELD Flames tests with the coflow burner were conducted with no coflow. That enabled a three-way comparison with the tests conducted with a coflow and those with the gas-jet burner (which of course were without a coflow).

As previously mentioned, the coflow burner is slightly shorter than the gas-jet burner, where the electrode mesh used for the E-FIELD Flames experiment was nominally 35 and 37 mm downstream of the gas-jet and coflow burners, respectively (based on the outlet of the inner tube in the case of the coflow burner). Given the two situations, the electrode mesh can appropriately be described as positioned nominally 36 mm downstream of the

two burners.

Unlike the other burners, the coflow burner was not equipped with a thermocouple because of concerns that it could adversely result in flow and thus flame asymmetry.

The final tests from E-FIELD Flames' first round on 17 May 2018 featured sooty 100% ethylene (C_2H_4) flames and a positively-charged electrode mesh which 'pushed' the flames toward the burner leading to significant soot deposition on the coflow burner. Figure 16 reveals that soot accumulated on the burner's inner tube to the extent that its shape was effectively altered and would impede the surrounding coflow. Meanwhile, the honeycomb coflow outlet was blackened in the central region near the inner tube, as seen in Figure 16, where that soot could have obstructed and reduced the coflow in the most important region.

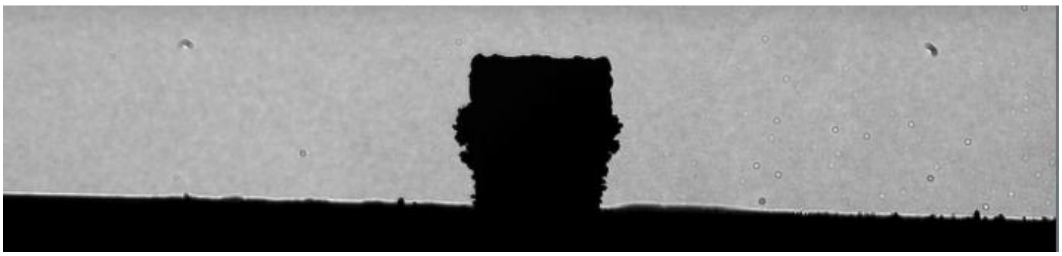


Figure 16. The coflow burner outlet imaged with the backlit HiBMs camera on 25 May 2018. The inner tube, from which the fuel exits in normal flame testing, has been coated with soot.

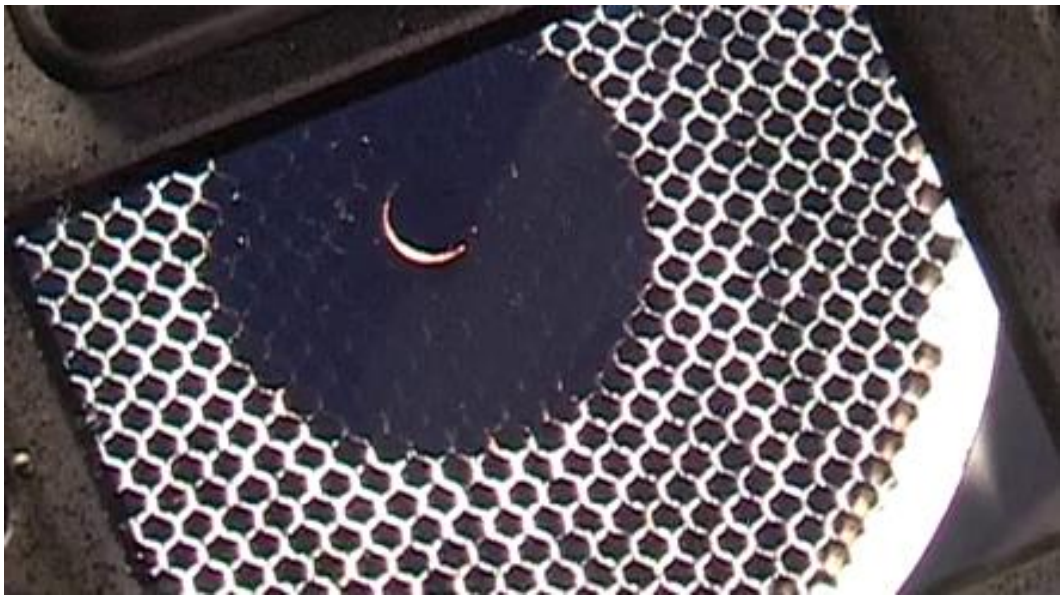


Figure 17. The coflow burner outlet (as seen in the display of a handheld digital camera) from a high-definition video downlink from the ISS on 19 June 2018 prior to the burner cleaning. The central portion of the burner's outlet, including the honeycomb is blackened by soot.

A procedure to clean the coflow burner was developed on the ground and implemented

by astronaut Ricky Arnold on 19 June 2018. During the procedure, for which example images can be seen in Figure 18, Arnold used a toothbrush, piece of insulated wire, flashlight, vacuum sweeper, and mask (for personal protection). After the cleaning, the CLD Flame Principal Investigator Marshall Long subsequently declared that "we are satisfied that the burner will be able to provide flow conditions similar to those that have been measured in our lab before the burner was placed in orbit."

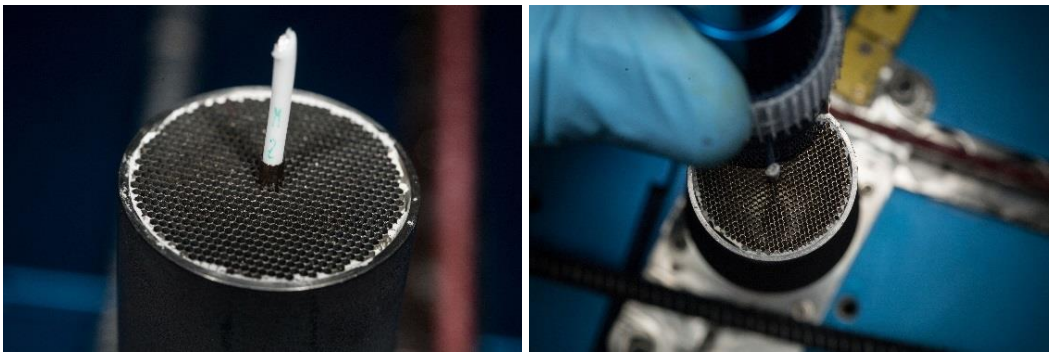


Figure 18. Images from the coflow burner cleaning carried out by astronaut Ricky Arnold on 19 June 2018. In the left image [ISS056e021501], a piece of insulated wire is used to ensure that the inner tube's interior is unobstructed by soot. In the right image [iss056e021495], a flashlight is used to verify that the honeycomb is not blocked by soot.

The spaceflight burner's flow profile was characterized by the CLD Flame team at Yale University prior to its launch, where measurements were made along the four different lines shown in Figure 19 where the velocities of the inner and annular flows were intended to be matched at 10, 20, 30, 40, and 50 cm/s. Hot-wire anemometry measurements were made for each of those velocity pairs at 0.5, 5, and 10 mm downstream of the burner's inner tube, where those results are respectively presented in Figures 20-24, 25-29, and 30-34. The black dashed lines in the subsequent plots of velocity profiles represent the inner and outer tubes of the burner.

Select measurements were also made for 11 mismatched velocity pairs at 0.5 mm downstream of the inner tube, where those are presented in Figures 35-45 (of which the first five are with no inner flow). Additional measurements may be found in the Dec. 2016 document titled ACME CLD burner hot wire anemometry measurements.

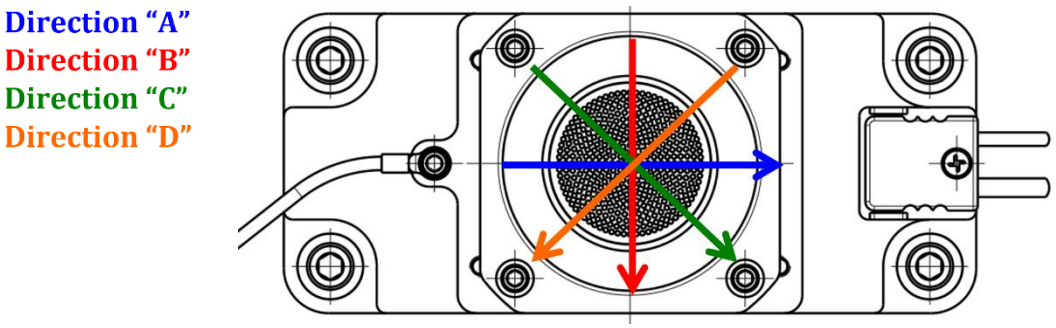


Figure 19. Lines along on which the velocity profiles were measured with the spaceflight coflow burner prior to launch.

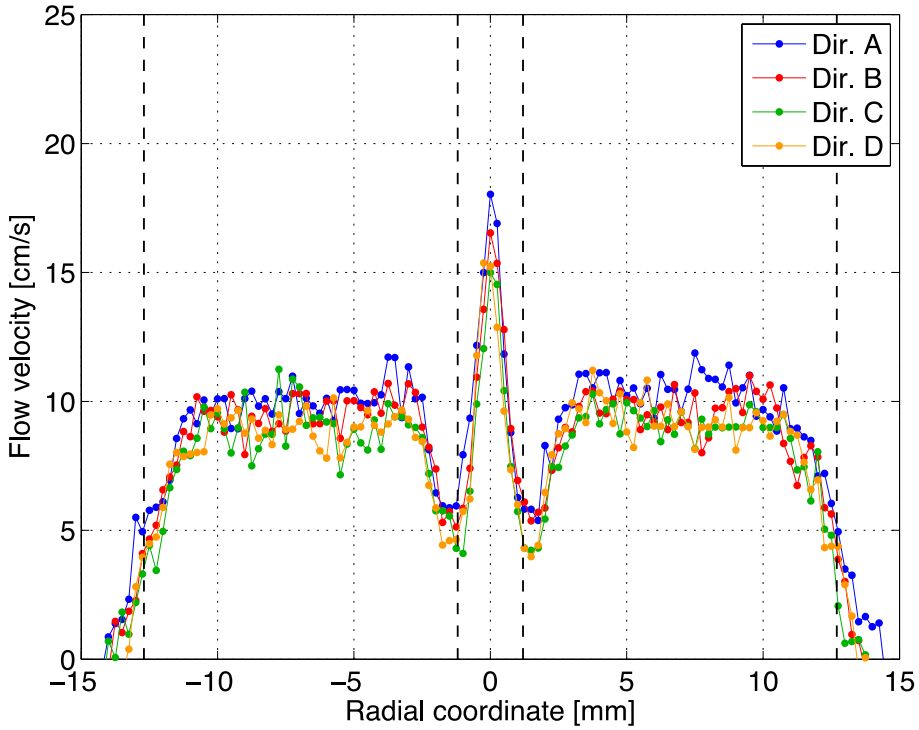


Figure 20. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 10 cm/s.

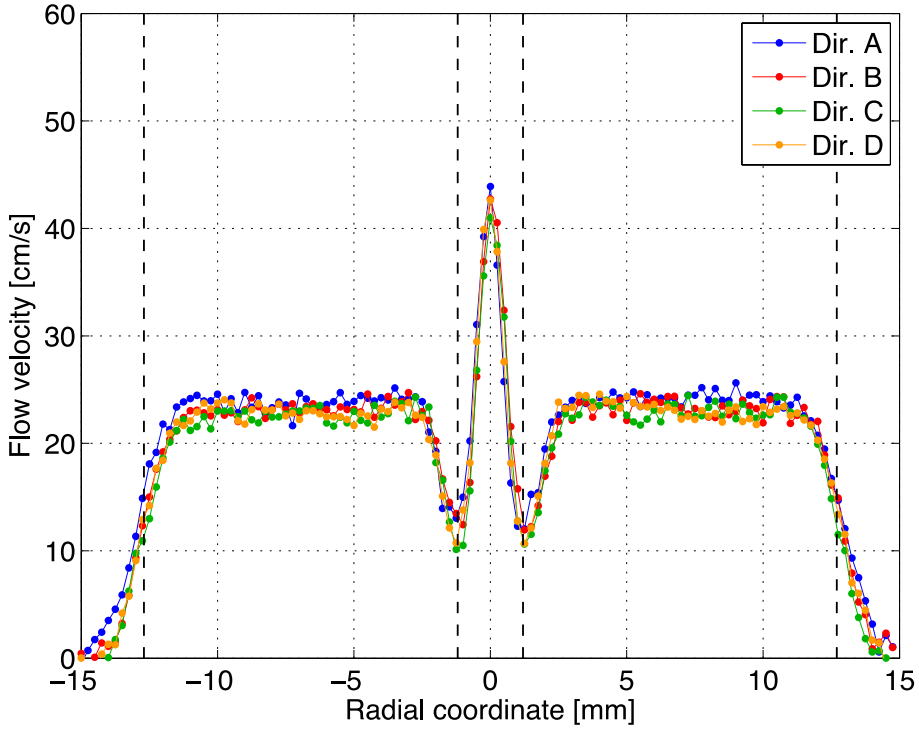


Figure 21. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 20 cm/s.

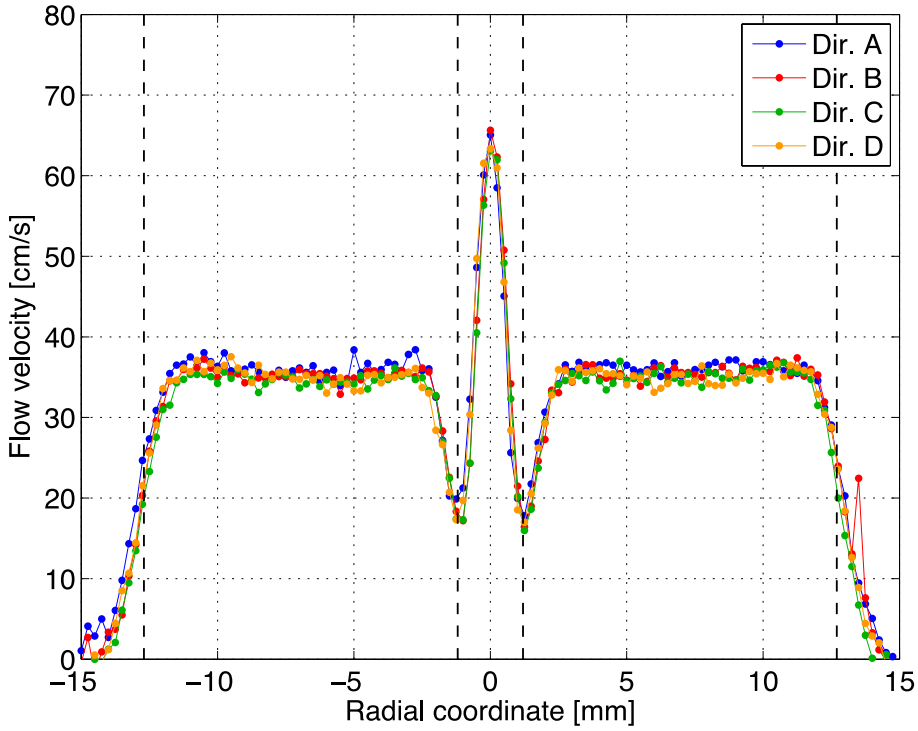


Figure 22. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 30 cm/s.

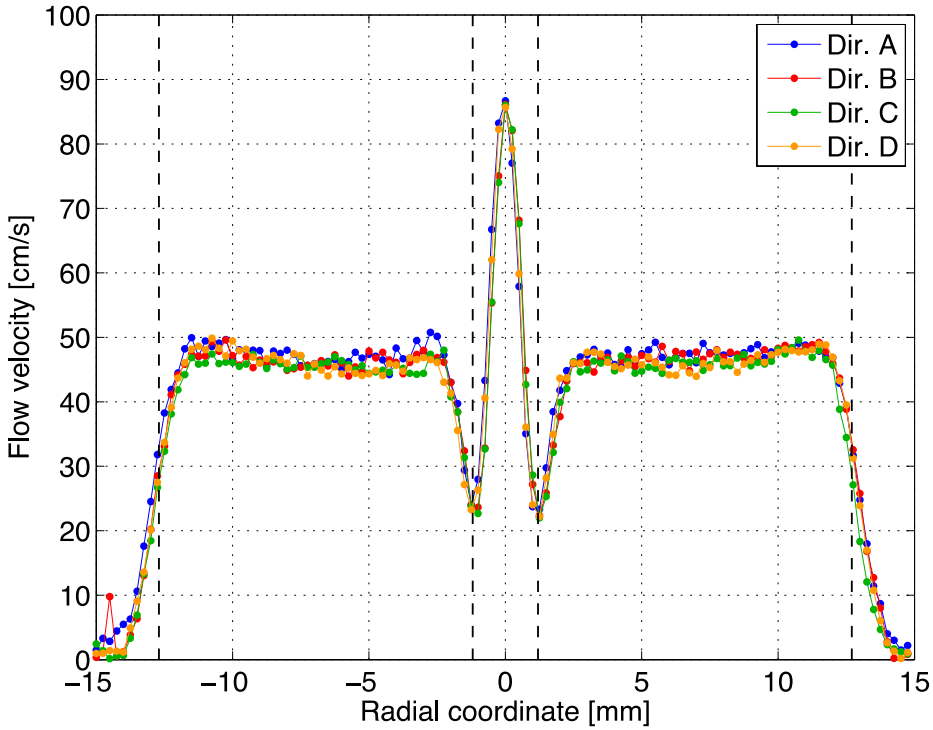


Figure 23. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 40 cm/s.

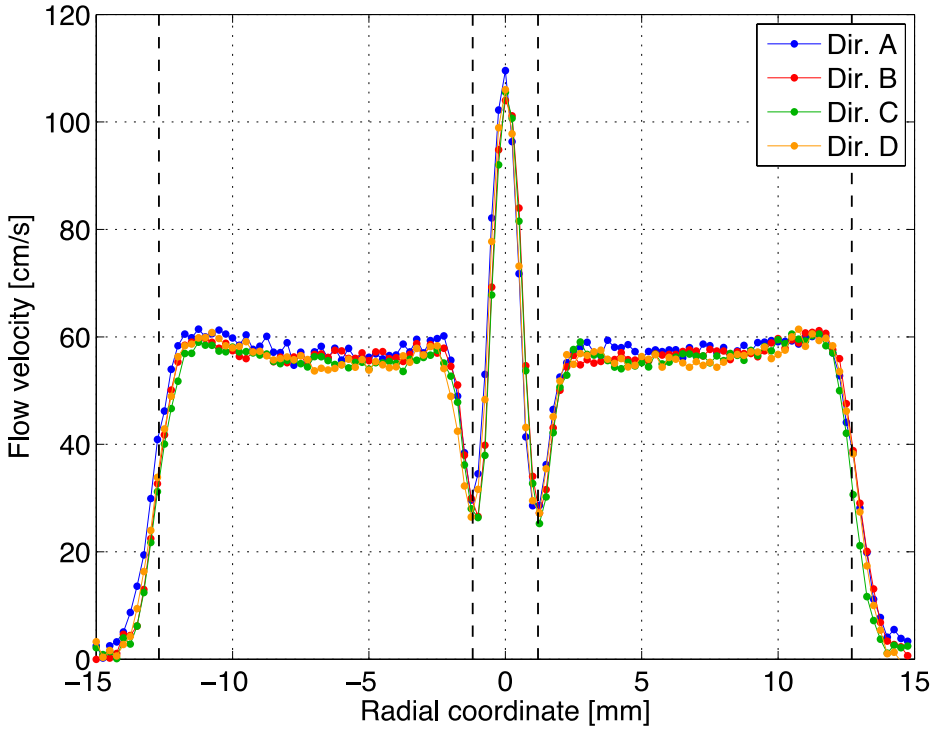


Figure 24. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 50 cm/s.

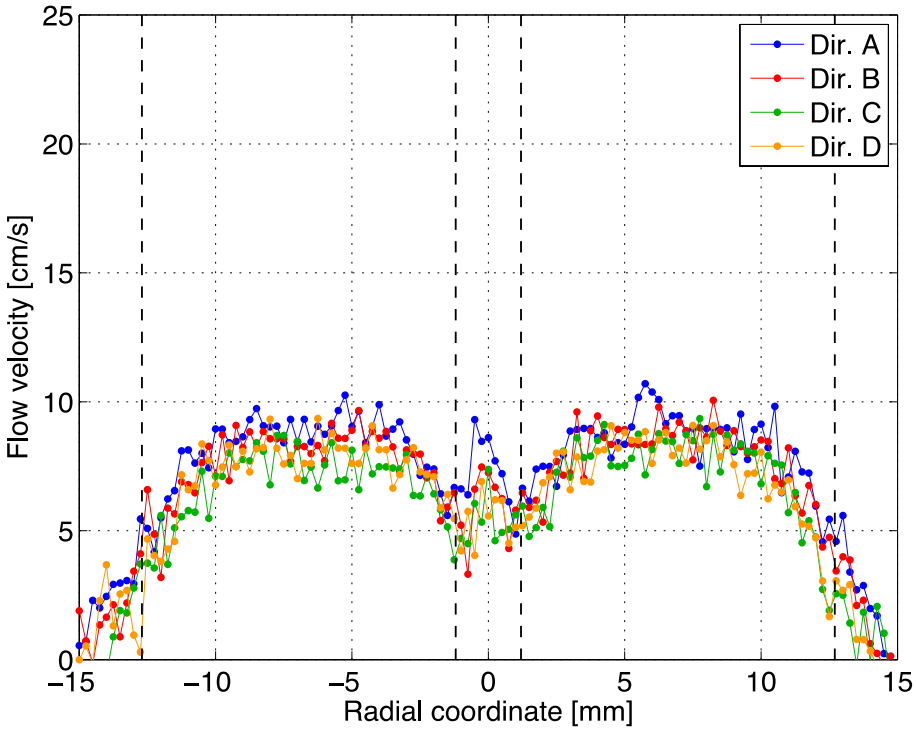


Figure 25. Velocity profiles 5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 10 cm/s.

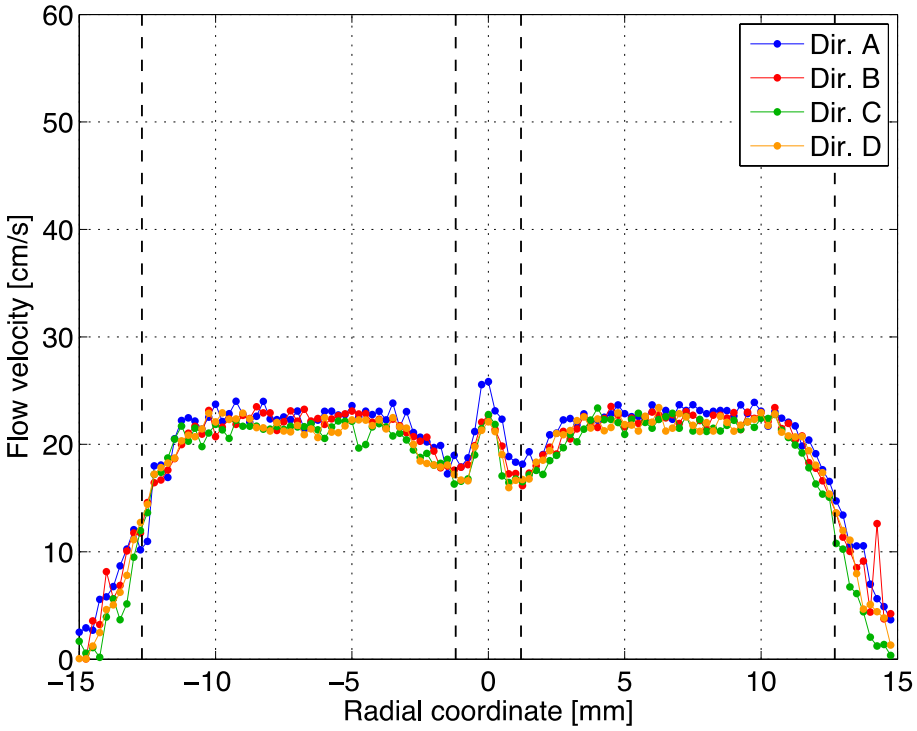


Figure 26. Velocity profiles 5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 20 cm/s.

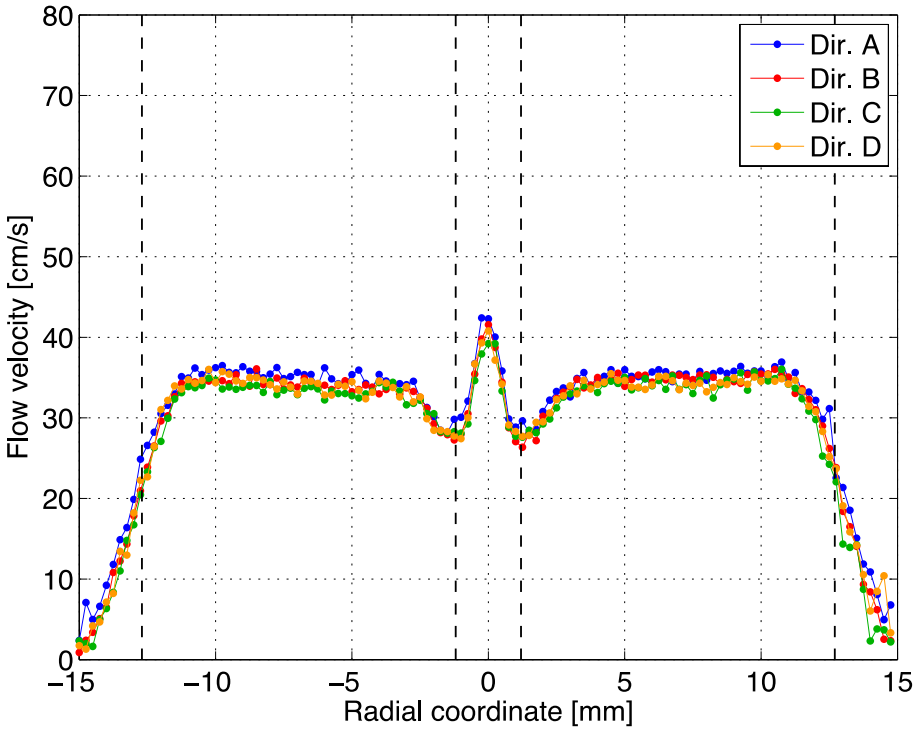


Figure 27. Velocity profiles 5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 30 cm/s.

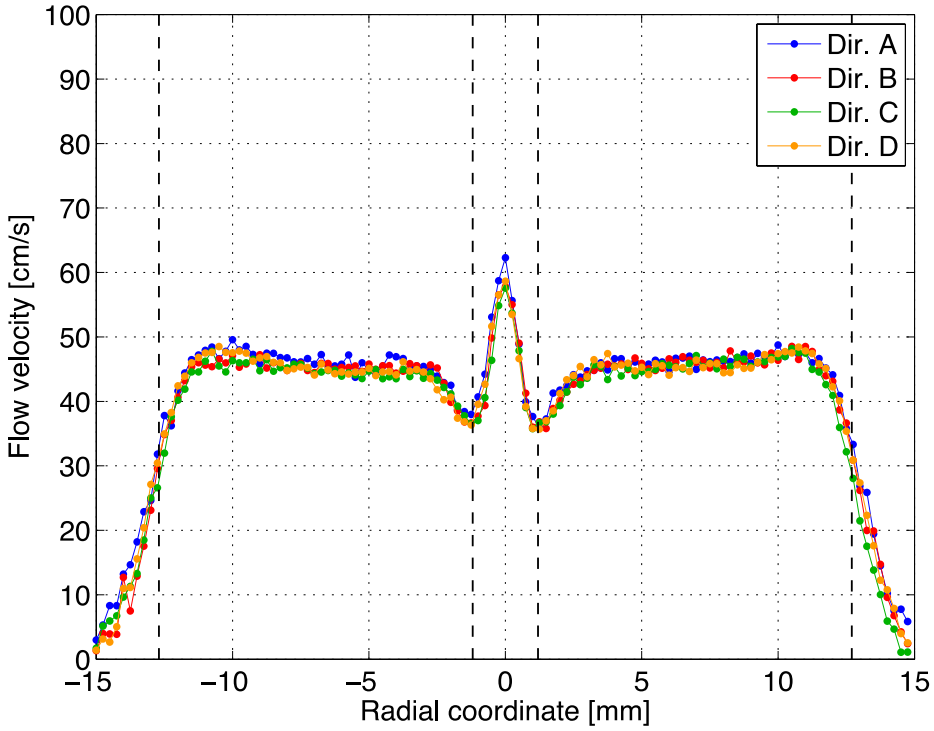


Figure 28. Velocity profiles 5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 40 cm/s.

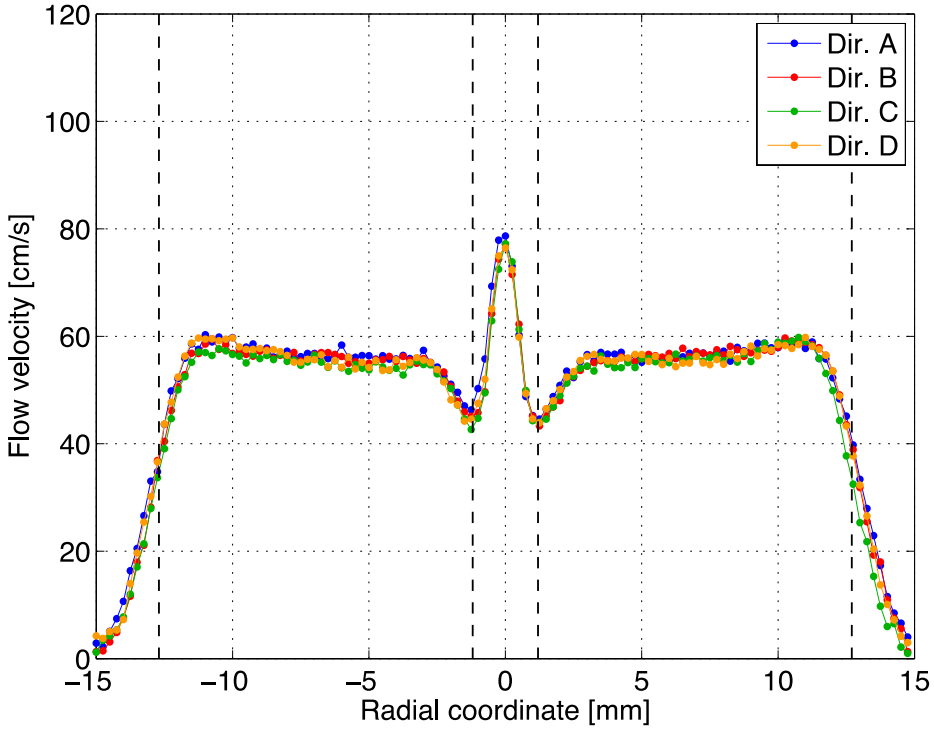


Figure 29. Velocity profiles 5 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 50 cm/s.

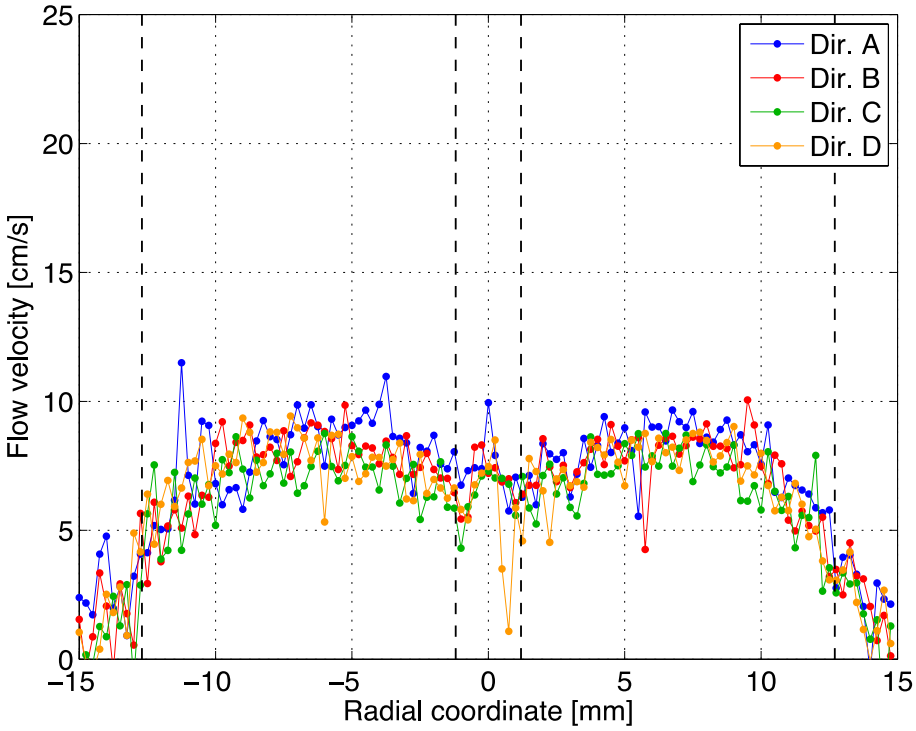


Figure 30. Velocity profiles 10 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 10 cm/s.

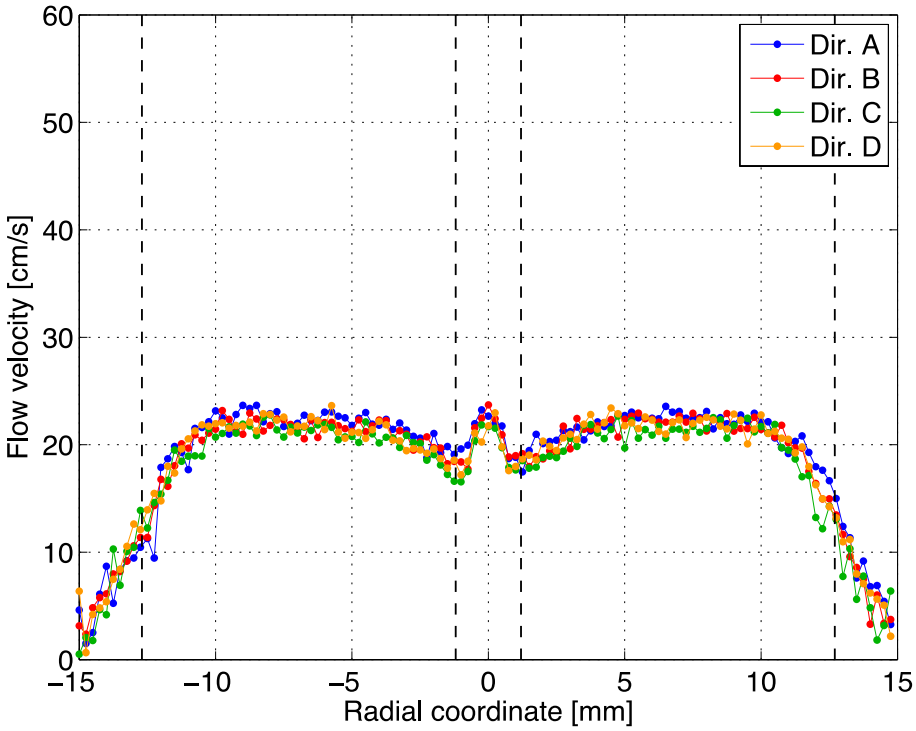


Figure 31. Velocity profiles 10 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 20 cm/s.

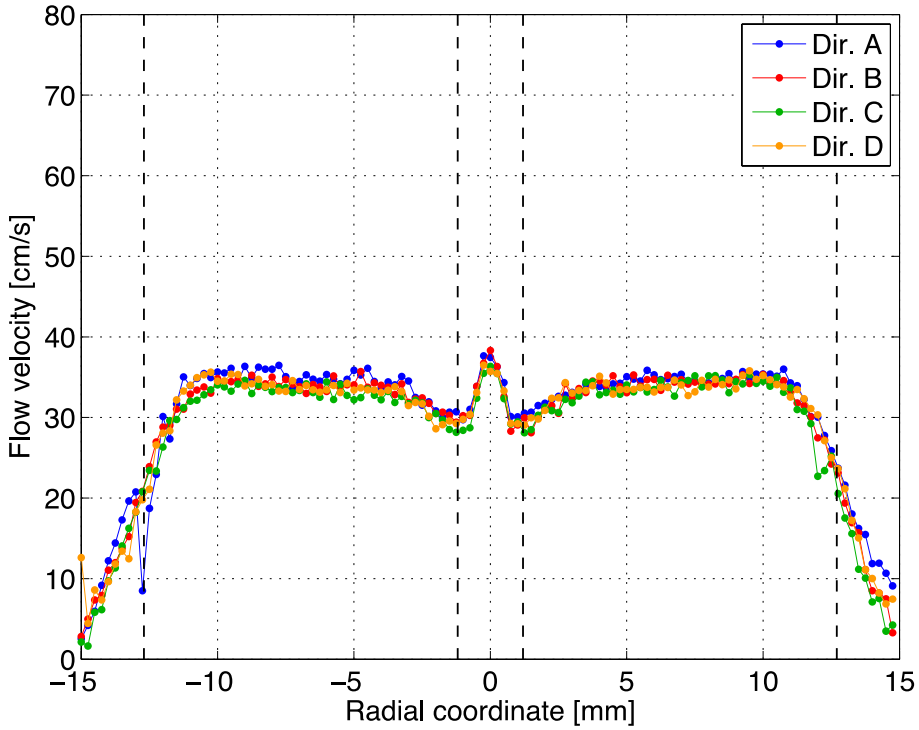


Figure 32. Velocity profiles 10 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 30 cm/s.

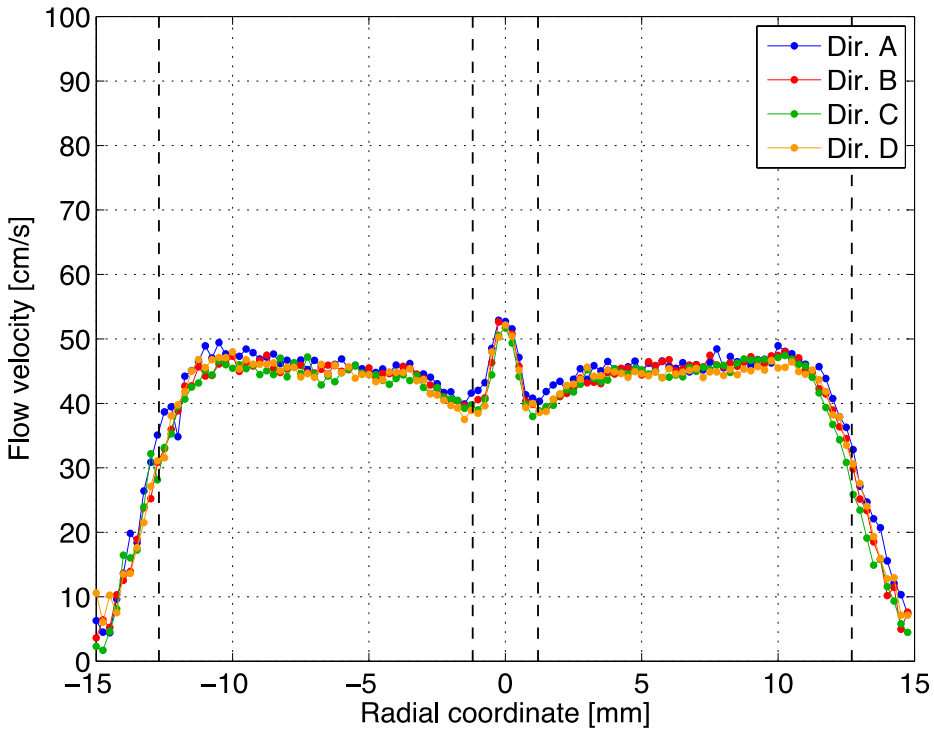


Figure 33. Velocity profiles 10 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 40 cm/s.

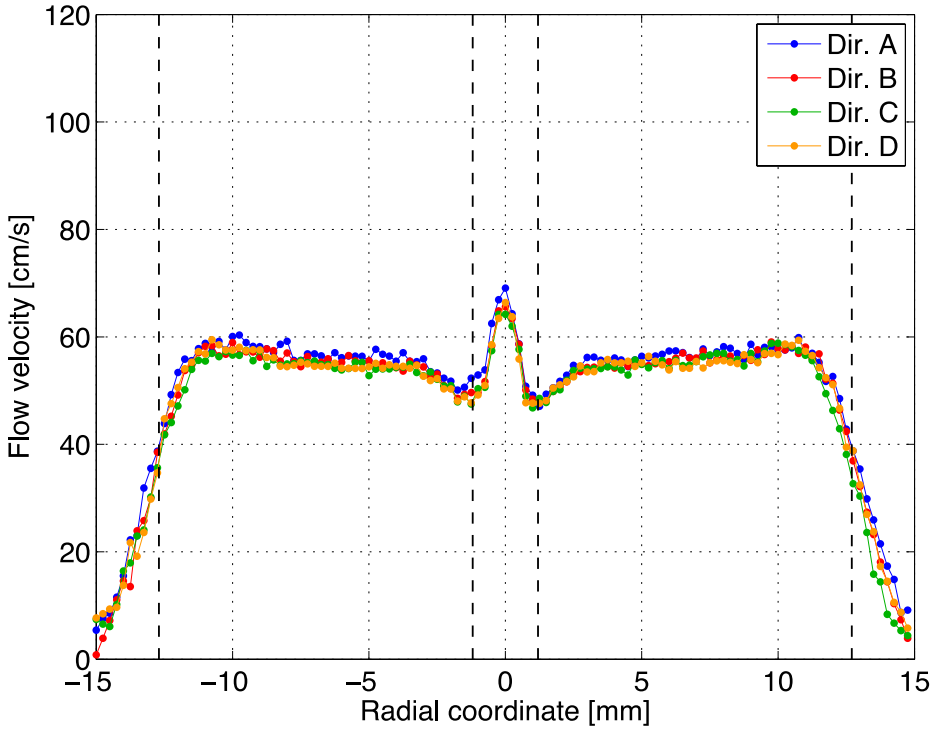


Figure 34. Velocity profiles 10 mm downstream of the coflow burner's inner tube with inner and annular velocities intended to be matched at 50 cm/s.

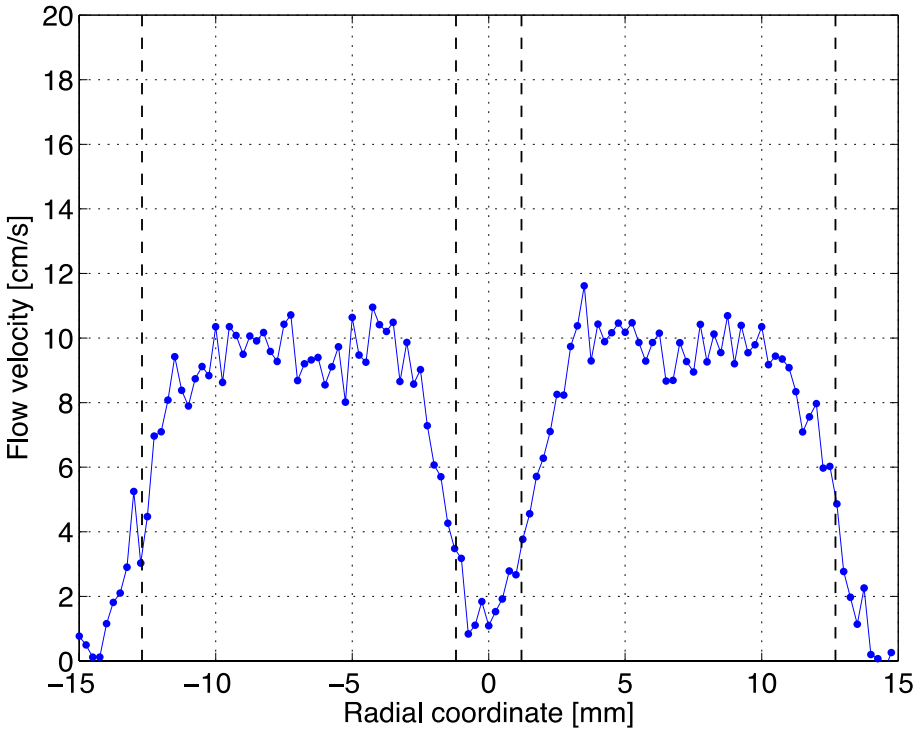


Figure 35. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 0 and 10 cm/s.

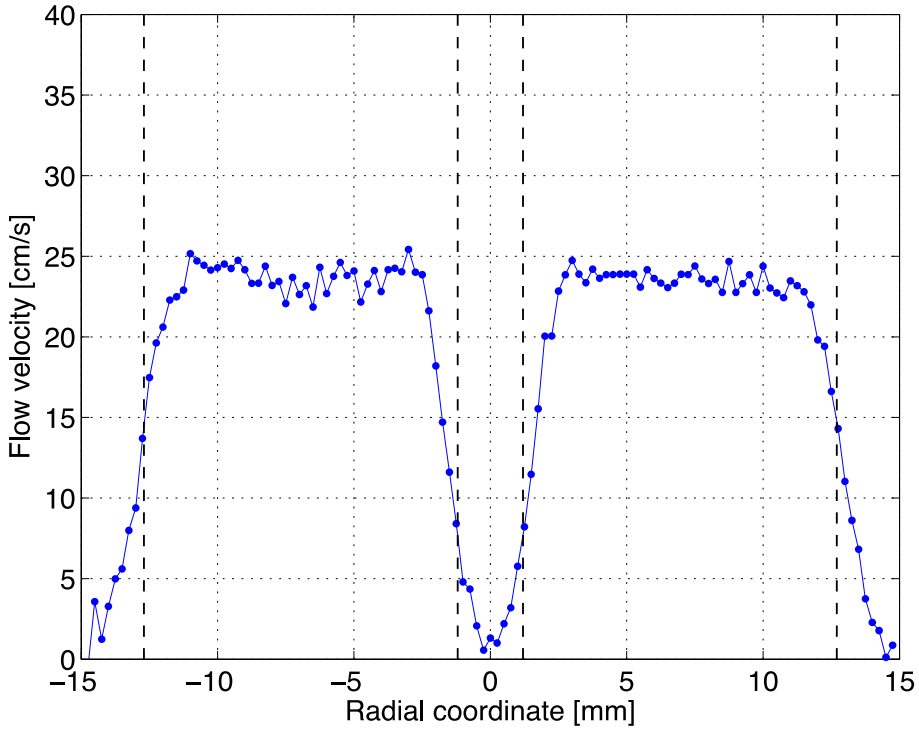


Figure 36. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 0 and 20 cm/s.

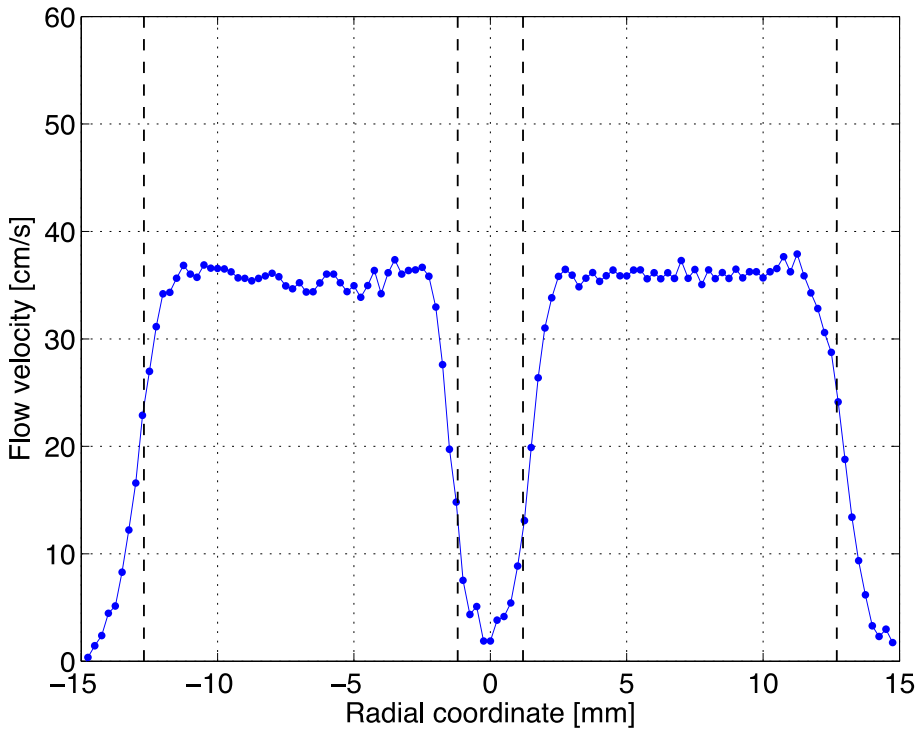


Figure 37. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 0 and 30 cm/s.

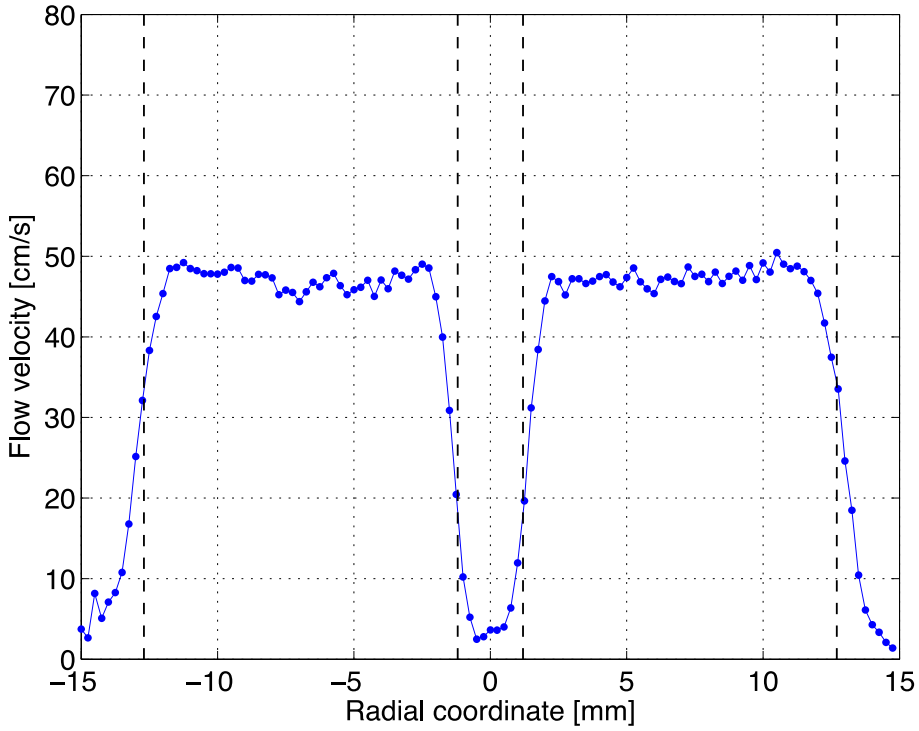


Figure 38. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 0 and 40 cm/s.

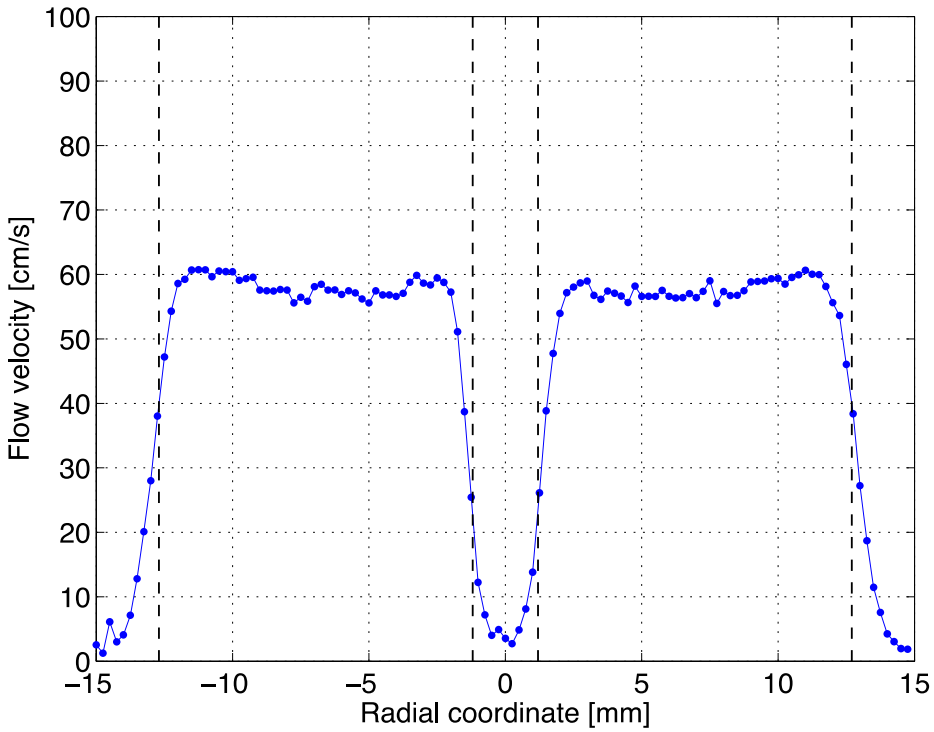


Figure 39. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 0 and 50 cm/s.

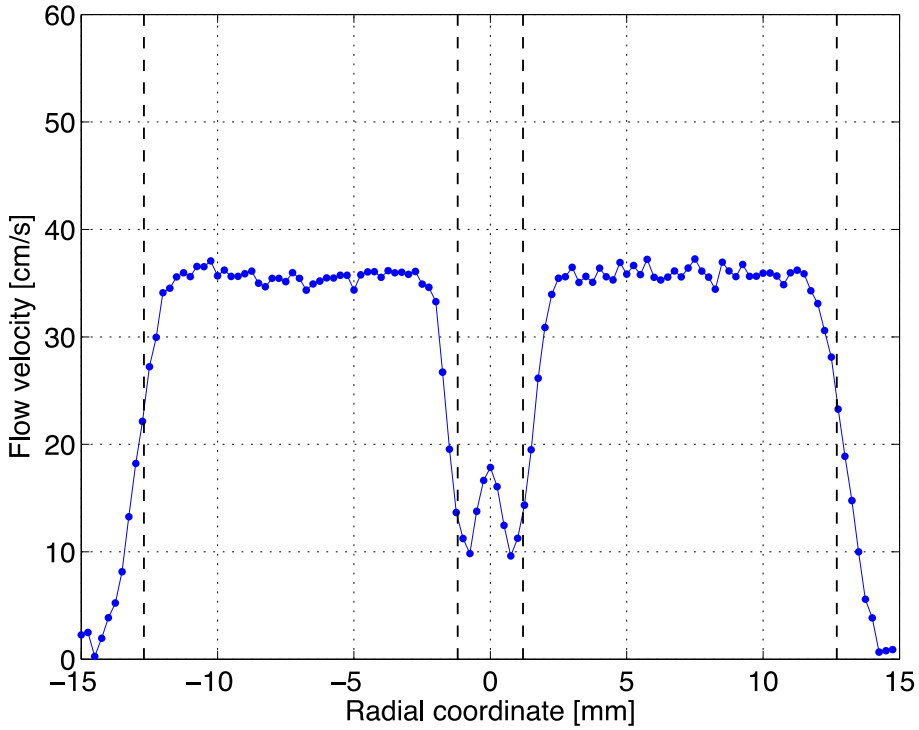


Figure 40. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 10 and 30 cm/s.

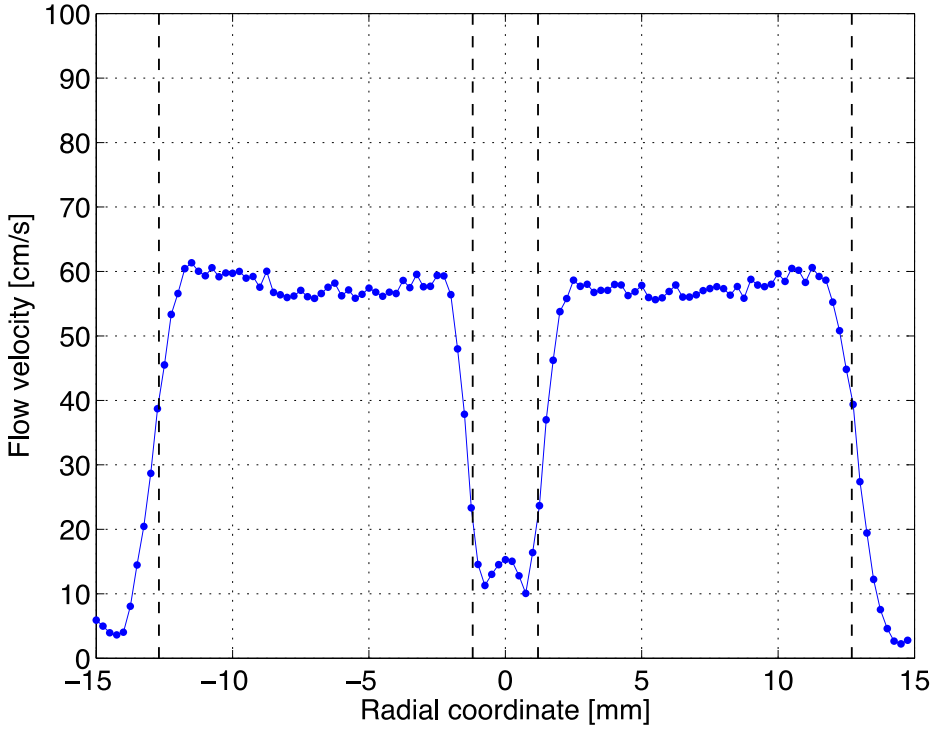


Figure 41. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 10 and 50 cm/s.

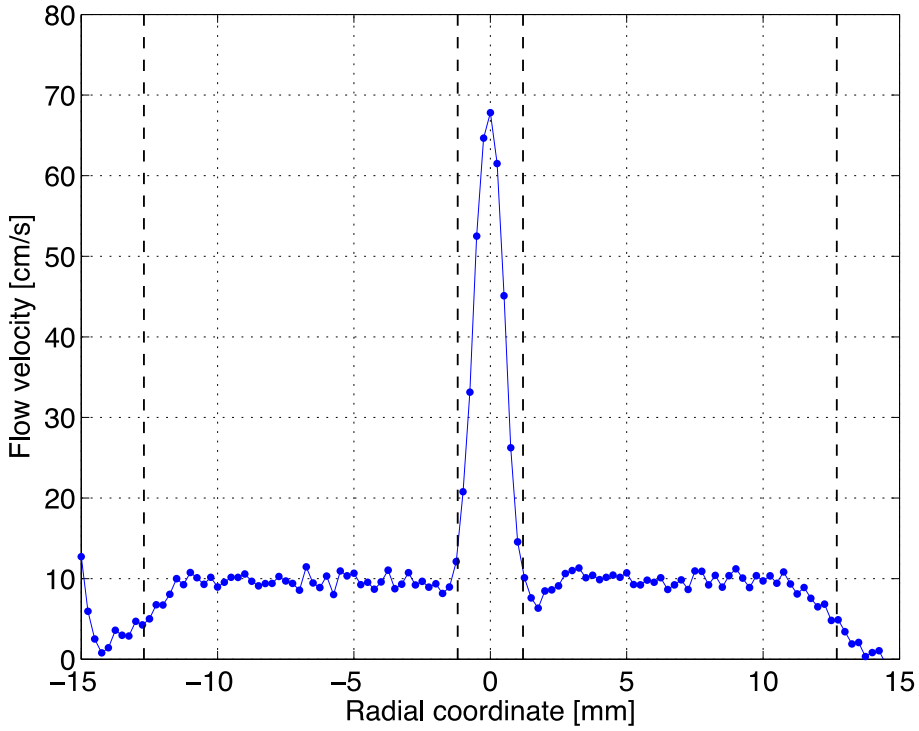


Figure 42. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 30 and 10 cm/s.

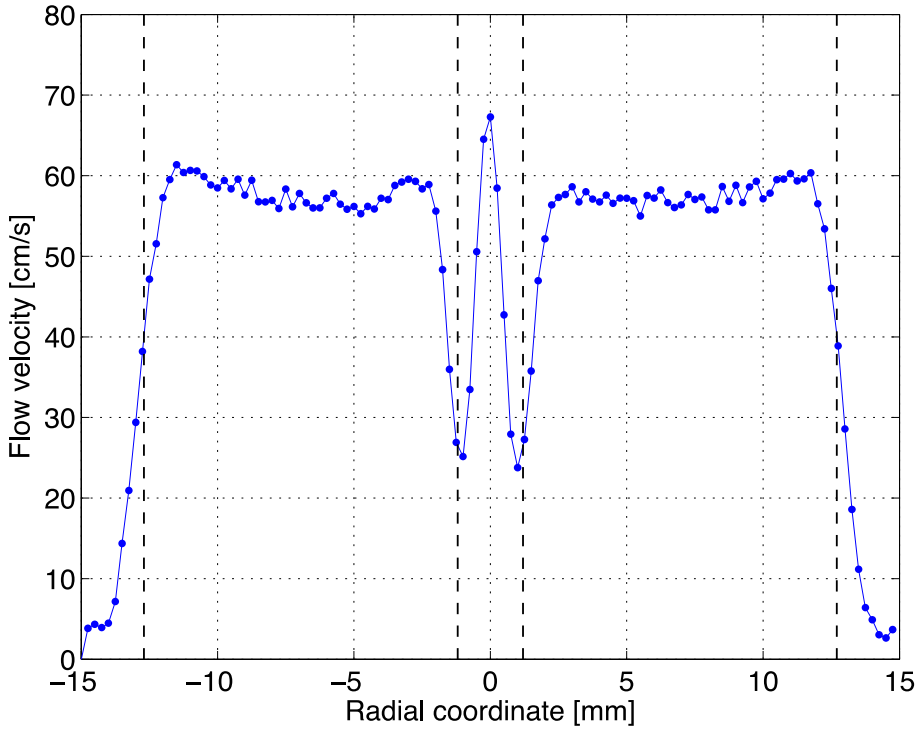


Figure 43. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 30 and 50 cm/s.

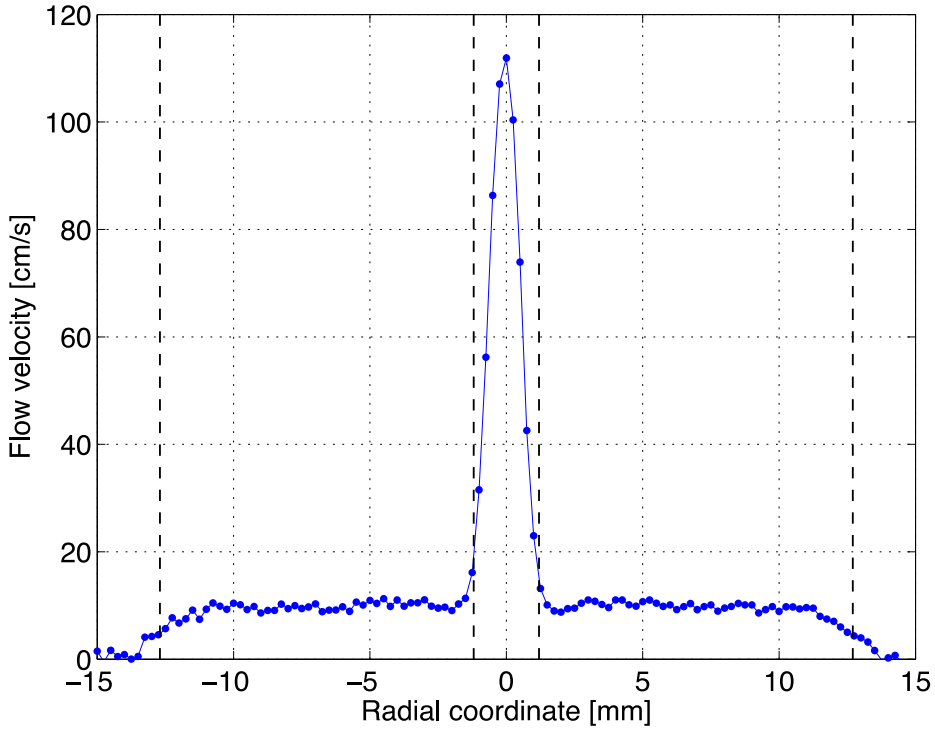


Figure 44. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 50 and 10 cm/s.

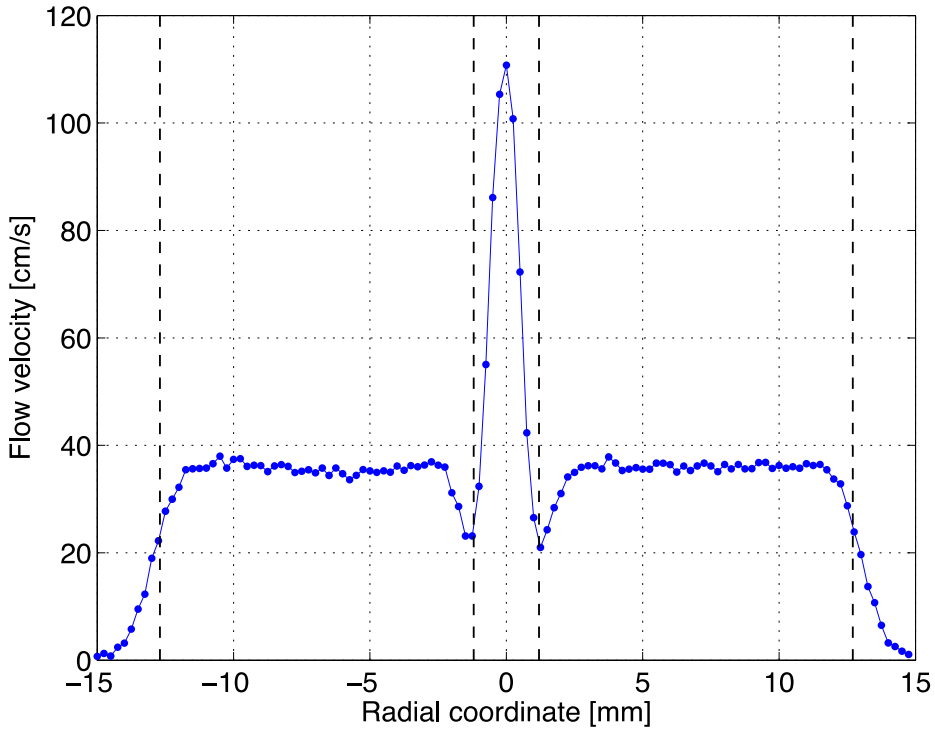


Figure 45. Velocity profiles 0.5 mm downstream of the coflow burner's inner tube with mismatched inner and annular velocities respectively intended to be 50 and 30 cm/s.

As previously mentioned, additional measurements and information about the pre-launch characterization of the coflow burner can be found in the Dec. 2016 document titled ACME CLD burner hot wire anemometry measurements.

3. Spherical Burners

The spherical burners have thus far been used for the CFI-G, Flame Design, and s-Flame experiments. In contrast to initial plans, spaceflight burners were only fabricated with a 6.35-mm porous sphere. It was initially expected that flames which were more spherical might be generated with larger diameter porous spheres, but 2.2-s drop testing indicated that 6.35-mm diameter spheres were as good or better than the larger sizes in creating spherical flames.

Multiple copies of the spherical burner were launched to the ISS because of concerns that they might be damaged and/or might prove to yield flames which aren't sufficiently spherically symmetric.

The porous spheres were fabricated from sintering 316 stainless steel with 10-micron media grade, for which the porosity is approximately 41% and the density is 4.6 g/cm3. More specifically, each sphere was formed by sintering together sintered disks, where the supply tube should extend to the sphere's center. Neglecting the interior of the supply tube, there is presumably no cavity within the porous sphere.

Each sphere was furthermore sinter-bonded to its stainless-steel supply tube. As a side effect, the tubing lost its temper and became very pliable because of the high temperature required for the bonding. Therefore, a close-fitting support tube, with an outer diameter of 1.8 mm, was added to the design. It doesn't quite extend to the porous sphere as can be seen in Figure 46. The gap between the support tube and the burner is often obvious in flame imaging because the supply tube is more reflective than the support tube.

Each burner was equipped with an extremely fine, sheathed, type-K thermocouple which was embedded as much as ~1 mm into the porous sphere. The thermocouple hole was 0.76 to 1.0 mm deep (and 0.25-0.30 mm in diameter) where the thermocouple was meant to be inserted fully, i.e., until reaching the hole's bottom, but that may not be its exact position. The junction is of course not at the extreme end of the sheathed thermocouple, but it is very fine and presumably close. If the porous sphere is imagined as a globe with the supply tube as the south pole, the thermocouple is inserted in the lower hemisphere, between the equator and pole as seen in Figure 46.

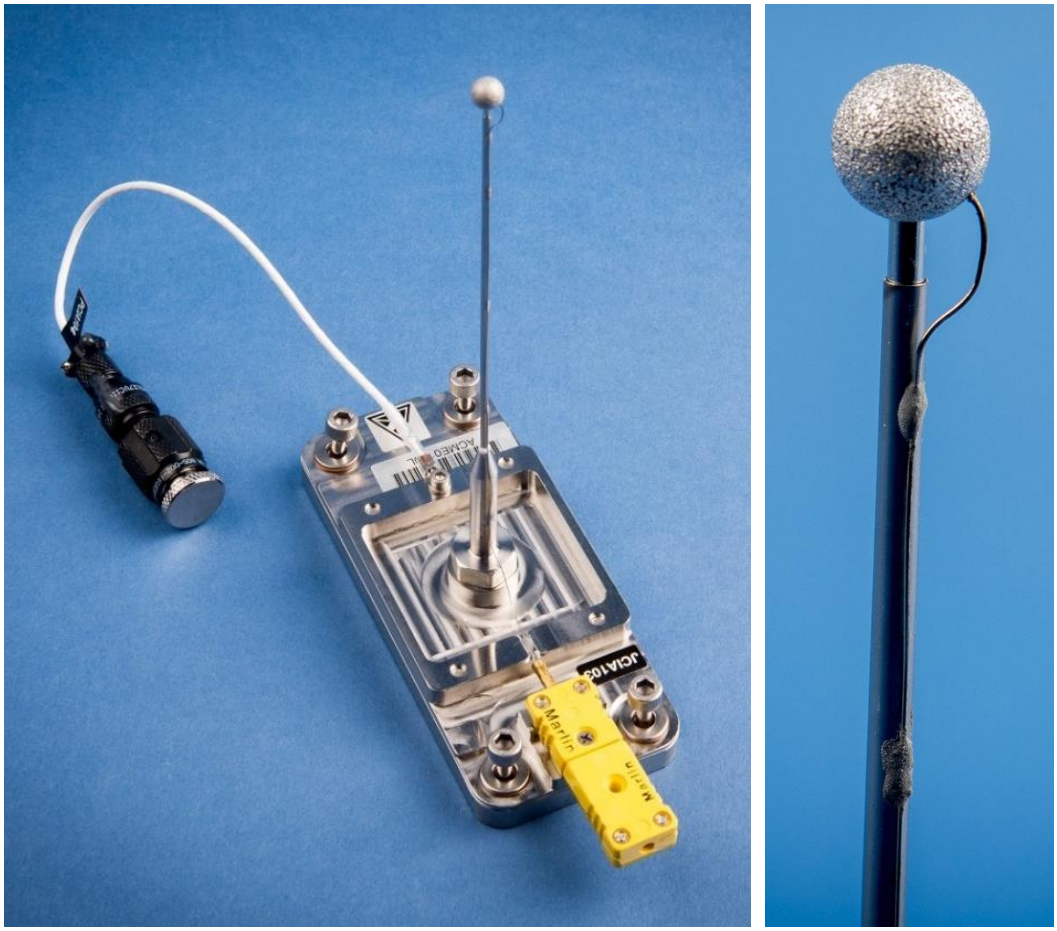


Figure 46. A spherical burner, prior to launch, showing the positioning of the slightly embedded thermocouple. The gap between the support tube and porous sintered sphere can also be seen. The white cable with the black connector on the left is for grounding, while the yellow connector is for the type-K thermocouple.

The spherical burners were first used for the Flame Design experiment, where the first flame tests were conducted on 25 April 2019. However, after several days of testing it was concluded by the investigators that the flames were insufficiently symmetric at high burner flow rates to meet the science objectives. At the high flow rates, the flame ceased to be spherically symmetric and its center didn't coincide with the burner's center. The results indicate that the sintered attachment of the sphere to the supply tube was more porous than the sintered sphere itself, where the flame was shifted toward the supply tube. Therefore, the spherical burner was replaced on 14 June 2019 where the shift was from serial number 1002 to 1003. The subsequent burner (s/n 1003) showed similar behavior, but to a significantly reduced degree. As a result, some of the early Flame Design testing was repeated after the burner swap. This burner (s/n 1003) has thus far been used for all subsequent spherical flame testing including the tests for the s-Flame and CFI-G experiments.

4. BRE Burners

In terms of both size and instrumentation, the two BRE burners are significantly different than the other existing ACME burners. The outlet diameters of the two burners are 25 and 50 mm, where the latter burner is shown in Figure 47. The overall burner diameters are even larger, for example because each burner is encompassed with 6 mm of ceramic insulation.



Figure 47. The 50-mm BRE burner prior to launch. The white cable with the black connector on the right is for grounding, while the yellow connector on the left was used to secure the cable for the burner thermocouple, where the measurements from the BRE burner's thermocouples and heat flux transducers were transmitted through the black cables. The two heat flux transducers can be seen in the perforated outlet plate as black 'circles.' The two light dots in the same 'row' as the two transducers are the tips of embedded thermocouples.

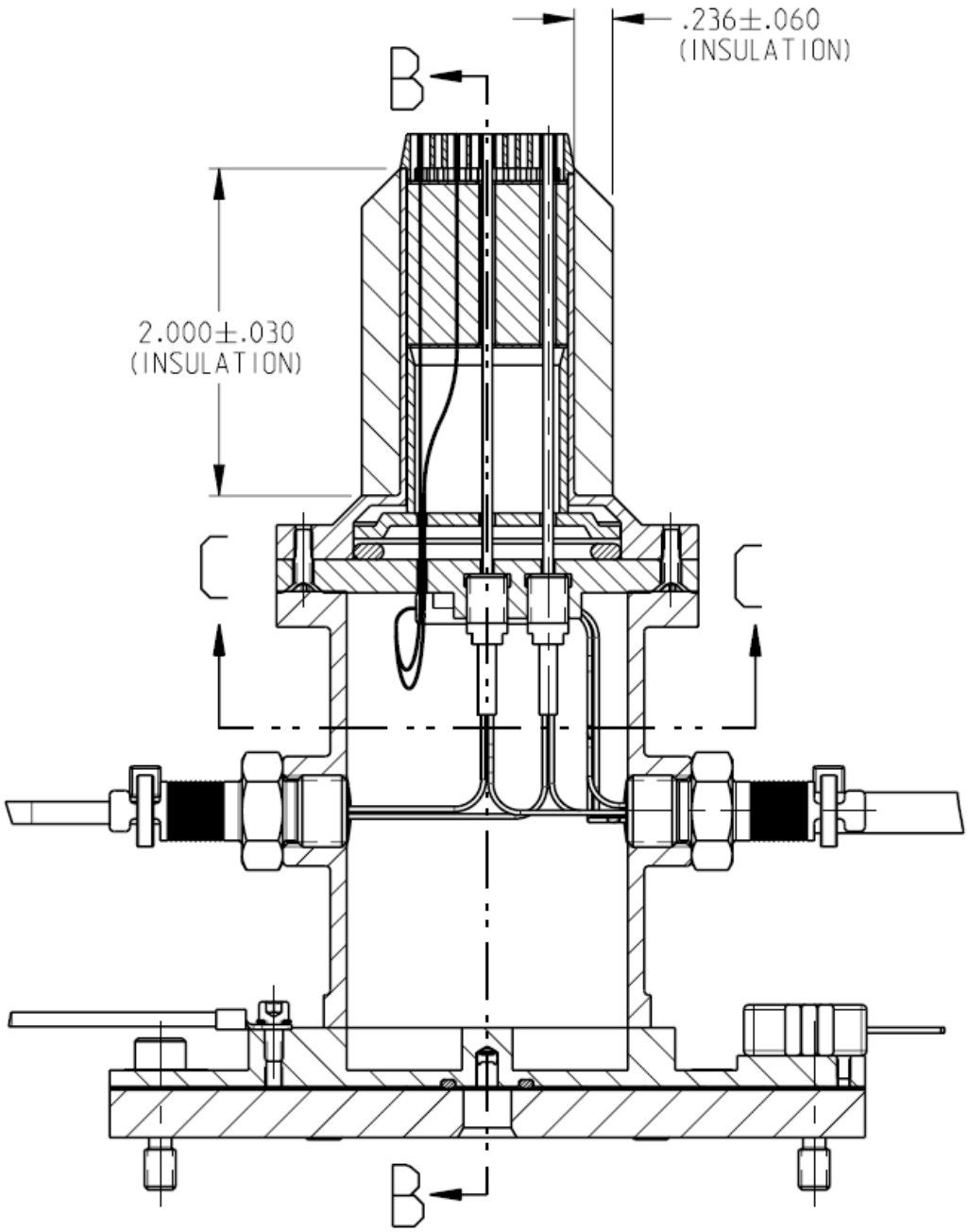


Figure 48. Assembly schematic for the 25-mm BRE burner, a detail from drawing S1101MFA1245. The dimensions for the ceramic insulation are in inches.

While the 25-mm BRE burner, depicted in Figure 48, has an outlet diameter corresponding to the coflow burner, it is nonetheless larger and more complex.

The perforated copper outlet plate of each BRE burner is 5.3 mm thick and has two sheathed type-K thermocouples (0.25 mm in diameter) embedded so their tips are at the blackened outer surface. This design allows the outlet plate to be used for calorimetric measurements of the burner heating by the flame. In this regard, note that the BRE burners - including their instrumentation - are not water-cooled, where that was not

feasible within ACME's constraints.

For flow control, each BRE burner has a ceramic honeycomb, 25 mm thick, upstream of the outlet plate. Each ceramic element is sandwiched between two layers of perforated brass which are 0.4 mm thick and have staggered holes which are 0.4 mm in diameter yielding a 22% open area. The perforated brass is visible in Figure 49.

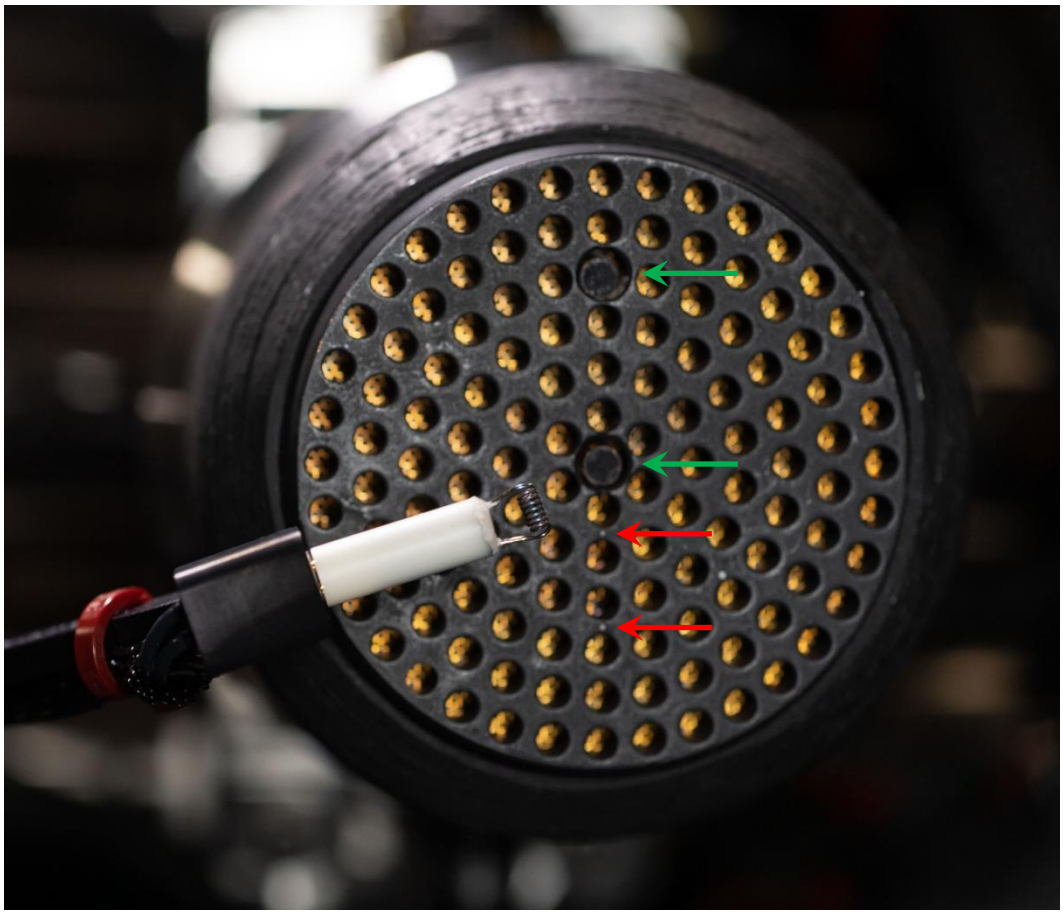


Figure 49. The 50-mm BRE burner with a coil-type igniter tip inserted above it during set-up by astronaut Chris Cassidy on 27 Aug. 2020. The center and offset heat flux transducers, which include thermocouples, are within openings in the perforated outlet plate and pointed out by green arrows. The tips of two additional thermocouples are visible as faint dots between holes in the black outlet below the transducers, as pointed out by the red arrows. [Cropped detail from ISS063e078811]

As seen in Figure 49, the BRE burners are each equipped with two Medtherm heat flux transducers, which each include a type-T thermocouple. The sensors pass through the holes in the outlet plate, where one is in burner's center and the other is offset, where their radial positions are indicated in Table 4. The 25- and 50-mm burners respectively have sensors with outer diameters of 1/16 and 1/8 inch, where the part numbers (as specified in the assembly drawings) are also provided in the table.

Table 4. Radial positions of BRE burner instrumentation

BRE burner	Thermocouples, type-K (qty 2)	Medtherm heat flux transducers, including a type-T thermocouple (qty 2)
25-mm	4.72 mm, 10.4 mm	0 mm, 9.46 mm (part # 4-10SB-2.688-0.43-0-36-22066KSA)
50-mm	5.72 mm, 13.3 mm	0 mm, 15.2 mm (part # 8-2.688-10SB-0-0-36-21427KSA)

BRE was a late addition to the ACME project and as a result it had to adapt to existing engineering design. As an important example, the BRE burner thermocouples were connected as if the burner was an alternate to the thermocouple rakes discussed in section 3.G. This is important because each of the rakes have 2 type-K thermocouples and 2 type-B thermocouples. Furthermore, the BRE wiring wasn't optimized to minimize mixed connections. The thermocouples and wiring are described in Table 5, where accurate measurements from the thermocouples will require accounting for the mismatched connections.

Table 5. Mismatched BRE thermocouple wiring (presented in the top-to-bottom order corresponding to Figures 49 and 50)

BRE burner position	Thermocouple type	Wiring type
Offset heat flux transducer	T	K (mismatched)
Center heat flux transducer	T	B (mismatched)
South thermocouple	K	K (matched)
Far south thermocouple	K	B (mismatched)

The flow from the two spaceflight BRE burners was characterized at the University of Maryland prior to the burners' launch. Flow profiles were measured 10 mm downstream from the burner outlet along four lines across the burner surface, namely N-S, NE-SW, E-W, and SE-NW, where the offset heat flux transducer is to the north (N) as shown in the subsequent figure. The flow profiles were made for the 25- and 50-mm burners with 3 and 12 slpm of nitrogen, respectively. At these flow rates, it can be seen in Figures 51 and 52, respectively, that flow from both burners was nominally axisymmetric. While it was nominally flat across the center of the 25-mm burner, the characterized flow was significantly depressed in the center of the 50-mm burner. However, the 50-mm burner was only used on 2 of the 17 days in which BRE flames were ignited. Furthermore, the 50-mm burner flow (i.e., of both fuel and nitrogen, if any) in the spaceflight testing was always less than 0.5 slpm total (on a nitrogen basis), in great contrast to the 12 slpm used in the characterization. As such, it is likely that the flow profile was more uniform than that seen in Figure 52.

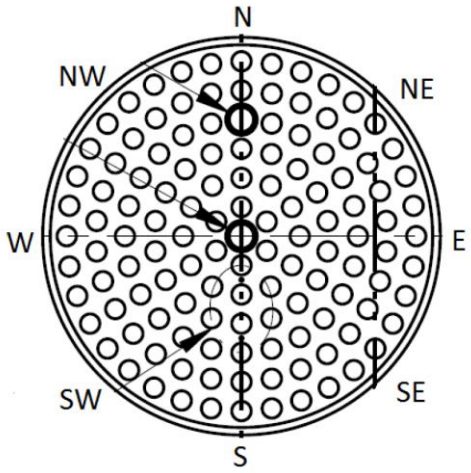


Figure 50. Drawing detail showing the arrangement of the four lines, N-S, NE-SW, E-W, and SE-NW, along which flow profiles were measured prior to launch. In this arrangement, the offset heat flux transducer is to the north (N), while the two separate thermocouples are to the south (S).

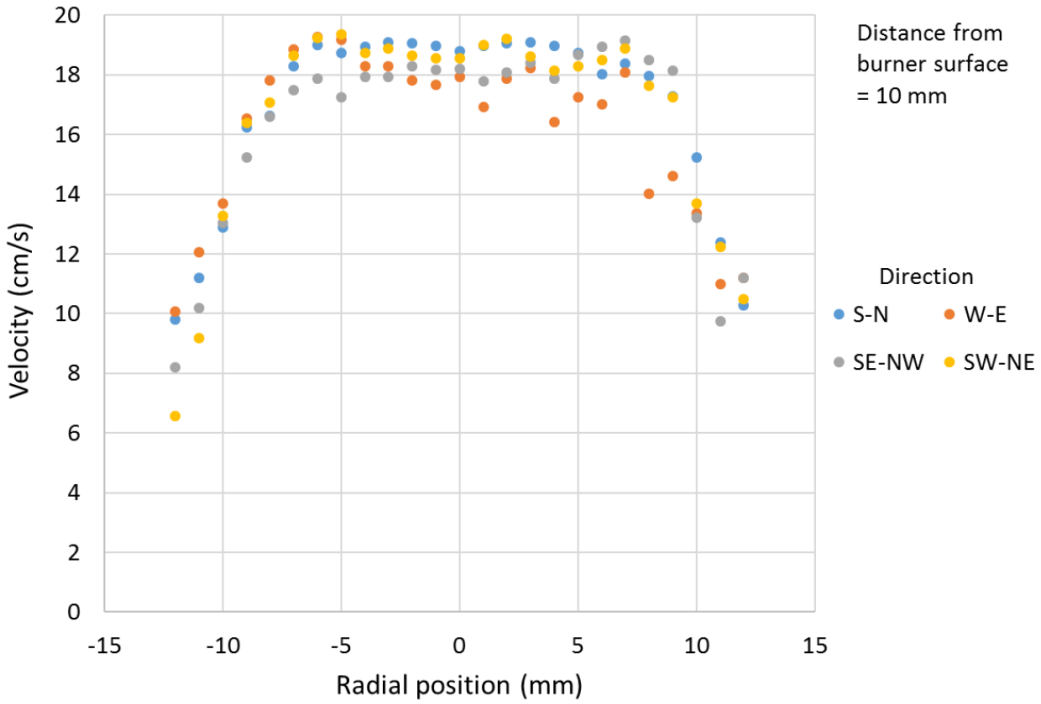


Figure 51. Flow profiles measured for the spaceflight copy of the 25-mm burner.

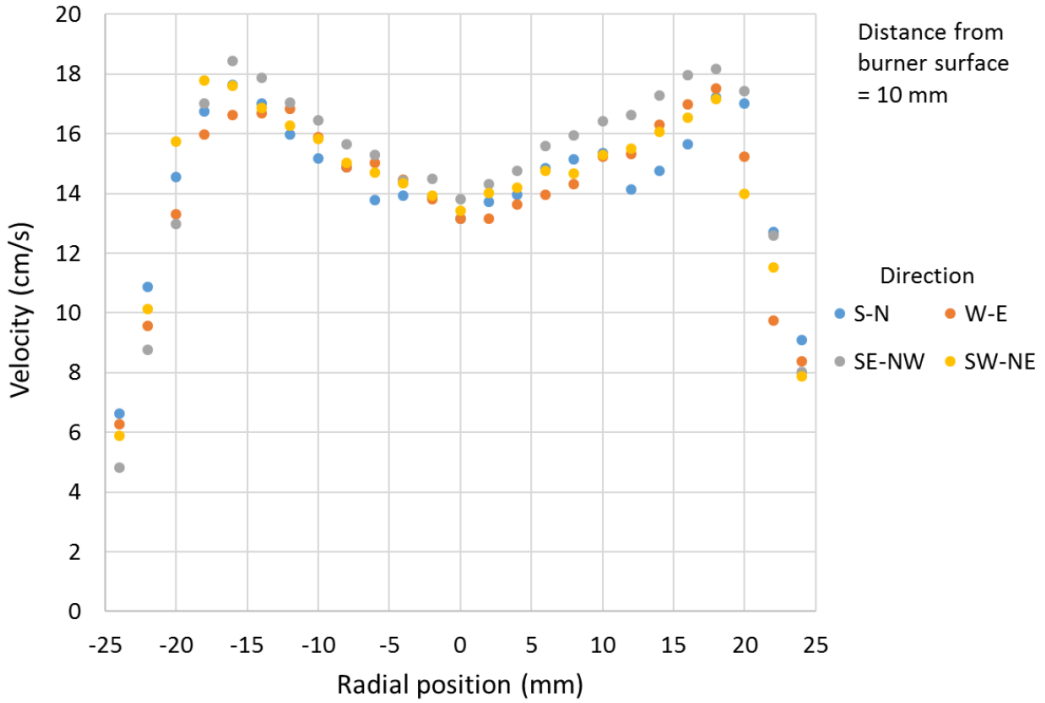


Figure 52. Flow profiles measured for the spaceflight copy of the 50-mm burner.

D. Igniter Tips

Two types of crew-exchangeable igniter tips have been used during ACME testing. Most tests were conducted with a tip featuring a simple loop of Kanthal A-1 wire (0.25 mm in diameter) which was resistively heated with specification of the voltage and duration. A pre-heat voltage and duration, e.g., prior to burner flow, can also be specified. The second type features a small coil of Kanthal A-1 wire (0.32 mm in diameter), where it was used in the final six s-Flame test days (in June-July 2020) and throughout the second round of BRE testing (in Sept. 2020 to Jan. 2021). The coil igniter tip will also be used in the CFI-G investigation.

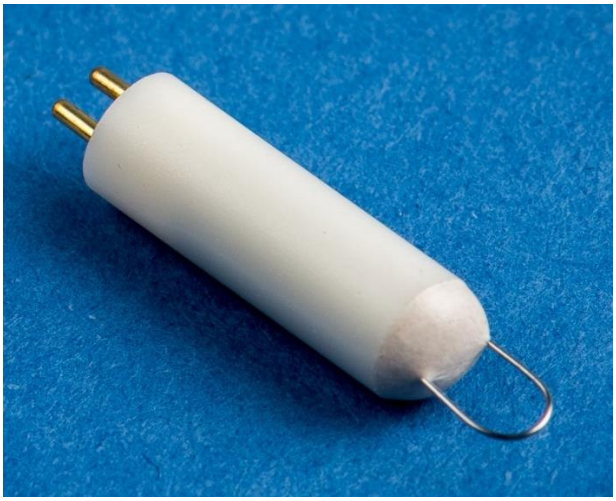


Figure 53. A standard ACME igniter tip prior to launch. The igniter tip's white ceramic insulator is 4.75 mm in diameter and the pin-to-pin spacing (as measured from their centers) is 2 mm.

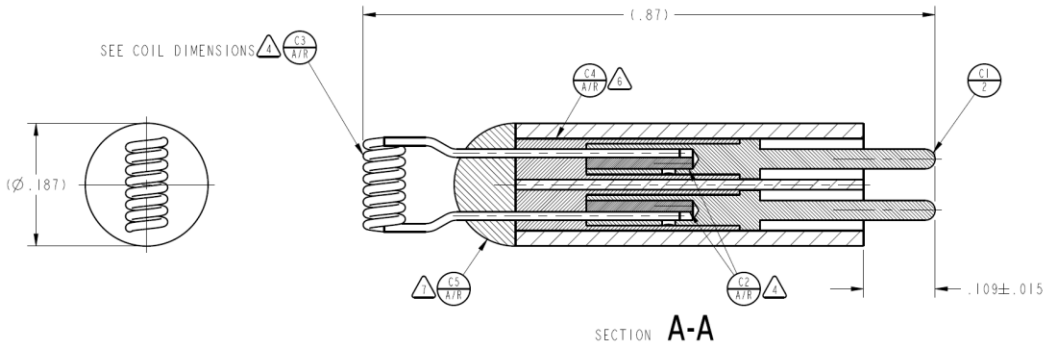


Figure 54. Coil type igniter from details in drawing S1101MFA1339, where the dimensions are in inches.

E. Optical Layout

This section graphically presents the layout of the cameras and non-imaging optical sensors. Note that the Low Light-Level Ultraviolet (LLL-UV) camera was only used for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment.

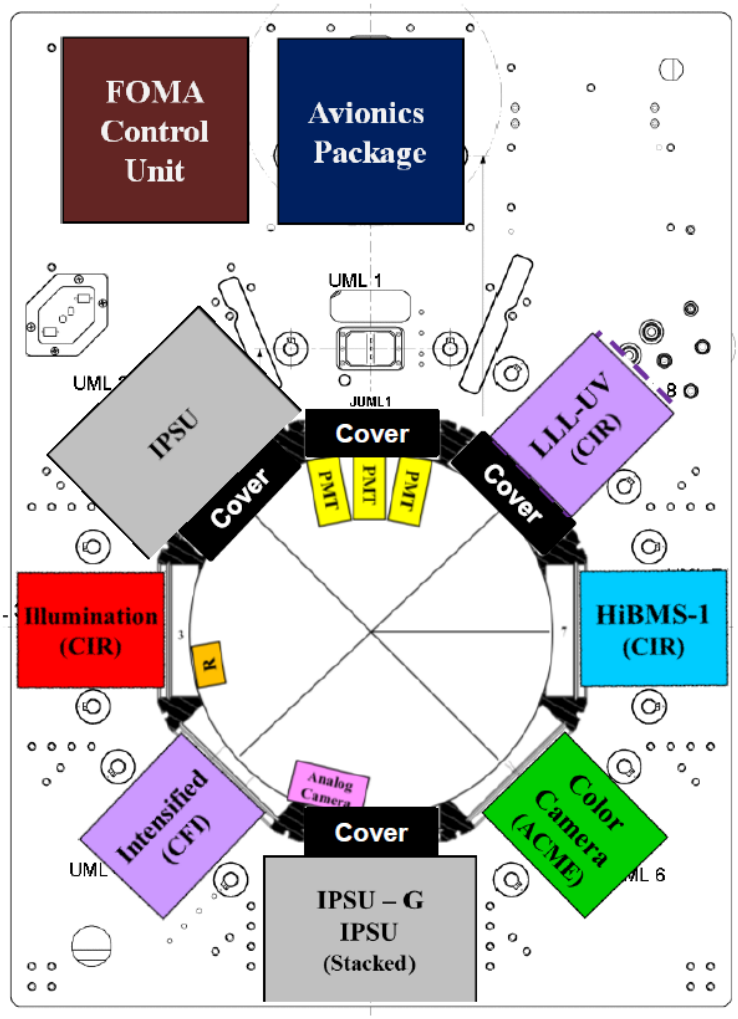


Figure 55: Full layout of the cameras and optical instrumentation, showing the location of the Fuel and Oxidizer Management Assembly (FOMA) control unit, avionics package, Low Light Level Ultraviolet (LLL-UV) camera, High Bit-depth Multispectral (HiBMs) camera and the opposing illumination package, the color ACME Data Camera (DataCam), the Image Processing and Storage Unit for Gigabit ethernet (IPSU-G) cameras (IPSU-G), the two standard IPSUs, the Cool Flame Investigation (CFI) intensified cameras, the internal window covers (as installed when not using the LLL-UV camera), array of three Photomultiplier Tubes (PMTs), analog operations camera, and the array of radiometers (simply marked 'R').

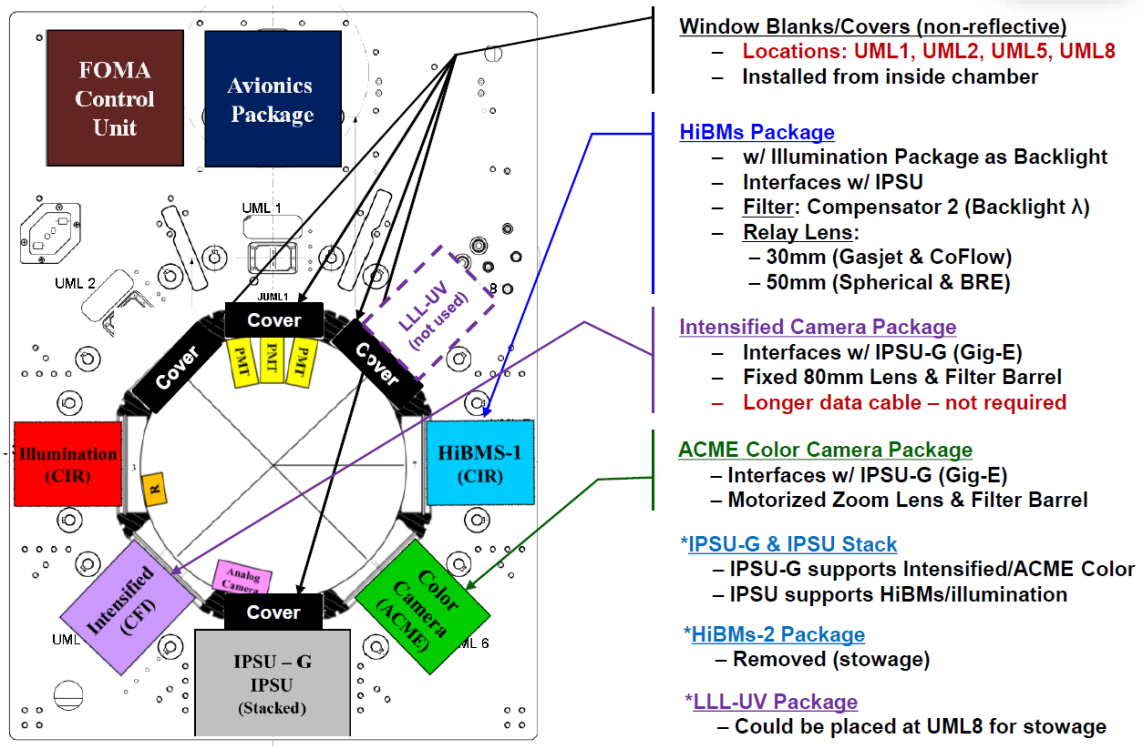


Figure 56. Optical arrangement for all ACME investigations except the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment.

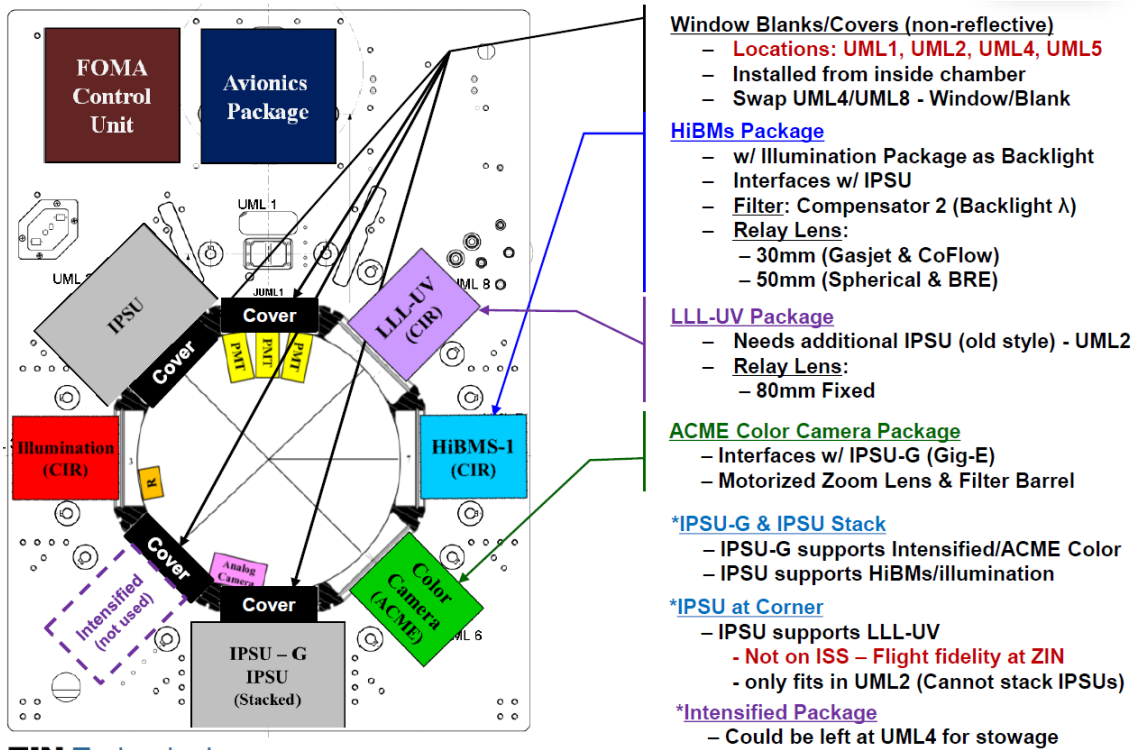


Figure 57. Optical arrangement for ACME's Structure and Response of Spherical Diffusion Flames (s-Flame) experiment, with the important exception that the CFI intensified camera was used for s-Flame and was not blocked with an internal window cover.

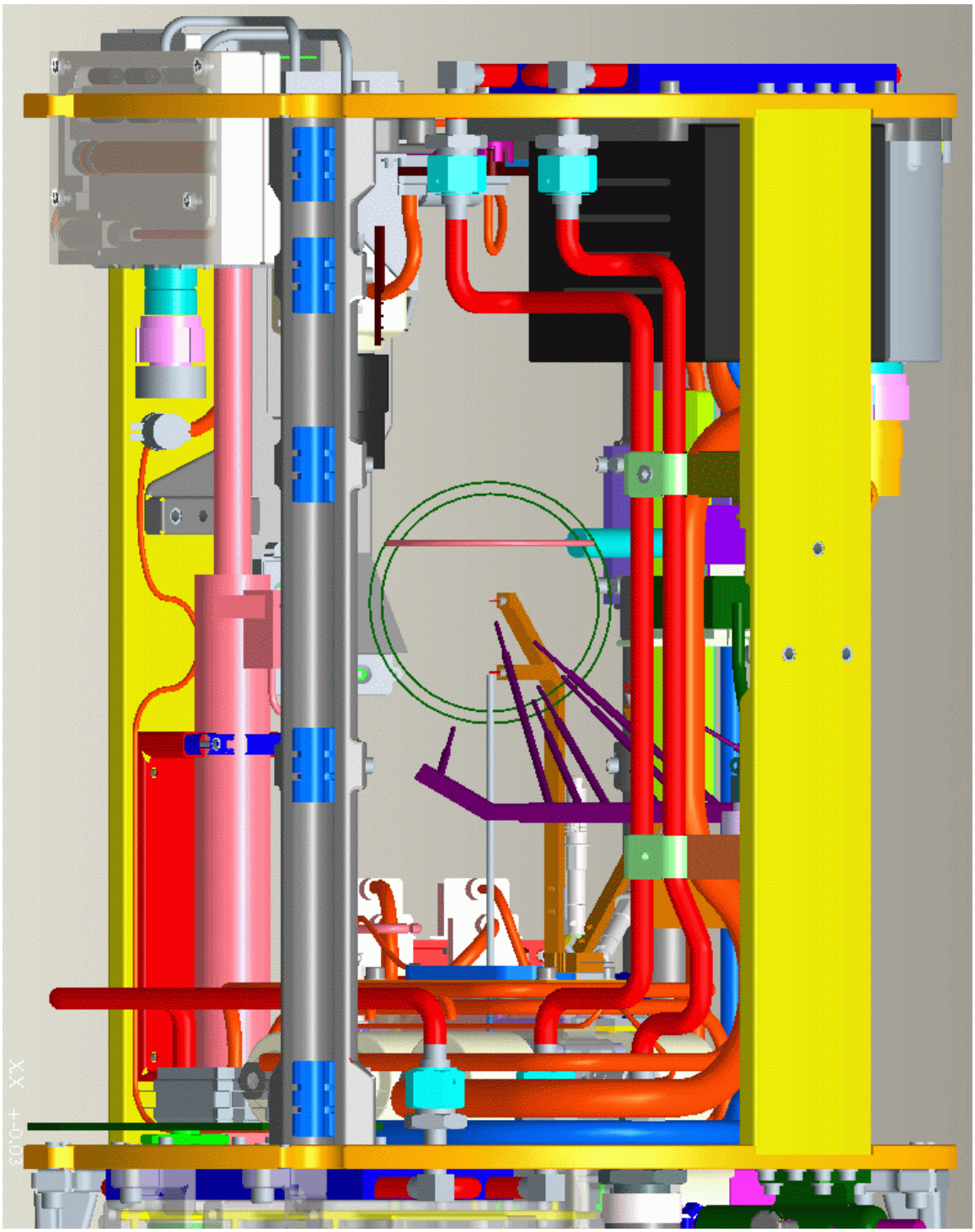


Figure 58. Solid model representation of the chamber insert as seen looking from the Low Light Level Ultraviolet (LLL-UV) camera, which has only been used for the s-Flame experiment. Similarly, the thermocouple rake, of which there are two versions, was only used for spherical flame experiments. To be clear, this figure is not meant to represent the camera's Field of View (FOV).

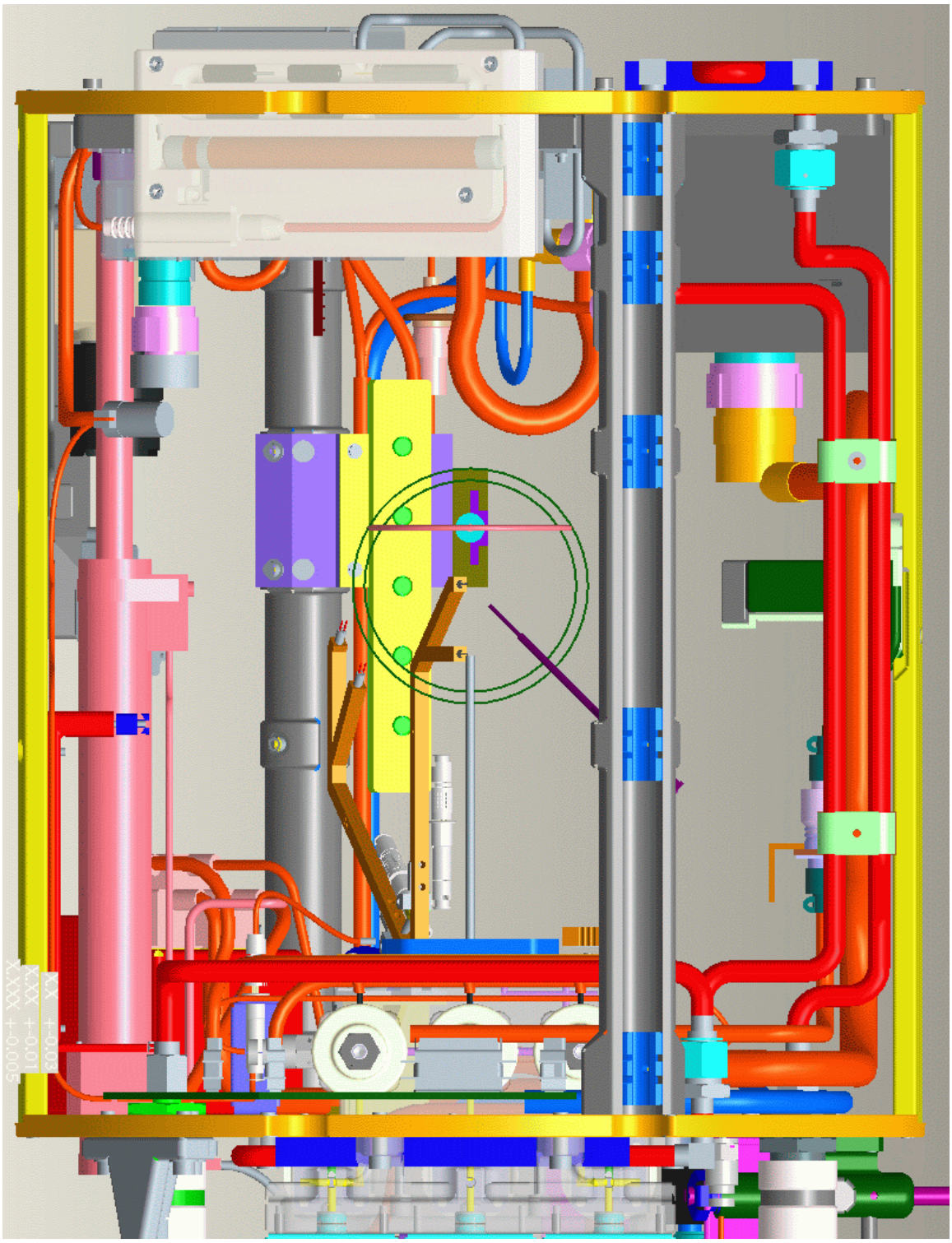


Figure 59. Solid model representation of the chamber insert as seen looking from the direction of the High Bit-depth Multispectral (HiBMs) camera toward the opposing illumination package. Close inspection will reveal that the illumination was significantly blocked in the E-FIELD Flames experiment by the bracket for the electrode mesh. The radiometer array is to the left of the burner and igniter arm (when inserted). To be clear, this figure is not meant to represent the camera's Field of View (FOV).

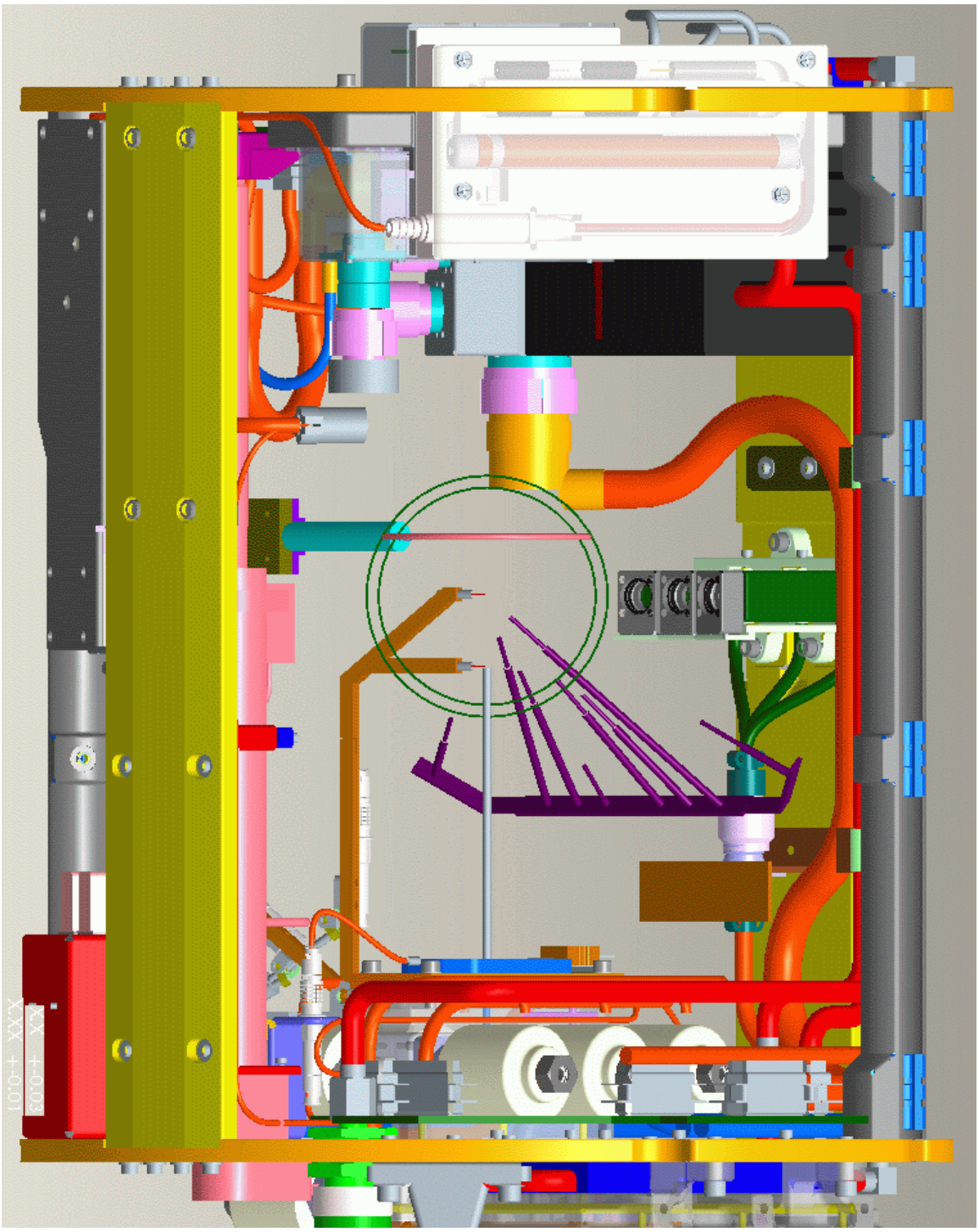


Figure 60. Solid model representation of the chamber insert as seen looking from the ACME Data Camera (DataCam). The thermocouple rake, of which there are two versions, was only used for the spherical flame experiments. To be clear, this figure is not meant to represent the camera's Field of View (FOV).

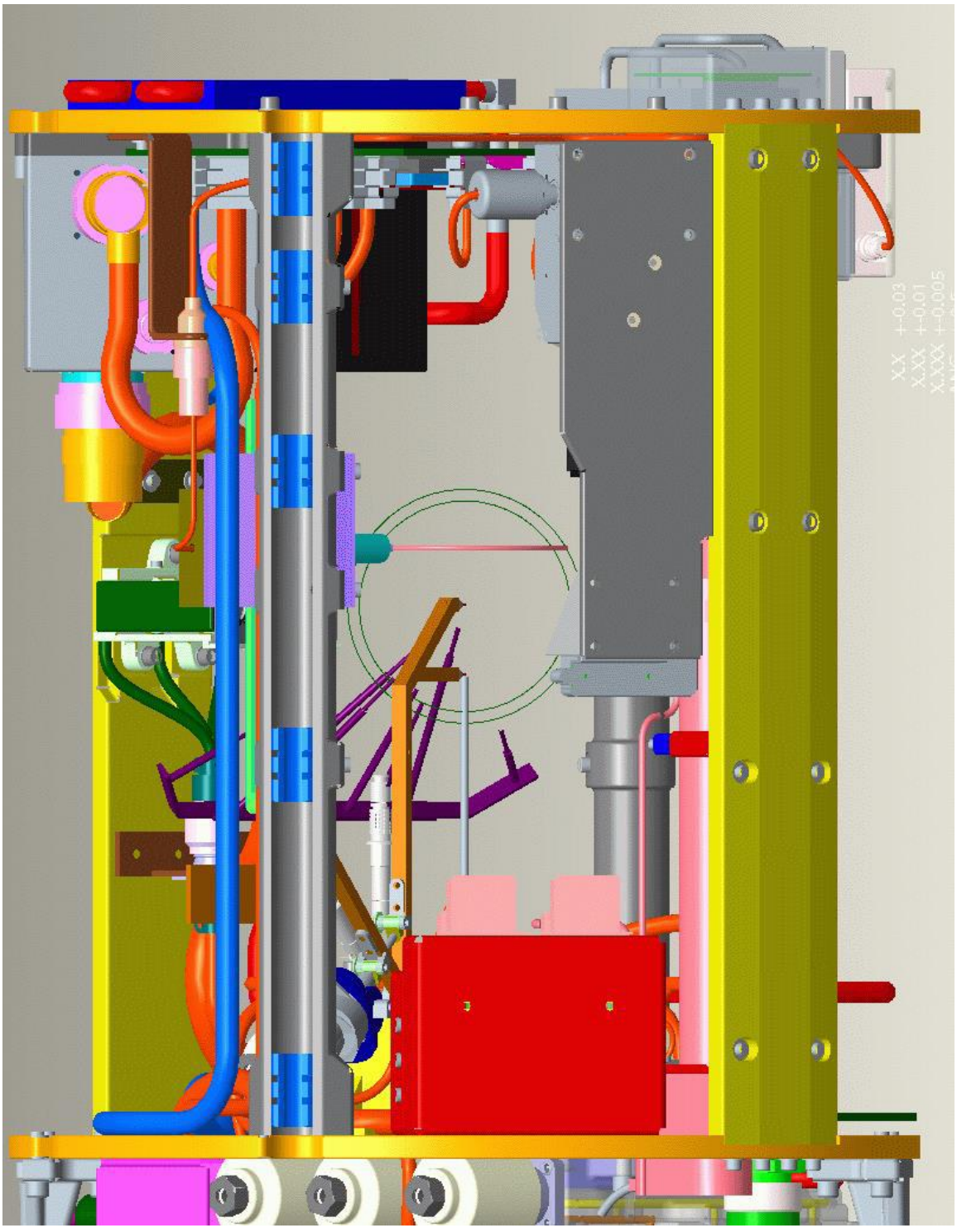


Figure 61. Solid model representation of the chamber insert as seen looking from the Cool Flames Investigation (CFI) intensified camera. The thermocouple rake, of which there are two versions, was only used for the spherical flame experiments. To be clear, this figure is not meant to represent the camera's Field of View (FOV).

F. Radiometer Fields of View

Detailed information about the ACME radiometers can be found in the following documents.

ACME-RPT-011 Radiometer Characterization Report
ACME-RPT-012 BRE Radiometer Characterization Report
CIR-RPT-4600 Cool Flames Radiometer Characterization Report

where the BRE radiometer array was only used in the first round of that experiment. However, the Fields of View (FOVs) are not clearly presented in those reports, so they are shown graphically in this section for the standard, offset, and BRE radiometers, but not the CFI-G radiometer array.

Note that the ACME burners vary in length, where the subsequent figures show a spherical burner, for which the porous sphere should be nominally centered at the intersection of the CIR window axes, the Photomultiplier Tube (PMT) FOVs, and standard radiometer array's central radiometer. The other ACME burners were shorter in length so that the burner-stabilized flames would be approximately centered relative to most of the optical instrumentation. More specifically, the outlets of the existing BRE, coflow, and gas-jet spaceflight burners are respectively about 10, 12, and 10 mm 'below' (i.e., upstream) of plane defined by the axes of CIR chamber windows.

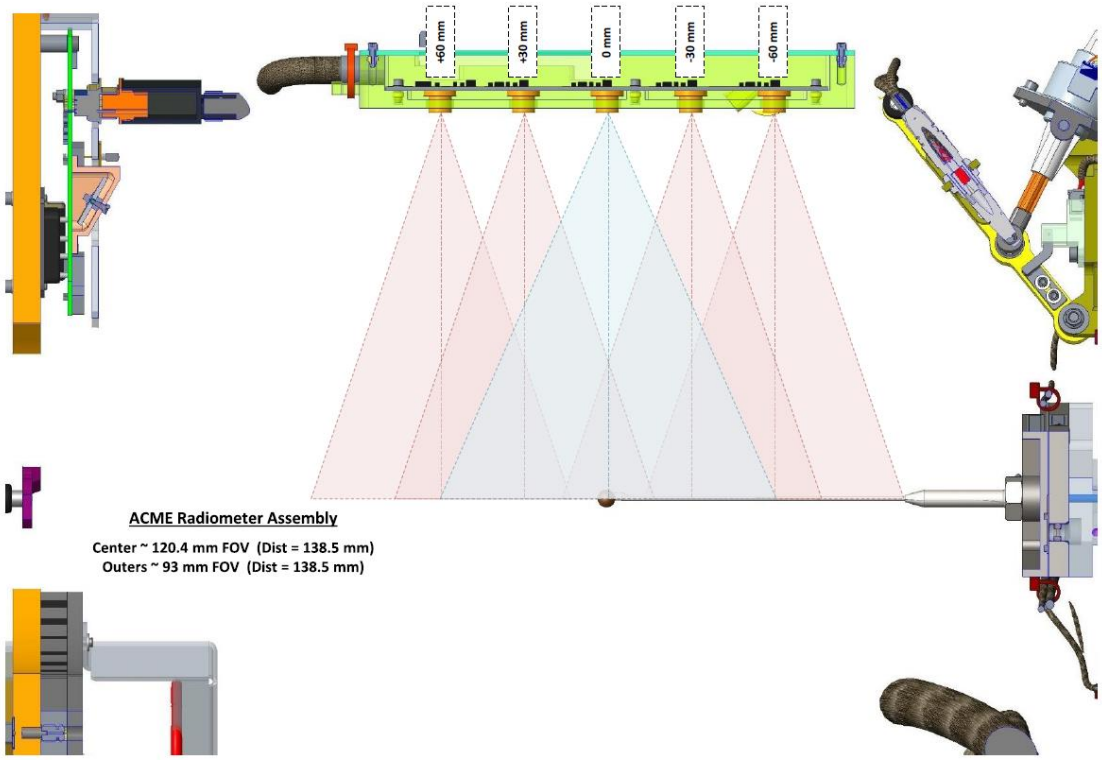


Figure 62. Representation of the FOVs of the five radiometers in the standard linear array. The thermopile detectors are of the same make and model, but the FOV is restricted by an additional aperture (not a lens) in front of each of the four outer radiometers.

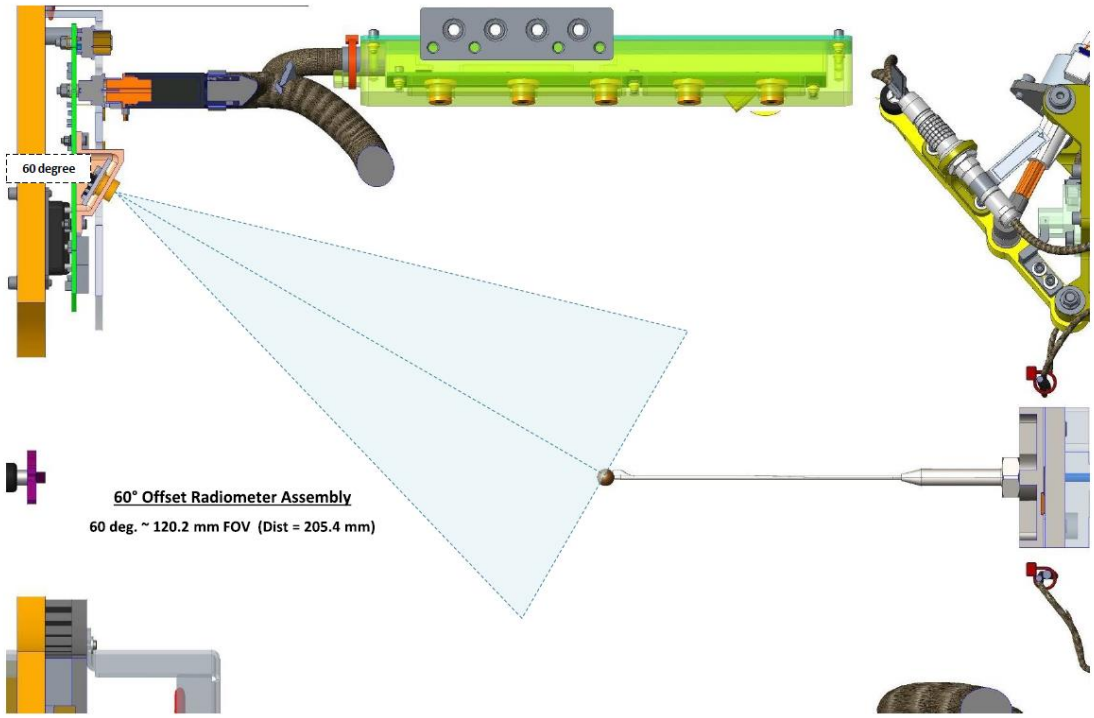


Figure 63. Representation of the offset radiometer's FOV, where that detector is always present. It was primarily intended to support the experiments with spherical flames and features a thermopile detector with a faster response than the other radiometers.

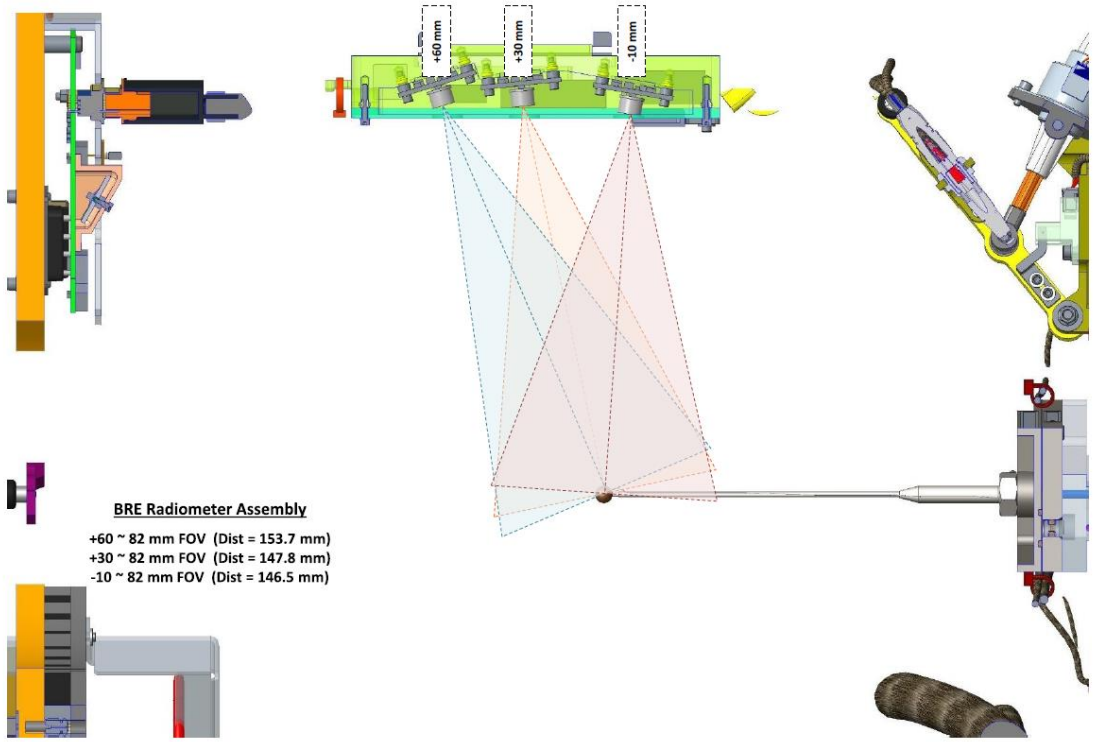


Figure 64. Representation of the FOVs for the Burning Rate Emulator (BRE) radiometer array, which was used in place of the standard radiometer array during BRE's first round of testing.

G. Thermocouples

The thermocouple rakes, of which there are two versions with different probe lengths (referred to as far and near or alternatively short and long), were developed for ACME's spherical flame experiments. The four thermocouple tips on each rake are nominally at the distances below from the center of the burner's porous sphere

Far/short	45, 60, 100, 100 mm
Near/long	13, 25, 50, 100 mm

In the far/short case, the two thermocouples at 100 mm distance are in orthogonal directions from the spherical burner.



Figure 65. Thermocouple rake dimensions.

The size of the exposed thermocouple beads was not measured, but it should be at least twice the 0.2-mm wire diameter.

The two innermost thermocouples (i.e., closest to the spherical burner) on each rake are type-B, while the outer thermocouples are type-K. In contrast, only the two thermocouples at 13 and 25 mm were intended to be type B, where this discrepancy is documented in deviation ACME-DEV-026.

There are also two separate thermocouples which are each at nominally 200 mm distance from the center of the spherical burner where these have thus far been always installed. However, they are located far upstream of the burner outlets and as such will be of limited value for experiments with the other burner types.

Thermocouple 2 (at 200 mm, on the relay side) was reported to be bent by about 10 degrees on 3 April 2018, about 10-15 degrees on 22 May 2018, and about 30 degrees on 19 June 2018. The next three reports indicated the same bend, and the procedures were changed to cover the tip

during crew operations. Photographs, where an example is shown in Figure 66, and crew inspection indicate that has remained bent about 30 degrees since that time. Given the 5 mm length of the bent exposed thermocouple tip, the effect on the 200 mm distance to the spherical burner center is small.

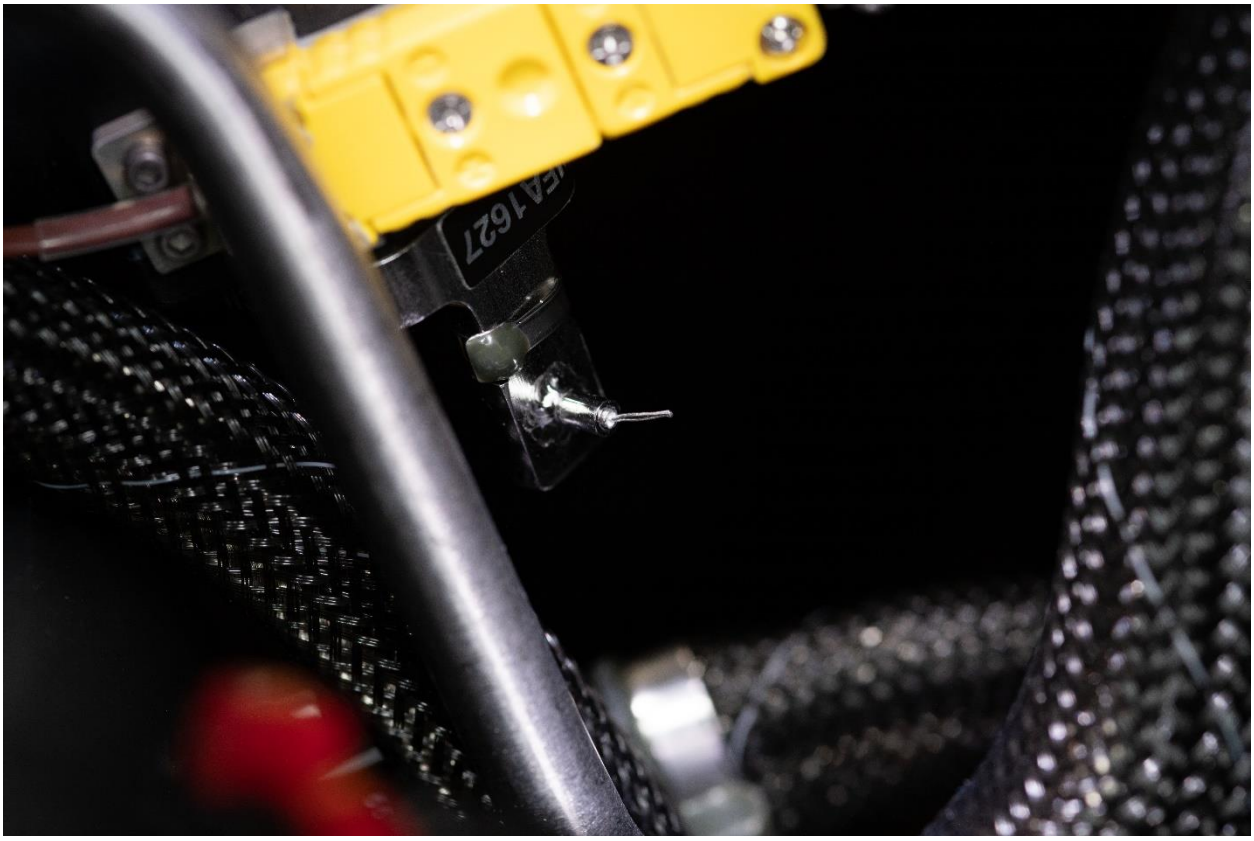


Figure 66. Bent relay-side thermocouple, meant to be 200 mm from the center of the spherical burner's porous sphere, as seen on 19 Sept. 2018 when the chamber insert was reconfigured for the second round of E-FIELD Flames testing. The bent length of exposed wire is 5 mm long. [ISS056e177654]

4. Engineering Documents

This section identifies a variety of engineering documents, some of which are relevant to interpreting ACME experiment results. Section A is broad in scope while B is limited to deviations and waivers.

A. General Documents

A list of general engineering documents including analyses, inspections, plans, procedures, requirements, tests, verifications, etc. is included in this section, where only a portion of these documents is available through the Physical Sciences Informatics Systems (PSI) website, http://psi.nasa.gov/. To avoid unnecessary repetition, only the latest revision for each document has been listed.

The great majority of documents subsequently listed are not required for interpreting the experimental results. But some can be critically important, where the following are some

examples:

ACME-RPT-010	A	Data Camera Characterization Report
ACME-RPT-011	-	Radiometer Characterization Report
ACME-RPT-012	-	BRE Radiometer Characterization Report
ACME-RPT-013	-	PMT Characterization Report

Understanding the design requirements may also provide to be useful, especially in comparison with the verification documents listed in this section and the deviation and waiver documents discussed in the subsequent section. Example documents that may be helpful in this regard include:

ACME-DOC-002	D	REQUIREMENTS COMPLIANCE MATRIX
ACME-DOC-009	-	BRE Requirements Compliance Matrix
ACME-REQ-001	K	Engineering Requirements Document

For interpretation of the experimental results, the most important verification reports are the two listed below. For example, key burner dimensions are provided in ACME-VER-001. Meanwhile, an informal document ACME Science Requirement Verifications (where the file is named ACME_science_reqt_verifications_...) indicates, at least in part, how science requirements specified in the ACME Integrated Science Requirements (ISRD) were verified and in what documents they are addressed.

ACME-VER-001	A	ACME Inspection Verification Report
ACME-VER-002	B	ACME ERD Verification Report

Number	**Rev.**	**Title**
ACME-ADP-001	-	ACME Acceptance Data Package
ACME-ADP-002	-	ACME Spares Acceptance Data Package
ACME-ADP-003	-	ACME Flight Spares Acceptance Data Package
ACME-ADP-004	-	ACME Spare Spherical burner and Igniter Acceptance Data Package
ACME-ADP-005	-	ACME Static Mixer Controller 2 S/N 1003 ADP
ACME-ANA-002	C	ACME Structural Analysis Report
ACME-ANA-003	-	ACME Thermal Analysis Report
ACME-ANA-004	C	ACME Derating Analysis
ACME-ANA-005	A	ACME Optics Bench Components Air Flow Analysis
ACME-ANA-006	-	ACME Avionics Package Power Draw Analysis
ACME-ANA-007	-	ACME Data Camera Power Draw Analysis
ACME-ANA-008	-	ACME Load Input Impedance Analysis
ACME-ANA-009	-	Circuit Isolation and Grounding Analysis
ACME-ANA-010	-	Non-Normal Output Voltage Levels Analysis
ACME-ANA-011	-	Normal Voltage Levels Analysis
ACME-ANA-012	-	Fluid Compatibility Analysis
ACME-ANA-013	A	E-Field Output Voltage Monitory Signal Analysis
ACME-ANA-014	-	E-Field Default Current Leakage
ACME-DOC-001	-	Baseline Concept Description (BCD)
ACME-DOC-002	D	REQUIREMENTS COMPLIANCE MATRIX
ACME-DOC-003	-	Requirements Definition Review (RDR) Presentation Package
ACME-DOC-004	-	Baseline System Description (BSD)
ACME-DOC-005	A	AMP CSCI Software Design Document (SDD)
ACME-DOC-006	-	Interface Design Description (IDD)

ACME-DOC-007	A	ACME Concept of Operations Document
ACME-DOC-008	-	ACME Software Data Dictionary
ACME-DOC-009	-	BRE Requirements Compliance Matrix
ACME-DOC-010	D	ACME Software Interface Design Document
ACME-DOC-011	-	ACME Material Identification / Usage List
ACME-DOC-012	A	ACME Operational Constraints
ACME-DOC-013	-	ACME Pressure Switch Jumper – Space Assurance Requirements Checklist
ACME-INS-001	B	Sharp Edges, Corners, Protrusions & Burrs Inspection
ACME-INS-001A-R2	-	Sharp Edges, Corners, Protrusions & Burrs Inspection Report
ACME-INS-001B-R1	-	Sharp Edges, Corners, Protrusions & Burrs Inspection Report
ACME-INS-001-R1	-	Sharp Edges, Corners, Protrusions & Burrs Inspection Report
ACME-INS-002	C	As-Built Inspection
ACME-INS-002B-R2	-	As-Built Inspection Report
ACME-INS-002C-R1	-	As-Built Inspection Report
ACME-INS-002-R1	-	As-Built Inspection Report
ACME-INS-003	-	As-Built Electrical Inspection
ACME-INS-004	A	Crew Procedures and Operations Products Safety Verifications Inspection
ACME-INS-004A-R2	-	Crew Procedures and Operations Products Safety Verifications Inspection Report
ACME-INS-004-R1	-	Crew Procedures & Operations Products Safety Verifications Inspection Report
ACME-INS-005	-	ACME ORU/Spares Sharp Edges, Corners, Protrusions & Burrs Inspection
ACME-INS-005-R1	-	ORU/Spares Sharp Edges, Corners, Protrusions & Burrs Inspection Report
ACME-INS-006	-	ACME ORU/Spares As-Built Inspection
ACME-INS-006-R1	-	ORU/Spares As-Built Inspection Report
ACME-INS-007	-	Spherical Burner and Igniter Tip ORU/Spares - Sharp Edges, Corners, Protrusions and Burrs Inspection
ACME-INS-007-R1	-	Spherical Burners Sharp Edges, Corners, Protrusions and Burrs Inspection Report
ACME-INS-008	-	Spherical Burner and Igniter Tip ORU/Spares As-Built Inspection
ACME-INS-008-R1	-	Spherical Burners As-Built Inspection Report
ACME-INS-009	-	Spherical Burner ORU/Spare As-Built Inspection
ACME-INS-009-R1	-	Spherical Burner ORU/Spare As-Built Inspection Report
ACME-MEM-001	-	ACME Software Classification
ACME-MEM-002	-	ACME Software Safety Litmus Test Report
ACME-MEM-003	-	ACME Valve and Gas Mixing Testing
ACME-MEM-004	A	Spherical Burner Assembly Fabrication, Test, and Verification
ACME-MEM-005	-	ACME Software Safety Criticality Assessment
ACME-MEM-006	-	ACME Burner Characterization
ACME-MEM-007	-	Water Cooling System Differential Pressure
ACME-MEM-008	-	ACME Software Functional Configuration Audit
ACME-MEM-009	-	ACME Software Physical Configuration Audit
ACME-MIUL-001	B	ACME-MIUL

ACME-MIUL-002	A	ACME Spares MIUL
ACME-MIUL-003	-	ACME Materials Identification/Usage List (MIUL)
ACME-MIUL-004	-	ACME resupply
ACME-MIUL-005	-	ACME Igniter Resupply MIUL
ACME-MIUL-006	-	Materials Identification/Usage List
ACME-PLN-001	-	Configuration Management Plan
ACME-PLN-002	F	Systems Engineering Management Plan (SEMP)
ACME-PLN-003	D	Systems Engineering Verification Plan
ACME-PLN-004	D	Software Management Plan
ACME-PLN-005	-	Software Quality Assurance Plan
ACME-PLN-006	A	EMI/EMC Control Plan
ACME-PLN-007	-	Limited-Life Items Control Plan
ACME-PLN-008	-	ACME Software Maintenance Plan
ACME-PLN-009	-	ACME Software Test Plan
ACME-PLN-010	-	Compliance Matrix against NPR 7150.2A for ACME Software
ACME-PLN-011	A	ACME Space Assurance Requirements Checklist
ACME-PLN-012	-	ACME Igniter Arm Assembly Test Plan
ACME-PLN-012-R1	-	ACME Igniter Arm Assembly Test Plan Results
ACME-PLN-013	A	ACME Storage Plan
ACME-PRO-001	-	ACME Main Processor Software Installation Procedure
ACME-PRO-002	-	ACME Ground Software Installation Procedure
ACME-PRO-003	A	ACME Zoom Lens Assembly Procedure
ACME-PRO-004	-	ACME Qualification Unit Nitrogen System Operating Procedure
ACME-PRO-005	-	ACME Create Software Update Procedure
ACME-PROP-001	-	Stepper Driver Enclosure Assembly, TFP Side, S1101MFA1323
ACME-PROP-002	-	Stepper Driver Enclosure Assembly, Igniter Side, S1101MFA1325
ACME-PROP-003	-	Housing Front Frame Assembly S1101MFA3110
ACME-PROP-004	-	Housing Left Frame Assembly S1101MFA3112
ACME-PROP-005	-	Hard Drive Front Panel Assembly, S1101MFA3303
ACME-PROP-006	-	Hard Drive Tray Housing Assembly, S1101MFA3306
ACME-PROP-007	-	Hard Drive Tray Back Panel Assembly, S1101MFA3308
ACME-PROP-008	-	Housing Top Frame Assembly, S1101MFA3122
ACME-PROP-009	-	Top Cover Airflow Restrictor PlateAssembly, S1101MFA3129
ACME-PROP-010	-	Breaker Cover Assembly, S1101MFA3131
ACME-PROP-011	-	Hard Drive Sleeve Assembly, S1101MFA3301
ACME-PROP-012	-	Housing Right Frame Assembly, S1101MFA3114
ACME-PROP-013	-	Housing Rear Frame Assembly, S1101MFA3116
ACME-PROP-014	-	Housing Rear Frame Interface Bracket Assembly, S1101MFA3119
ACME-PROP-015	-	Housing Baseplate Assembly, S1101MFA3101
ACME-PROP-016	-	Housing Top Cover Assembly, S1101MFA3127
ACME-PROP-017	A	Hard Drive Tray Assembly, S1101MFA3305
ACME-PROP-018	-	Power Distribution Heatsink Assembly, S1101MFA3401
ACME-PROP-019	-	CUBE BASE ASSEMBLY - S1101MFA0202
ACME-PROP-020	-	Hard Drive Tray Assembly - S1101EFA3310
ACME-PROP-021	-	AVP Housing Assembly - S1101MFA3100
ACME-PROP-022	A	Igniter Upper Arm Assembly, 0 MM Position, S1101MFA1313
ACME-PROP-023	A	Igniter Upper Arm Assembly, 10 MM Position, S1101MFA1315

ACME-PROP-024	A	Igniter Upper Arm Assembly, 30 MM Position, S1101MFA1317
ACME-PROP-025	-	Igniter Tip Assembly, S1101MFA1319
ACME-PROP-026	A	E-Field Mesh Assembly, S1101MFA1700
ACME-PROP-027	-	Optical Component Assembly
ACME-PROP-028	A	E-Field Power Supply Assembly, S1101MFA1708
ACME-PROP-029	-	E-Field Supply Base Assembly, S1101MFA1709
ACME-PROP-030	C	ACME-PROP-030 Cube Assembly, S1101MFA0200
ACME-PROP-031	-	Generic Disassembly
ACME-PROP-032	-	Modified Cube PC Board, S1101EFA0214
ACME-PROP-033	-	Modified Cube Power Board, S1101EFA0215
ACME-PROP-034	-	Modified Cube 25 Channel Input Board, S1101EFA0216
ACME-PROP-035	-	Modified Cube 32 Channel Output Board, S1101EFA0217
ACME-PROP-036	-	Modified Cube 16 Channel Input Board, S1101EFA0218
ACME-PROP-037	-	E-Field Negative Supply Interface Assembly
ACME-PROP-038	-	E-Field Positive Supply Interface Assembly, S1101EFA1719
ACME-PROP-039	-	Modified Cube Assembly, S1101MFA0201
ACME-PROP-040	-	Modified Cube Housing Assembly, S1101MFA0212
ACME-PROP-041	-	Modified Cube Housing, S1101MFP0213
ACME-PROP-042	-	Generic Assembly
ACME-PROP-043	-	Modified Camera Motor
ACME-PROP-044	-	Data Camera Enclosure Harness - S1101EFA2214
ACME-PROP-045	-	Hard Drive SATA Data Cable, S1101EFA3506
ACME-PROP-046	C	Mass Flow Controller Assembly Generic
ACME-PROP-047	D	Modified Mass Flow Controller, S1101MFA1140
ACME-PROP-048	-	EMS Locating Pin Assembly, S1101MFA1020
ACME-PROP-049	-	Hard Drive Assembly S1101MFA3300
ACME-PROP-050	E	Generic Burner Assembly
ACME-PROP-051	B	AVP Assembly S1101MFA3000
ACME-PROP-052	-	Card Cage Assembly S1101MFA3200
ACME-PROP-053	B	ACME Data Camera Package S1101MFA2000
ACME-PROP-054	A	Camera Control Assembly S1101MFA2100
ACME-PROP-055	B	Camera Enclosure S1101MFA2200
ACME-PROP-056	B	Filter Barrel Assembly S1101MFA2300
ACME-PROP-057	B	Burner Stem Assembly
ACME-PROP-058	-	Stepper Driver Assembly S1101MFA1322
ACME-PROP-059	A	Radiometer Assembly S1101MFA1520
ACME-PROP-060	A	BRE Radiometer Assembly S1101MFA1550
ACME-PROP-061	-	Data Camera Control to Filter Barrel Harness Assembly Jack Screws Per ECO-407
ACME-PROP-062	A	Input Manifold Assembly, S1101MFA1100
ACME-PROP-063	A	Switching Manifold Assembly, S1101MFA1104
ACME-PROP-064	A	BRE 50mm Burner Assembly, S1101MFA1230
ACME-PROP-065	A	BRE 25mm Burner Assembly, S1101MFA1245
ACME-PROP-066	B	Coflow Burner Assembly, S1101MFA1258
ACME-PROP-067	C	Analog Camera, S1101MFA1400
ACME-PROP-068	-	PMT Assembly, S1101MFA1500
ACME-PROP-069	-	TFP Assembly, S1101MFA1529
ACME-PROP-070	-	TFP Array Assembly

ACME-PROP-071	A	Cold Junction Compensation Assembly, S1101MFA1601
ACME-PROP-072	-	Thermocouple Array Assembly
ACME-PROP-073	B	200mm Thermocouple Array
ACME-PROP-074	-	E-Field Positive Supply Interface Assembly Rework
ACME-PROP-075	-	Main Vibe Plate Testing Process Plan
ACME-PROP-076	-	Secondary Vibe Plate 1 Testing Process Plan
ACME-PROP-077	-	Secondary Vibe Plate 2 Testing Process Plan
ACME-PROP-078	-	TFP to Stepper Harness-Cable Driver Board Coating
ACME-PROP-079	-	E-Field Cover Assembly S1101MFA1730
ACME-PROP-080	-	E-FIELD POSITIVE SUPPLY INTERFACE POPULATE C8
ACME-PROP-081	C	Chamber Insert Assembly, S1101MFA0100
ACME-PROP-082	A	ACME Flight Packaging
ACME-PROP-083	B	E-Field Power Supply Assembly
ACME-PROP-084	-	Analog Camera LED Assembly, S1101MFA1406
ACME-PROP-085	-	MFC COTS Connector Replacement, ACME-NCR-072
ACME-PROP-086	-	Flight Hardware Label Application
ACME-PROP-087	B	Igniter Tip Holder Base Assembly, S1101MFA1332
ACME-PROP-088	-	GN2 Relief Valve Assembly
ACME-PROP-089	-	ACME Accumulator Assembly
ACME-PROP-090	-	ACME Igniter Tip Tempering Process
ACME-PROP-091	A	¼ Spherical Burner Stem Assembly S110MFA1217
ACME-PROP-092	-	S1101MFA1339 Coiled Igniter Assembly
ACME-PROP-093	-	Modified Camera Motion Control Board Assembly
ACME-REQ-001	K	Engineering Requirements Document
ACME-REQ-002	-	Main Processor Software Requirements Specification
ACME-REQ-003	-	Ground Software Software Requirements Specification
ACME-REQ-004	A	ACME Main Processor (AMP) and ACME Ground Software (AGS) Sof
ACME-REQ-005	-	ACME Coil Igniter Engineering Requirement Document
ACME-RPT-001	-	Breadboard Testing Summary Report
ACME-RPT-002	-	Lessons Learned Report
ACME-RPT-003	-	Mass Properties Control Report
ACME-RPT-004	-	Radiometer Concept Feasibility Report
ACME-RPT-005	-	ACME-TST-035A Test Report
ACME-RPT-006	-	ACME-TST-037 Rev – Test Report
ACME-RPT-008	-	ACME-TST-038 Rev- Test Report
ACME-RPT-009	-	ACME AVP Post Workmanship Vibration Functional Test Report
ACME-RPT-010	A	Data Camera Characterization Report
ACME-RPT-011	-	Radiometer Characterization Report
ACME-RPT-012	-	BRE Radiometer Characterization Report
ACME-RPT-013	-	PMT Characterization Report
ACME-RPT-014	-	ACME GIDEP Report
ACME-SDP-001	A	Phase 0/1 Flight Safety Data Package
ACME-SDP-002	A	Phase 2 Flight Safety Data Package
ACME-SDP-003	C	Phase 3 Flight Safety Data Package
ACME-SPROP-001	A	ACME Flight Software Installation
ACME-SPROP-002	A	ACME Linux Installation
ACME-SPROP-003	-	ACME Flight Config and Scripts Installation

ACME-SPROP-004	-	ACME Ground Software Release Procedure
ACME-SPROP-005	-	ACME Ground Software Installation Procedure
ACME-SPROP-006	-	ACME Main Processor Build Procedure
ACME-TST-001	-	ACME Engineering Model Test Plan
ACME-TST-002	-	Flow System Model Breadboard Test Report
ACME-TST-002-R1	-	Flow System Model Breadboard Test Report, As Run Results
ACME-TST-003	-	Breadboard System Level Test
ACME-TST-004	-	E-Field Mesh Assembly Test Plan
ACME-TST-005	B	E-Field Power Supply Assembly Test Procedure
ACME-TST-005-R1	-	E-Field Power Supply Assembly Test Report
ACME-TST-005-R2	-	E-Field Power Supply Assembly Test Report
ACME-TST-006	-	Hard Drive Tray Assembly/ SATA Cable Test Plan
ACME-TST-007	-	ACME EMI E-Field HV Assembly Test Plan
ACME-TST-008	A	ACME Hall Effect Sensor Board
ACME-TST-009	B	ACME Avionics Package Test Plan
ACME-TST-009-R1	-	ACME Avionics Assembly As Run Test Results
ACME-TST-009-R2	-	ACME Avionics Assembly As Run Test Results
ACME-TST-010	A	ACME STEPPER MOTOR ASSEMBLY/SUBASSEMBLY TEST PROCEDURE
ACME-TST-011	-	ACME Power Distribution Board Assembly Test Procedure
ACME-TST-012	-	ACME Flow system Test Procedure
ACME-TST-013	C	ACME Battery Vacuum Test Procedure
ACME-TST-013-R1	-	ACME Battery Vacuum Test Results
ACME-TST-014	-	ACME TC Cold Junction Compensation Board Test Procedure
ACME-TST-015	A	ACME Analog Camera LED Board Test Procedure
ACME-TST-016	-	ACME Filter Board Test Procedure
ACME-TST-017	-	ACME Camera Stepper Driver Board Test Procedure
ACME-TST-018	-	ACME Cube Assembly Test
ACME-TST-018-R1	-	ACME Cube Assembly Test Results
ACME-TST-019	D	Data Camera/Control Enclosure Test Procedure
ACME-TST-019C-R1	-	Data Camera/Control Enclosure As Run Test Results
ACME-TST-019C-R2	-	Data Camera/Control Enclosure Test Results
ACME-TST-020	-	E-Field HV assembly EMI Retest
ACME-TST-021	-	ACME Camera Control Power Distribution Board Test Procedure
ACME-TST-022	A	ACME Radiometer Board Test Procedure
ACME-TST-022A-R1	-	BRE Radiometer Post Workmanship Vibe Functional Test
ACME-TST-022A-R2	-	Post Qual MEFL Vibe BRE Radiometer Functional Results
ACME-TST-022A-R3	-	Post Vibe BRE Radiometer Functional Test Results”
ACME-TST-023	A	Motion Control Board Interface Assembly Test Procedure
ACME-TST-024	A	VGA Interface Board Test Procedure
ACME-TST-025	-	E-Field HV Assembly Test 3
ACME-TST-026	A	Down Stream Relay Board Test Procedure
ACME-TST-027	A	Analog Camera Assembly Test Procedure
ACME-TST-028	A	Upstream Relay Board Test Procedure
ACME-TST-029	-	ACME Flight Cube Board Test
ACME-TST-030	A	Mass Flow Controller Assembly Test Procedure
ACME-TST-030A-R1	-	Mass Flow Controller Workmanship Vibration Functional Test Results 1

ACME-TST-030A-R2	-	Mass Flow Controller Workmanship Vibration Functional Test Results 2
ACME-TST-030A-R3	-	Mass Flow Controller Workmanship Vibration Functional Test Results 3
ACME-TST-030A-R4	-	Post Qual MEFL Vibe MFC Functional Results
ACME-TST-030A-R5	-	Post Acceptance MEFL Functional Test
ACME-TST-030A-R6	-	Post Acceptance MEFL Vibe Functional Test
ACME-TST-031	A	Upstream Relay Connector Board Test Procedure
ACME-TST-032	A	Modified Data Camera Power Board Test Procedure
ACME-TST-033	A	Avionics PC/104 Stack Functional Test Procedure
ACME-TST-034	-	Cold Junction Compensatin Board Calibration Procedure
ACME-TST-035	A	ACME Qualification Unit Proof Pressure Test
ACME-TST-036	B	ACME Chamber Insert Assembly Test
ACME-TST-036-R1	-	ACME Chamber Insert Assembly Test Results
ACME-TST-037	A	ACME WCS Proof Pressure and Leak Test
ACME-TST-037-R1	-	ACME WCS Proof Pressure and Leak Test Results
ACME-TST-038	-	ACME Engineering Model Test Plan
ACME-TST-039	A	ACME Electromagnetic Interference (EMI) Test Plan
ACME-TST-039-R1	-	Flight Model EMI Test Results
ACME-TST-040	C	Workmanship Vibration Test Plan
ACME-TST-040-R1	-	ACME Workmanship Vibration Test Results
ACME-TST-041	B	Mass Flow Controller Tuning Procedure
ACME-TST-042	A	ACME Thermal Vacuum Cycle Test Plan
ACME-TST-043	A	ACME Thermal Cycle Test Plan and Procedure
ACME-TST-043-R1	-	ACME Flight Unit Thermal Cycle Test Results
ACME-TST-044	D	ACME Maximum Expected Flight Load (MEFL) Vibration Test Plan
ACME-TST-044A-R4	-	ACME Coil Igniter MEFL Vibe Test Report
ACME-TST-044C-R5	-	ACME Composite Vibration Test Report for Mass Flow Controller S1101MFA1121 sn 1003
ACME-TST-044-R1	-	ACME MEFL Vibration Test Results
ACME-TST-044-R2	-	ACME Spares MEFL Vibration Test Results
ACME-TST-044-R3	-	Spare Spherical Burner MEFL Vibe Test Results
ACME-TST-045	C	ACME Burner Thermocouple and Heat Flux Sensor Assembly Test
ACME-TST-045-R1	-	Post MEFL Vibe Functional Test
ACME-TST-045-R2	-	Post MEFL Vibe Functional Test
ACME-TST-046	A	ACME Data Camera Power Draw Test
ACME-TST-046-R1	-	Data Camera Package Power Draw Test Results
ACME-TST-047	-	ACME Avionics Power Draw Analysis
ACME-TST-048	A	ACME Avionics Power Draw Test
ACME-TST-048-R1	-	ACME AVP and CIA Power Draw Test Results
ACME-TST-049	-	ACME Data Camera Power Draw Analysis
ACME-TST-050	A	ACME Electrical Bonding Test
ACME-TST-050-R1	-	ACME Electrical Bonding Test Results
ACME-TST-050-R2	-	ACME Electrical Bonding Test Results
ACME-TST-051	-	AVP/CIA and DCP Isolation and Grounding Test
ACME-TST-051-R1	-	Avionics Package/Chamber Insert and Data Camera Package Isolation and Grounding Test Results
ACME-TST-052	-	ACME Gas System Pressure and Leak Test

ACME-TST-052-R1	-	ACME Flight Unit Gas Pressure and Leak Test Results
ACME-TST-053	-	ACME System Functional Test
ACME-TST-053-R1	-	Pre-WCS Pressure and Leak Functional Test Results
ACME-TST-053-R10	-	Pre-EMI Functional Test
ACME-TST-053-R11	-	Post-EMI Functional Test
ACME-TST-053-R12	-	Pre Integration Verification Functional Test Report
ACME-TST-053-R2	-	EM AVP Pre-Vibe Functional Test Results
ACME-TST-053-R3	-	Flight AVP Pre-Vibe Functional Test Results
ACME-TST-053-R4	-	Post Qualification MEFL Vibe Functional Test
ACME-TST-053-R5	-	Post Acceptance MEFL Vibe Functional Test
ACME-TST-053-R6	-	26V Level Functional Test Results
ACME-TST-053-R7	-	29V Level Functional Test Results
ACME-TST-053-R8	-	Pre-Thermal Cycle Functional Test Results
ACME-TST-053-R9	-	Post-Thermal Cycle Functional Test Results
ACME-TST-054	A	ACME In-Lab Verification Test Plan
ACME-TST-054-R1	-	ACME In-Lab Verification Results
ACME-TST-055	-	ACME/CIR Integration Verification Test Plan
ACME-TST-055-R1	A	ACME CIR GIU Verification Results
ACME-TST-055-R2	-	ACME Spares Verification Test Results
ACME-TST-056	-	Normal Voltage Levels Analysis
ACME-TST-056-R1	-	ACME Normal Voltage Level Test Results
ACME-TST-057	-	Bonding of External Connector Shells
ACME-TST-057-R1	-	ACME Bonding of External Connector Shells Test Report
ACME-TST-058	-	Burner Resistance Test Procedure
ACME-TST-058-R1	-	Burner Resistance Test Results
ACME-TST-058-R2	-	1.3mm Gas Jet Burner Flight Spare Resistance Test Results
ACME-TST-058-R3	-	2.1mm Gas Jet Burner Flight Spare Resistance Test Results
ACME-TST-059	-	ACME AVP Additional Bond Paths Test
ACME-TST-059-R1	-	ACME AVP Additional Bond Paths Test Report
ACME-TST-060	-	ACME Cooling Water Thermal Expansion Test
ACME-TST-060-R1	-	ACME Cooling Water Thermal Expansion Test Report
ACME-TST-061	A	ACME DataCam Characterization Test Plan and Procedure
ACME-TST-062	A	E-Field Characterization Test Procedure
ACME-TST-062-R1	-	ACME E-Field Characterization Test Results
ACME-TST-063	-	ACME Software Update Test Plan
ACME-TST-064	-	ACME/CIR Oxygen Mass Flow Controller Tuning Procedure
ACME-TST-064-R1	-	Oxygen Mass Flow Controller S1101MFA1121 SN 1003 Tuning Test Report
ACME-TST-065	-	ACME Pressure Switch Jumper Rack-Level Functional Test Procedure
ACME-TST-065-R1	-	ACME Pressure Switch Jumper Rack-Level Functional Test Report
ACME-TST-066	-	ACME Pressure Switch Jumper Bonding/Grounding Test
ACME-TST-066-R1	-	ACME Pressure Switch Jumper Bonding and Grounding Test Report
ACME-VDD-001	-	Software Version Definition Document
ACME-VER-001	A	ACME Flight Verification Report #1
ACME-VER-002	B	ACME Flight Verification Report #2
ACME-VER-003	A	ACME Flight Verification Report #3
ACME-VER-004	A	ACME Flight Verification Report #4
ACME-VER-005	B	ACME Soot Analysis Verification Report

ACME-VER-006	-	ACME Chamber Insert Assembly Coolant Fill Verification Report
ACME-VER-007	-	ACME Igniter Tips S/N 1019 – 1028 and Igniter Tip Case S/N 1002 Packing Verification Report
ACME-VER-008	-	ACME TFP Arrays S/N 1005 and 1006 Packing Verification Report
ACME-VER-009	-	ACME Coil Igniter Verifications
ACME-VER-010	-	ACME Spherical Burner S/N 1005 Packing Verification Report
ACME-VER-011	-	ACME Coil Igniters S/N 1001 - 1010 and ACME Igniter Tip Holder S/N 1003 Packing Verification Report
ACME-VER-012	-	ACME 1.3 mm and 2.1mm Gas Jet Burners Packing Verification Report
ACME-VER-013	-	ACME Spherical Burner S/N 1003 Packing Verification Report
ACME-VER-016	-	ACME Pressure Switch Jumper S/N 2001 Hazard Verification Report
ACME-VER-017	-	ACME Pressure Switch Jumper S/N 2001 Packing Verification Report

B. Deviations and Waivers

Engineering design requirements have their source in the spacecraft, CIR facility, safety, and of course the experiments' science. But all of them could not be readily addressed. A list of deviation and waiver documents is subsequently provided, including the reason for each request.

ACME-DEV-002

Engineering model will not be exposed to high oxygen concentrations during the course of testing and validation, so this requirement is not necessary for this build

ACME-DEV-004

In order to avoid scrapping and remaking the S1101MFD2206 Zoom Lens Housing, a deviation is required to replace item C13, QTY 6, on S1101MFA2200, from NAS1101E00-3 Screw, Machine-Flat Fillister Head .060-80 UNJF-3A x .188 Length to NAS1101E01-3 Machine-Flat Fillister Head .073-64 UNJF-3A x .188 Length on S/N 1001 and 1002. Also, Revision C of S1101MFD2206, per ACME-ECO-182, will not be incorporated on S/N 1001 and 1002.

ACME-DEV-005

This deviation will enable us to use these parts for the GIU, without having to scrap and rework the Thermocouple rakes.

ACME-DEV-006

This deviation enables the two brackets currently fabricated to be used for EM and flight with minimal rework.

ACME-DEV-007

The NCR and Deviation allow for the use of the covers without having to refabricate parts, at a significant impact to schedule.

ACME-DEV-008

To protect the expensive and delicate flow straightener from undue damage and stress from machining.

ACME-DEV-009

Two assembly fasteners will not engage the locking features of their locking inserts.

ACME-DEV-011

Two of the inserts on this assembly, as identified in ACME-NCR-055 and in the attached PDFs, are missing their locking features. This deviation allows for the use of item C19

Loctite on the two fasteners that no longer have their second locking feature (the first is torque) due to the faulty inserts on the subassembly parts.

ACME-DEV-014

The positions of the two required thermocouples are at 5.72mm and 13.3mm, instead of the required 5 and 15mm.

ACME-DEV-015

A formaldehyde = 440 nm x 100 nm (FWHM) filter will be used in place of the 450 nm bandpass filter described in the requirement.

ACME-DEV-018

The required test matrix received by ZIN engineering from the ACME project scientist excluded gases required in the science requirements document. This disparity would require the engineering team to fill bottles that would not be utilized by science.

ACME-DEV-020

For both S1101MFA1245 Rev.- (25mm BRE) and S1101MFA1230 Rev. A (50mm BRE_ the offset heat flux sensor position radius does not meet the requirement as stated. Actual offset heat flux sensor positions are as follows:

S1101MFA1245 - 9.46mm instead of the required 10mm.

S1101MFA1230 - 15.24mm instead of the required 15mm.

ACME-DEV-025

The manufacturer for the Platinum wire (Goodfellow P/N PT005130) cannot meet the 5 micron requirement, and Zin does not have the capability to accurately measure the wire. The wire diameter for the 3 TFP Calibration Array Assemblies (S1101MFA1546, S1101MFA1557, S1101MFA1558) will instead be known to ±20 microns, the tolerance on the wire per the manufacturer.

ACME-DEV-026

For the purposes of consistency, the two TFP Arrays (Thermocouple Near Array Assembly S1101MFA1605 S/N 1002 and Thermocouple Far Array Assembly S1101MFA1609 S/N 1002) were designed identically, with two Type B thermocouples in the close positions (12 mm and 25mm for the Near, 45mm and 60mm for the Far) and two Type-K thermocouples in the other positions. Type B thermocouples have a high-tolerance range of 870-1700 C and Type K are 0-1250 C. Type B thermocouples do not work in the range of 0-100 C. The inner positions on the far rake (the 45 and 60mm positions) should have been Type K, instead of Type B. Since they are Type B, they do not meet the low end of the temperature range per the requirement. Additionally, at lower temperatures, the ±1 °C measurement also cannot be guaranteed due to the use of the Type B thermocouples.

ACME-DEV-027

The 13mm and 25mm thermocouple locations on Thermocouple Near Array Assembly, S1101MFA1605, have Type B thermocouples installed, as noted by the project scientist in 3.4.4.6. Type B thermocouples do not meet the required tolerance range at the high end of the measurement range per 3.4.4.7. Standard Limit Type B thermocouples have a tolerance range of ±0.5% of measurement At the top end of the measurement range, 1700 C, that tolerance is ±8.5 C, well outside of the required ±3 C. The 45mm and 60mm positions also have Type B thermocouples installed. Their disposition is discussed in ACME-DEV-026, and have the same tolerance range as the 13mm and 25mm positions. The 50mm, 100mm and 200mm positions have Type K thermocouples installed, manufactured by Marlin Mfg. The 50 and 100 mm positions have special limit tolerances, while the 200mm positions have standard limit tolerances. Per Marlin's data sheets, the tolerances on these measurements are based on the temperature measurement:

Special limits: 0 - 285 C : ± 1.1 C
285 - 1250 C: ± .4% (which at the upper range is ± 5 C)
Standard limits: 0 - 285 C: : ± 2.2 C
285 - 1250 C: ± .75% (which at the upper range is ± 9.4 C)

At all temperature ranges, neither the standard limit or special limit Type K thermocouple can meet the ± 1 C requirement.

ACME-DEV-028

For the Static Mixer QD:

The part number chosen by ACME is 941-08-2-T-04-V-D. This differs from the required part number in the -04- term, defining a 1/4" ID hose connection, instead of 1/2" and in the final D term, which specifies a cap. Additionally, no keying option was selected, so the YY field is blank.

For the Manifold #4 (Fuel) QD:

The part number chosen by ACME is 941-04-2-T-04-V-D. This differs from the requirement in the final D term, which specifies a cap. Additionally, no keying option was selected, so the YY field is blank.

ACME-DEV-031

The outer diameter of the outer support tube is 28.8% of porous sphere diameter.

ACME-DEV-032

The wire used on pin 57 of PPIL1-C (Assembly S1101EFA3501) is 22 not 20 AWG as the requirement states. The requirement detail was missed before the ACME harness was designed and built. The wire on PPIL-2 pin 26 (Assembly S1101DEF3500) is the correct 16 AWG size.

ACME-DEV-033

The Cube Assembly (13 of S1101MFA0100) and the Analog Camera Assembly ((item 52 of S1101MFA0100) are not directly bonded to the CIR structure but are bonded to chassis through their power returns to their respective single point ground locations.

ACME-DEV-035

Review of the drawings shows the requirement is met in the S1101EFA0309 - CIA Signal Harness Assembly and the S1101EFA1413 - Analog Camera to Downstream Relay Harness Assembly - both of which reside in the Chamber Insert Assembly (CIA). However in the Avionics Package (AVP) for the S1101EFA3501 harness, for this short run for the CIA video (5”) and the S1101EFA3502/S1101EFA3501 harness for the Data Camera Video (14”) twisted 22 AWG - M22759/11-22 – was used.

ACME-DEV-036

The E-Field Voltage Monitoring signal doesn't meet the .12% tolerance requirement referenced in the description of ISRD trace 1.7.1 below a magnitude of 3595 volts as calculations from ACME-ANA-013 illustrate In addition the signal does not meet the "1% or 20 volt -whichever is larger" tolerance referenced in the description of ISRD Trace 2.5.1, 2.5.4 between 80 and 20 volts. This is also illustrated in ACME-ANA-013. Using this same reasoning for the verification activity of the - Electric Potential – Accuracy requirement of section 2.51, 2.54 from the ISRD altering the above equations give:

For the low side: .99 * Vout < ((.999 * Vout) -.719) solving this gives Vout > 80 volts
For the high side 1.01 * Vout > ((1.001 * Vout + .719) solving for Vout also gives Vout > 80 volts.

ACME-DEV-038

The BRE Burner flux sensor position is out of the required tolerance. The tolerance is ± 0.05mm (.002 in).

S1101MFA1230 50mm BRE Burner: Both positions are at -.012 in.
S1101MFA1245 25mm BRE Burner: Center positions is at -.008 in, and the outer position is at -.014 in.

ACME-DEV-039

ACME Does not use Nitrogen Bottles

ACME-DEV-040

The TFP motor (S1101MFA1529 all current and subsequent serial numbers) cannot reach the minimum translation speed of 100mm/s. It also cannot achieve an uncertainty of .1mm/s from commanded to actual translation speed.

ACME-DEV-041

Commercially sourced Silicon Carbide (SiC) fibers are 14 Micron diameter. There is no given tolerance band on the requirement so 14 microns is not within the requirement.

ACME-DEV-046

A BG-7 Schott Glass Filter included on the ACME Data Camera Package meets all the specifications described in the requirement, except for the 1mm thickness. The BG-7 Filter that was purchased and installed has a thickness of 2mm.

ACME-DEV-047

The long dimension of the FOV is not aligned with the CIR chamber axis.

ACME-DEV-049

Igniter speed uncertainty was measured as being 2.54 mm/s not the required 0.1mm/s or less.

ACME-DEV-050

ACME Igniter Motor testing met the first portion of the requirement but failed to retract the igniter from the "free area" of the ACME burner wiith in 0.5 seconds of turning off. Instead of 0.5 seconds, removal of the igniter from the free area was measured to have happened in 0.6 secs.

ACME-WAI-012

The 9.5mm (3/8 inch) and 12.7mm (1/2 inch) burners were included in the requirements as contingencies for the 6.4mm (1/4 inch) burner.

ACME-WAI-013

Only two of the originally required gas jet burner sizes are used in the required test matrix for ACME. (1.3mm and 2.1mm I.D.)

ACME-WAI-016

One instance of the hall effect sensor boards (S1101EFA2216 s/n 1006) was excluded from workmanship vibration testing.

ACME-WAI-019

ACME engineering will not be performing ignition reliability tests in 1g using gaseous fuels and therefore cannot meet this requirement.

ACME-WAI-021

The science requirements document requires flow rates that are not being used in the flight required test matrix.

ACME-WAI-022

The spherical burner positioned at 30mm upstream from the CIR window axis and the gas jet burner positioned at 30mm upstream of the CIR window axis is not needed for flight.

ACME-WAI-023

The E-Field Supply Base and Downstream Relay Board Cover both impinge on the required exclusion zone.

ACME-WAI-029

The parts were fabricated on a calibrated CNC machine, but the inspection prints were incompletely provided.

ACME-WAI-030

Request a waiver of the thermal vacuum cycle test for ACME flight unit the following reasons:

- The GRC testing facility was found to not be adequate due to lack of engineering best practices.
- The engineering model (that is built to flight drawings) survived a mission sequence test in which it was exposed to a vacuum while operating several times, giving the engineering team high confidence that the flight unit will behave in the same manner. (ACME-RPT-008)
- The flight unit has successfully passed thermal cycle testing with no adverse effects (ACME-TST-043-R1)
- The thermally sensitive component (analog camera) that drove the need for vacuum thermal cycle testing is an orbital replaceable unit and can be swapped out in the unlikely event of a failure on orbit. (See ACME risk 034)

ACME-WAI-034

The video circuitry utilizes a transmitter that is powered by the ACME secondary system and meets the balance signal requirement but has has a 1ohm impedance to supply "ground" (ie. to the system secondary single point ground) although this meets the requirement in point 2 (that it be referenced to the structure at a single point) it cannot at the same time meet the requirement to be isolated from structure (and thus single point ground) by > 6k.

ACME-WAI-037

A portion of the Cube Assembly case (13 of S1101MFA0100) is not directly bonded to the CIR structure but is bonded to through the primary power returns respective single point ground location at the EPCU. This bond path was tested for the < 0.1 ohm value which it did not meet, thus it does not meet the requirement of ACME-PVP-7.2.1a Part 1.

ACME-WAI-042

The ACME DataCam Failed to meet the success criteria for FCF optics bench air flow testing as described in requirement 5.1.1.3.1 FCF-PLN-4149 Rev.- ACME Payload verification plan. Per ACME-ANA-005 section 3.2: The analytical required flow rate for the DataCam given it's known power draw was determined to be 1.85CFM. However, the minimum allowed flow rate to achieve a pressure drop of 0.4 inches of H2O was determined by test to be 6.08CFM. Therefore the DataCam does not meet the requirement as stated.

ACME-WAI-043

The ACME AVP Failed to meet the success criteria for FCF optics bench air flow testing as described in requirement 5.1.3.1 FCF-PLN-4149 Rev.- ACME Payload verification plan. Per ACME-ANA-005 section 3.3: The analytical required flow rate for the AVP given it's known power draw was determined to be 10.78CFM. However the maximum allowed flow rate to achieve a pressure drop of 0.4 iches of H20 was determined by test to be 8.68CFM. Therefore the AVP does not meet the requirement as stated.

ACME-WAI-044

The Advanced Combustion Microgravity Experiment (ACME) does not meet the SSP 30237 requirements for radiated susceptibility (RS03), conducted susceptibility (CS01) and conducted emission CE(07) as documented in electromagnetic interference (EMI) test

report GRC-EMI-RPT-381. The radiated susceptibility (RS03) symptoms were higher than normal (< 10%) radiometer signal readings mostly in the horizontal polarity, between 1 and 14 GHz. The conducted susceptibility (CS01) symptoms were higher than normal deviating signal readings of the thermocouples photomultipliers and the pressure transducers, all less than +/-1% full scale. The transient of the CE07 test exceeded the limit by as much as 4V at .4ms and was within the envelope within 2ms.

ACME-WAI-045

The Data Package uses external harnesses whose backshells to case or bulkhead mount resistances were required to be <2.5 milliohms. Not all the backshells get down to that value. The Results were as follows:

801-040-07M19-26SA-422B to PIPG3 Backshell 2.00 milliohms pass
PCAM2 to J33 8.53 milliohms fail
PCAM5 to J34 1.30 milliohms pass
PCAM4 to J75 2.30 milliohms pass
PCAM 1 to J36 4.60 milliohms fail
PCAM3 to J76 3.24 milliohms fail

ACME-WAI-048

An image was taken in order to check for reflections, using the ACME Flight Data Camera Package installed on the CIR GIU. The chamber was dark and only the ignitor, with a halogen bulb instead of the ignitor wire, was powered at a setting close to the expected max. The resulting image shows the spherical burner is in the center (it is dark but can be seen) and the bright ignitor is coming in from the left. Between the 2 you can see the round reflection off the window directly opposite from the Data Camera. Note that the opposite window is the Intensified Camera view, looking at the flame from the other side. There are no other significant internal reflections from the hardware, leaving the only concern for reflections being the opposite window.

Deviations and waivers which were identified by Stocker as affecting science in August 2016 are topically listed here. The degree to which the deviations and waivers were expected to positively or adversely affect the science at that time is indicated using stoplight indicators for good (G), neutral (N), cautionary (Y), or bad (R). The issue of back-reflection from opposed windows noted in regard to ACME-WAI-048 was subsequently addressed with the addition of internal window covers shown in the figures in section 3.C Optical Layout. Meanwhile, ignition concerns associated with ACME-WAI-019 were partially mitigated by the development of a second type of hot-wire igniter tip, specifically with a spring-like coil design.

Ambient environment

(Y) ACME-WAI-023 – impingement of hardware by up to 8 mm into exclusion zone

BRE burners

(Y) ACME-DEV-014 – minor offset in thermocouple positions

(Y) ACME-DEV-020 and ACME-DEV-038 – minor recessed position of the heat flux gages (i.e., where the gages aren't flush with the outlet plate surface)

DataCam diagnostics

(Y) ACME-DEV-015 – formaldehyde instead of 450-nm bandpass filter

(Y) ACME-DEV-025 – ±20 instead of ±5 micron uncertainty in platinum wire diameter (for absolute light calibration)

(G) ACME-DEV-041 – SiC fibers have 14 instead of 15 micron diameters

(R) ACME-DEV-046 – blue-green BG7 filter is 2 instead of 1 mm thick adversely affecting the transmission and thus the quantitative determination of temperature, soot volume fraction, and CH* concentrations
(R) ACME-DEV-047 – camera is not on its side with its long dimension along flame (chamber) axis
(R) ACME-WAI-048 – back reflection from opposite window

Electric field

(Y) ACME-DEV-036 – voltage uncertainty greater than specified
(Y) ACME-WAI-044 – high susceptibility of radiometer, thermocouple, and pressure measurements to EMI, and high conducted transient emissions (relative to FCF specifications)

Gas-jet burners (*E-FIELD Flames only*)

(N) ACME-WAI-013 – 2 instead of 7 burner sizes, i.e., matching ops plan

Ignition

(R) ACME-WAI-019 - no 1g testing
(Y) ACME-DEV-040, ACME-DEV-049, and ACME-DEV-050 – motion is slower than specified (e.g., retracts in 0.6 instead of 0.5 s) and there is greater than specified uncertainty in the speed

Operations

(N) ACME-DEV-018 – exclusion of specified gas bottles not used in ops plan
(N) ACME-WAI-021 – exclusion of flow ranges, i.e., mass flow controllers, not used in ops plan

Spherical burners

(N) ACME-WAI-012 – 1 instead of 3 burner sizes (i.e., just ¼" spheres)
(N) ACME-WAI-022 – center sphere position only (i.e., no offset burner option for large flames)
(Y) ACME-DEV-031 – tube O.D. of 28.8% instead of ≤25% of the sphere diameter

Thermocouple rake measurements (*spherical flames only*)

(Y) ACME-DEV-026 – type B instead of K thermocouples for 45 and 60-mm positions
(Y) ACME-DEV-027 – greater than specified measurement uncertainty

5. Drawings

An extensive list of drawings associated with the ACME project is included in this section, where only a portion of these drawings is available through the Physical Sciences Informatics Systems (PSI) website, http://psi.nasa.gov/. They are organized by category as can be seen in the Table of Contents, where the most important for interpreting the experimental data are likely the burner drawings in section F. Note that drawings for the Combustion Integrated Rack (CIR), the facility in which ACME was implemented, are not listed here.

Meanwhile, many photographs of the hardware are available at www.flickr.com/photos/space-flames. Additional photographs can be found at other websites, such as www.facebook.com/space.flames. While neither of those sites are specifically focused on ACME, they are both administered by Dennis Stocker, the ACME Project Scientist.

A. Top Level Assembly

S1101MFB0001	ACME Flight Drawing Tree
S1101MFS0002	ACME Pneumatic Schematic
S1101MFA0003	ACME Chamber Insert Assembly On-Orbit Configuration
S1101MFS0004	ACME / CIR Integrated Pneumatic Schematic
S1101MFA0100	ACME Chamber Insert Assembly Launch Configuration
S1101MFT0101	Generic IMS Label

B. Common Electrical

S1101MFA0200	Cube Assembly
S1101MFA0201	Modified Cube
S1101MFA0202	Cube Base Assembly
S1101MFD0203	Cube Base
S1101MFA0204	Cube wiring cover assembly
S1101MFD0205	Cube wiring cover
S1101MFD0206	Cube Insulator
S1101MFD0207	Cube Fastener Insulating Sleeve
S1101MFD0208	Cube Rear Cover
S1101EFA0209	Cube Power Harness Assembly
S1101EFA0210	Cube Communication Harness Assembly
S1101EFA0211	Cube Signal Harness Assembly
S1101MFA0212	Modified Cube Housing Assembly
S1101MFP0213	Modified Cube Housing
S1101EFA0214	Cube PC Board
S1101EFA0215	Cube Power Board
S1101EFA0216	Cube 25 Channel Input Board
S1101EFA0217	Cube 32 Channel Output Board
S1101EFA0218	Cube 16 Channel Input Board
S1101EFA0300	Downstream Relay Board Assembly
S1101EFS0300	Downstream Relay Board Schematic
S1101EFF0300	Downstream Relay Board Fabrication
S1101EFL0300	Downstream Relay Board Artwork
S1101MFD0301	Downstream Relay Board Cover
S1101EFA0302	Upstream Relay Connector Board Assembly
S1101EFS0302	Upstream Relay Connector Board Schematic
S1101EFF0302	Upstream Relay Connector Board Fabrication
S1101EFL0302	Upstream Relay Connector Board Artwork
S1101EFA0303	PMT/TC to Upstream Relay Board Harness Assembly
S1101EFA0304	Cube to Upstream Relay Board Harness Assembly
S1101EFA0305	Cube to Downstream Relay Board Harness Assembly
S1101EFA0306	IRR to Bracket Signal Harness Assembly
S1101EFA0307	IRR to Bracket Power Harness Assembly
S1101EFA0308	CIA Power Harness Assembly
S1101EFA0309	CIA Signal Harness Assembly
S1101EFA0310	E-Field Power Supply to Relay Board and Burner Harness Assembly
S1101EFA0311	E-Field Power Supply to Mesh Harness Assembly
S1101MFD0312	Upstream Relay Board Cover
S1101EFA0313	Upstream Relay Board Assembly
S1101EFS0313	Upstream Relay Board Schematic
S1101EFF0313	Upstream Relay Board Fabrication
S1101EFL0313	Upstream Relay Board Artwork
S1101MFA0314	Upstream Relay Board Stack Assembly
S1101EFC0315	CIA Interconnect Diagram
S1101MFA0316	Low Pressure Switch Assembly

S1101MFA0317	High Pressure Switch Assembly
S1101MFD0318	Pressure Switch Mounting Plate
S1101MFA0319	Pressure Switch Mounting Plate Assembly
S1101MFP0320	Pressure Switch Fastener
S1101MFA0321	ACME Cap Bag 1
S1101MFA0322	ACME Cap Bag 2
S1101MFA0323	ACME Cap Bag 3
S1101EFA0324	Pressure Switch Disable

C. Quick Disconnects (QDs) and Hoses

S1101MFA0400	Fuel Hose Assembly
S1101MFA0401	Oxygen Hose Assembly
S1101MFA0402	Inert Hose Assembly
S1101MFA0403	Green Water Cooling Hose Assembly
S1101MFA0404	Gray Water Cooling Hose Assembly
S1101MFA0405	GN2 Relief Valve Assembly
S1101MFA0406	Accumulator Assembly

D. Experiment Mounting Structure (EMS)

S1101MFA1000	Experiment Mounting Structure Assembly
S1101MFA1001	Upstream Endcap Assembly
S1101MFD1002	Upstream Endcap
S1101MFA1003	Downstream Endcap Assembly
S1101MFD1004	Downstream Endcap
S1101MFA1005	Igniter Side Frame Rail Assembly
S1101MFD1006	Igniter Side Frame Rail
S1101MFA1007	E-Field Side Frame Rail Assembly
S1101MFD1008	E-Field Side Frame Rail
S1101MFD1009	Frame Rail Bearing
S1101MFP1010	Bearing Retaining Screw
S1101MFA1011	Upper Component Mount Assembly
S1101MFD1012	Upper Component Mount
S1101MFA1013	Lower Component Mount Assembly
S1101MFD1014	Lower Component Mount
S1101MFD1015	IRR Electrical Connector Bracket
S1101MFA1016	Radiometer Bracket Assembly
S1101MFD1017	Radiometer Bracket
S1101MFD1018	Extraction Handle
S1101MFP1019	Bearing Retaining Screw
S1101MFA1020	EMS Locating Pin Assembly
S1101MFD1021	Locating Pin Housing
S1101MFD1022	Locating Pin
S1101MFD1023	Locating Pin Handle
S1101MFD1024	Spring Stop
S1101MFP1025	Modified Socket Head Cap Screw, .190-32 UNRF-3A X .750
S1101MFA1026	Burner Connector Bracket Assembly
S1101MFD1027	Burner Connector Bracket

S1101MFA1028	Rail Lock Assembly
S1101MFA1029	Rail Lock Bracket Assembly
S1101MFD1030	Rail Lock Bracket
S1101MFP1031	Rail Lock Screw

E. Gas Supply

S1101MFA1100	Input Manifold Assembly
S1101MFA1101	Input Manifold Insert Assembly
S1101MFD1102	Input Manifold
S1101MFD1103	Input Manifold Cold Plate Spacer Pad
S1101MFA1104	Switching Manifold Assembly
S1101MFA1105	Switching Manifold Insert Assembly
S1101MFD1106	Switching Manifold
S1101MFA1107	Solenoid Coil Assembly
S1101MFD1108	Solenoid Coil Nut
S1101MFD1109	Solenoid Coil Set Screw
S1101EFA1110	Input Manifold Harness Assembly
S1101EFA1111	Switching Manifold Harness Assembly
S1101MFD1112	Mass Flow Controller Mounting Plate
S1101MFD1113	Mass Flow Controller Cover
S1101MFA1114	Mass Flow Controller Cover Bracket Assembly
S1101MFD1115	Mass Flow Controller Cover Bracket
S1101MFP1116	Mass Flow Controller Mounting Screw
S1101EFA1117	Oxygen Mass Flow Controller Harness Assembly
S1101EFA1118	Fuel Mass Flow Controller Harness Assembly
S1101EFA1119	Inert Mass Flow Controller Harness Assembly
S1101MFA1120	Oxygen Mass Flow Controller Assembly, 0.5 SLPM
S1101MFA1121	Oxygen Mass Flow Controller Assembly, 20 SLPM
S1101MFA1122	Fuel Mass Flow Controller Assembly, 0.1 SLPM
S1101MFA1123	Fuel Mass Flow Controller Assembly, 0.5 SLPM
S1101MFA1124	Inert Mass Flow Controller Assembly, 0.1 SLPM
S1101MFA1125	Inert Mass Flow Controller Assembly, 5 SLPM
S1101MFA1126	Inert Mass Flow Controller Assembly, 0.5 SLPM
S1101MFA1127	*OBSOLETE*
S1101MFA1128	*OBSOLETE*
S1101MFA1129	*OBSOLETE*
S1101MFA1130	*OBSOLETE*
S1101MFD1131	Mass Flow Controller Interface Cover
S1101MFA1132	Fuel Mass Flow Controller Assembly, 0.05 SLPM
S1101MFA1133	Fuel Mass Flow Controller Assembly, 0.2 SLPM
S1101MFA1134	Fuel Mass Flow Controller Assembly, 1 SLPM
S1101MFA1135	Inert Mass Flow Controller Assembly, 0.01 SLPM
S1101MFA1136	Inert Mass Flow Controller Assembly, 0.02 SLPM
S1101MFA1137	Inert Mass Flow Controller Assembly, 0.05 SLPM
S1101MFA1138	Inert Mass Flow Controller Assembly, 2 SLPM
S1101MFA1139	Inert Mass Flow Controller Assembly, 1 SLPM NOT REQUIRED FOR FLIGHT

S1101MFA1140-001	Modified HFC-D-302 0-0.01 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-002	Modified HFC-D-302 0-0.02 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-003	Modified HFC-D-302 0-0.05 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-004	Modified HFC-D-302 0-0.1 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-005	Modified HFC-D-302 0-0.2 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-006	Modified HFC-D-302 0-0.5 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-007	Modified HFC-D-302 0-1 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-008	Modified HFC-D-302 0-2 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-009	Modified HFC-D-302 0-5 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1140-010	Modified HFC-D-302 0-20 SLM Mass Flow Controller (Teledyne Hastings)
S1101MFA1141	Mass Flow Controller Interface Cover Assembly
S1101MFA1142	Inert Mass Flow Controller Assembly, 0.2 SLPM
S1101MFA1143	Fuel Mass Flow Controller Assembly, 2.0 SLPM

F. Burners

S1101MFA1200	Burner Base Assembly
S1101MFD1201	Burner Top Plate
S1101MFD1202	Burner Insulator
S1101MFD1203	Burner Bottom Plate
S1101MFD1204	Burner Mounting Screw Washer
S1101MFD1205	Burner Mounting Screw Bushing
S1101MFD1206	Burner Assembly Bushing
S1101MFA1207	1.3 mm ID Gas Jet Burner Assembly, 10mm Position
S1101MFA1208	1.3 mm ID Gas Jet Burner Stem Assembly, 10mm Position
S1101MFD1209	1.3 mm ID Gas Jet Burner Stem, 10mm Position
S1101MFD1210	.062" OD Burner Tube Collar
S1101MFA1211	2.1 mm ID Gas Jet Burner Assembly, 10mm Position
S1101MFA1212	2.1 mm ID Gas Jet Burner Stem Assembly, 10mm Position
S1101MFD1213	2.1 mm ID Gas Jet Burner Stem, 10mm Position
S1101MFD1214	.095" OD Burner Tube Collar
S1101EFA1215	Burner Harness Assembly
S1101MFA1216	1/4" Spherical Burner Assembly
S1101MFA1217	1/4" Spherical Burner Stem Assembly
S1101MFA1218	1/4" Spherical Burner Tube Assembly
S1101MFD1219	1/4" Spherical Burner Tube
S1101MFA1220	3/8" Spherical Burner Assembly
S1101MFA1221	3/8" Spherical Burner Stem Assembly
S1101MFD1222	.083" OD Burner Collar
S1101MFA1223	3/8" Spherical Burner Tube Assembly
S1101MFD1224	3/8" Spherical Burner Tube
S1101MFA1225	1/2" Spherical Burner Assembly

S1101MFA1226	1/2" Spherical Burner Stem Assembly
S1101MFD1227	.120" OD Burner Tube Collar
S1101MFA1228	1/2" Spherical Burner Tube Assembly
S1101MFD1229	1/2" Spherical Burner Tube
S1101MFA1230	BRE 50mm Burner Assembly
S1101MFA1231	BRE 50mm Base Assembly
S1101MFD1232	BRE 50mm Base
S1101MFD1233	BRE Flow Tube
S1101MFD1234	BRE 50mm Bottom Housing
S1101MFD1235	BRE 50mm Sensor Mounting Plate
S1101MFD1236	BRE 50mm Top Housing
S1101MFD1237	BRE 50mm Burner Outlet
S1101MFD1238	BRE 50mm Flow Screen
S1101MFD1239	BRE 50mm Flow Straightener
S1101MFD1240	BRE 50mm Top Spacer
S1101MFD1241	BRE 50mm Baffle
S1101MFD1242	BRE Thermocouple Plug
S1101MFD1243	BRE Sensor Plug
S1101EFA1244	BRE 50mm Harness Assembly
S1101MFA1245	BRE 25mm Burner Assembly
S1101MFA1246	BRE 25mm Base Assembly
S1101MFD1247	BRE 25mm Base
S1101MFD1248	BRE 25mm Bottom Housing
S1101MFD1249	BRE 25mm Sensor Mounting Plate
S1101MFD1250	BRE 25mm Top Housing
S1101MFD1251	BRE 25mm Burner Outlet
S1101MFD1252	BRE 25mm Flow Screen
S1101MFD1253	BRE 25mm Flow Straightener
S1101MFD1254	BRE 25mm Top Spacer
S1101MFD1255	BRE 25mm Baffle
S1101MFD1256	BRE 25mm Sensor Gasket
S1101EFA1257	BRE 25mm Harness Assembly
S1101MFA1258	Coflow Burner Assembly
S1101MFD1260	Coflow Burner Base
S1101MFD1261	Coflow Burner Baffle
S1101MFD1262	Coflow Burner Outer Tube
S1101MFA1263	Coflow Burner Inner Tube Assembly
S1101MFD1264	Coflow Burner Inner Tube
S1101MFD1265	Coflow Burner Mesh Screen
S1101MFD1266	Coflow Burner Flow Straightener
S1101MFD1267	Coflow Burner Outlet
S1101MFD1268	Coflow Burner Collar
S1101MFD1269	Burner Stem Cover
S1101MFD1270	1/4" Spherical Burner Support Tube
S1101MFD1271	3/8" Spherical Burner Support Tube
S1101MFD1272	1/2" Spherical Burner Support Tube
S1101MFD1273	Burner Lower Foam
S1101MFD1274	Burner Upper Foam

S1101MFD1275	Coflow Burner Outlet Honeycomb
S1101MFD1276	Coflow Burner Mesh Screen
S1101MFD1277	Coflow Burner Lower Spacer
S1101MFD1278	Coflow Burner Upper Spacer

G. Igniter

S1101MFA1300	Igniter Assembly
S1101MFA1301	Igniter Mounting Bracket Assembly
S1101MFD1302	Igniter Mounting Bracket
S1101MFA1303	Igniter Base Assembly
S1101MFD1304	Igniter Base
S1101MFD1305	Igniter Motor Pivot
S1101MFD1306	Igniter Motor Tip
S1101MFD1307	Igniter Motor Tip Spacer
S1101MFA1308	Igniter Lower Arm Assembly
S1101MFD1309	Igniter Lower Arm
S1101MFD1310	Igniter Tip Connector Bracket
S1101MFD1311	Igniter Home Switch Engagement Bracket
S1101MFP1312	Igniter Pivot Bushing
S1101MFA1313	Igniter Upper Arm Assembly, 0mm Position
S1101MFD1314	Igniter Upper Arm, 0mm Position
S1101MFA1315	Igniter Upper Arm Assembly, 10mm Position
S1101MFD1316	Igniter Upper Arm, 10mm Position
S1101MFA1317	Igniter Upper Arm Assembly, 30mm Position
S1101MFD1318	Igniter Upper Arm, 30mm Position
S1101MFA1319	Igniter Tip Assembly
S1101MFD1320	Igniter Tip Insulator
S1101MFD1321	Igniter Upper Arm Insulator
S1101MFA1322	Stepper Driver Assembly
S1101MFA1323	Stepper Driver Enclosure Assembly, TFP Side
S1101MFD1324	Stepper Driver Enclosure, TFP Side
S1101MFA1325	Stepper Driver Enclosure Assembly, Igniter Side
S1101MFD1326	Stepper Driver Enclosure, Igniter Side
S1101EFA1327	Altered Stepper Driver, Igniter
S1101EFA1328	Altered Stepper Driver, TFP
S1101MFD1329	Stepper Driver Insulator
S1101EFA1330	Igniter to Stepper Driver Harness Assembly
S1101EFC1331	TFP Stepper Driver Assembly Connection Diagram
S1101MFA1332	Igniter Tip Holder Base Assembly
S1101MFD1333	Igniter Tip Holder Base
S1101MFD1334	Igniter Tip Holder Cover
S1101MFD1335	Igniter Tip Holder Cover Stop
S1101MFA1336	Igniter Tip Holder Cover Tab Assembly
S1101MFD1337	Igniter Tip Holder Cover Tab
S1101MFA1338	Igniter Tip Holder Assembly
S1101MFA1339	Coiled Igniter Tip Assembly

H. Analog Camera System

S1101MFA1400	Analog Camera
S1101MFA1401	Analog Camera Base Assembly
S1101MFD1402	Analog Camera Base
S1101MFA1403	Analog Camera Wiring Enclosure Assembly
S1101MFD1404	Analog Camera Wiring Enclosure
S1101MFA1405	Analog Camera Turning Mirror Assembly
S1101MFA1406	Analog Camera LED Assembly
S1101MFA1407	Analog Camera LED Housing Assembly
S1101MFD1408	Analog Camera LED Housing
S1101MFD1409	Analog Camera LED Housing Cover
S1101EFA1410	Analog Camera LED Board Assembly
S1101EFS1410	Analog Camera LED Board Schematic
S1101EFF1410	Analog Camera LED Board Fabrication
S1101EFL1410	Analog Camera LED Board Artwork
S1101MFD1411	Analog Camera Turning Mirror
S1101EFA1412	Analog Camera Internal Harness Assembly
S1101EFA1413	Analog Camera to Downstream Relay Harness Assembly
S1101MFD1414	Analog Camera Insulating Pad

I. Diagnostic System

S1101MFA1500	PMT Assembly
S1101MFA1501	PMT Bracket Assembly
S1101MFD1502	PMT Bracket
S1101MFD1503	PMT Bracket Clamp
S1101MFA1504	PMT Filter Housing Assembly 1
S1101MFA1505	PMT Filter Housing Assembly 2
S1101MFA1506	PMT Filter Housing Assembly 3
S1101MFA1507	PMT Filter Housing Insert Assembly
S1101MFD1508	PMT Filter Housing
S1101MFD1509	PMT Filter Housing Retaining Ring
S1101MFD1510	PMT Filter Spacer
S1101MFD1511	PMT Filter Gasket
S1101EFA1512	PMT Harness Assembly
S1101MFD1513	PMT Connector Bracket
S1101MFD1514	PMT Bracket Pad
S1101MFA1520	Radiometer Assembly
S1101MFA1521	Radiometer Housing Assembly
S1101MFD1522	Radiometer Housing
S1101MFD1523	Radiometer Cover
S1101EFA1524	Radiometer Board Assembly
S1101EFS1524	Radiometer Board Schematic
S1101EFF1524	Radiometer Board Fabrication
S1101EFL1524	Radiometer Board Artwork
S1101EFA1525	Radiometer Harness Assembly
S1101MFA1526	60 Degree Radiometer Assembly
S1101EFA1527	60 Degree Radiometer Board Assembly

S1101EFS1527	60 Degree Radiometer Board Schematic
S1101EFF1527	60 Degree Radiometer Board Fabrication
S1101EFL1527	60 Degree Radiometer Board Artwork
S1101MFD1528	60 Degree Radiometer Housing
S1101MFA1529	TFP Assembly
S1101MFA1530	TFP Motor Base Assembly
S1101MFD1531	TFP Motor Base
S1101MFD1532	TFP Motor Clamp
S1101MFA1533	TFP Arm Base Assembly
S1101MFD1534	TFP Arm Base
S1101MFA1535	TFP Arm Assembly
S1101MFD1536	TFP Arm
S1101MFA1537	TFP Array Assembly
S1101MFD1538	TFP Array Bracket
S1101MFA1539	TFP Array Cover Assembly
S1101MFD1540	TFP Array Cover
S1101EFA1541	TFP Motor to Stepper Driver Harness Assembly
S1101MFA1542	TFP Array Cover Base Assembly
S1101MFD1543	TFP Array Cover Base
S1101MFP1544	Modified TFP Cover Fastener
S1101MFD1545	Screw Retaining Plate
S1101MFA1546	TFP Calibration Array Assembly
S1101MFD1547	TFP Calibration Array Bracket
S1101MFA1548	35 Micron TFP Array Assembly
S1101MFA1549	140 Micron TFP Array Assembly
S1101MFA1557	35 Micron TFP Calibration Array Assembly
S1101MFA1558	140 Micron TFP Calibration Array Assembly
S1101MFA1550	BRE Radiometer Assembly
S1101MFA1551	BRE Radiometer Housing Assembly
S1101MFD1552	BRE Radiometer Housing
S1101EFA1553	BRE Radiometer Board Assembly
S1101EFS1553	BRE Radiometer Board Schematic
S1101EFF1553	BRE Radiometer Board Fabrication
S1101EFL1553	BRE Radiometer Board Artwork
S1101EFA1554	BRE Radiometer Harness Assembly
S1101MFD1555	BRE Radiometer Cover
S1101MFD1556	BRE Radiometer Shield

J. Measurement System

S1101EFA1600	TC Cold Junction Compensation Board Assembly
S1101EFS1600	TC Cold Junction Compensation Board Schematic
S1101EFF1600	TC Cold Junction Compensation Board Fabrication
S1101EFL1600	TC Cold Junction Compensation Board Artwork
S1101MFA1601	Cold Junction Compensation Assembly
S1101MFD1602	Cold Junction Compensation Board Heatsink
S1101MFD1603	Cold Junction Compensation Board Insulator
S1101MFD1604	Cold Junction Compensation Board Cover

S1101MFA1605	Thermocouple Near Array Assembly
S1101MFA1606	Thermocouple Near Rake Assembly
S1101EFA1607	Thermocouple Near Rake Harness Assembly
S1101MFA1608	Thermocouple Near Mounting Arm Assembly
S1101MFA1609	Thermocouple Far Array Assembly
S1101MFA1610	Thermocouple Far Rake Assembly
S1101EFA1611	Thermocouple Far Rake Harness Assembly
S1101MFA1612	Thermocouple Far Mounting Arm Assembly
S1101MFA1613	Thermocouple Rake Mounting Bracket Assembly
S1101MFD1614	Thermocouple Rake Mounting Bracket
S1101MFD1615	Thermocouple Rake Mounting Arm
S1101MFD1616	Thermocouple Sleeve, 13mm Position
S1101MFD1617	Thermocouple Sleeve, 25mm Position
S1101MFD1618	Thermocouple Sleeve, 50mm Position
S1101MFD1619	Thermocouple Sleeve, 100mm Position
S1101MFD1620	Thermocouple Sleeve, 45mm Position
S1101MFD1621	Thermocouple Sleeve, 60mm Position
S1101MFA1622	Cold Junction Compensation Board Heatsink Assembly
S1101MFA1623	200mm Thermocouple Assembly, Igniter Side
S1101MFA1624	200mm Thermocouple Mounting Bracket Assembly, Igniter Side
S1101MFD1625	200mm Thermocouple Mounting Bracket, Igniter Side
S1101MFD1626	200mm Thermocouple Support Sleeve
S1101MFA1627	200mm Thermocouple Assembly, Relay Side
S1101MFA1628	200mm Thermocouple Mounting Bracket Assembly, Relay Side
S1101MFD1629	200mm Thermocouple Mounting Bracket, Relay Side
S1101MFD1630	200mm Thermocouple Strain Relief Base
S1101MFP1631	Modified Far Field Thermocouple Fastener
S1101MFA1632	Thermocouple Rake Cover Assembly
S1101MFA1633	Thermocouple Rake Bottom Cover Assembly
S1101MFD1634	Thermocouple Rake Bottom Cover
S1101MFA1635	Thermocouple Rake Top Cover Assembly
S1101MFD1636	Thermocouple Rake Top Cover
S1101MFD1637	Thermocouple Rake Cover Mounting Bracket
S1101MFD1638	Thermocouple Rake Cover Alignment Bracket
S1101MFD1639	200mm Thermocouple Cover
S1101MFA1640	200mm Thermocouple Cover Assembly

K. E-Field System

S1101MFA1700	E-Field Mesh Assembly
S1101MFD1701	E-Field Mesh Support Ring
S1101MFD1702	E-Field Mesh Screen
S1101MFD1703	E-Field Mesh Stem
S1101MFD1704	E-Field Mesh Insulator
S1101MFA1705	E-Field Mesh Bottom Clamp Assembly
S1101MFD1706	E-Field Mesh Bottom Clamp
S1101MFD1707	E-Field Mesh Top Clamp
S1101MFA1708	E-Field Power Supply Assembly

S1101MFA1709	E-Field Supply Base Assembly
S1101MFD1710	E-Field Supply Base
S1101MFA1711	E-Field Supply Passive Circuit Mount Assembly
S1101MFD1712	E-Field Supply Passive Circuit Mount
S1101MFD1713	E-Field Supply Passive Circuit Base
S1101MFD1714	E-Field Supply Passive Circuit Cover
S1101MFD1715	E-Field Negative Supply Interface Cover
S1101EFA1716	E-Field Negative Supply Interface Assembly
S1101EFS1716	E-Field Negative Supply Interface Schematic
S1101EFF1716	E-Field Negative Supply Interface Fabrication
S1101EFL1716	E-Field Negative Supply Interface Artwork
1717	*vacant*
S1101MFD1718	E-Field Positive Supply Interface Cover
S1101EFA1719	E-Field Positive Supply Interface Assembly
S1101EFS1719	E-Field Positive Supply Interface Schematic
S1101EFF1719	E-Field Positive Supply Interface Fabrication
S1101EFL1719	E-Field Positive Supply Interface Artwork
1720	*vacant*
S1101EFC1721	E-Field Assembly Connection Diagram
S1101MFD1722	E-Field Mesh Connector Bracket
S1101MFP1723	Modified E-Field Positive Supply Module
S1101MFP1724	Modified E-Field Negative Supply Module
S1101MFA1725	E-field Cover Plate Assembly
S1101MFD1726	E-Field Cover Plate
S1101MFA1727	E-Field Connector Support Assembly
S1101MFD1728	E-Field Connector Support
S1101MFD1729	E-Field Connector Retainer Plate
S1101MFA1730	E-Field Cover Assembly
S1101MFD1731	E-Field Mesh Lower Foam
S1101MFA1732	E-Field Negative Power Supply Assembly
S1101MFA1733	E-Field Positive Power Supply Assembly
S1101MFP1734	Modified 10KV Negative Supply Module
S1101MFP1735	Modified 10KV Positive Supply Module
S1101MFD1736	E-Field Mesh Upper Foam

L. Cooling System

S1101MFA1800	Cold Plate Assembly
S1101MFP1801	Altered Cold Plate
S1101MFD1802	Cooling Tube Assembly, Supply to Downstream Cooling Plate
S1101MFA1803	Cooling Tube Assembly, Return to Upstream Cooling Plate
S1101MFD1804	Cooling Tube Assembly, Upstream Cooling Plate to Downstream Cooling Plate
S1101MFA1805	Downstream Cooling Tube Bracket Assembly
S1101MFD1806	Cooling Tube Bracket
S1101MFD1807	Upstream Cooling Tube Bracket Assembly
1808	*not used*
S1101MFD1809	Cooling Tube Bracket Clamp

M. ACME Data Camera Package

S1101MFA2000	ACME Data Camera Package
S1101MFT2000	ACME Data Camera Package Labels
S1101EFA2001	Data Camera Control to Camera Enclosure Harness Assembly
S1101EFA2002	AVP to Data Camera Harness Assembly
S1001EFA2003	Data Camera Control to Filter Barrel Harness Assembly
S1101MFA2004	Data Camera Connector Cover Assembly
S1101MFD2005	Data Camera Connector Cover
S1101EFC2006	Data Camera Package Interconnect Diagram
S1101EFA2007	IPSU-G to ACME Camera Interface Harness Assembly
S1101MFA2100	Camera Control Assembly
S1101MFA2101	Camera Control Base Assembly
S1101MFD2102	Camera Control Base
S1101MFA2103	Camera Control Board Mount Assembly
S1101MFD2104	Camera Control Board Mount
S1101MFA2105	Camera Control Left Side Assembly
S1101MFD2106	Camera Control Left Side
S1101MFA2107	Camera Control Right Side Assembly
S1101MFD2108	Camera Control Right Side
S1101MFD2109	Camera Control Air Inlet Screen
S1101MFD2110	Camera Control Air Inlet Screen Cover
S1101MFD2111	Camera Control Air Outlet Screen
S1101MFD2112	Camera Control Air Outlet Screen Cover
S1101MFD2113	Camera Control Bench Seal
S1101MFD2114	Camera Control Handle Driver Pin NCR-025
S1101MFD2115	Camera Control Handle Spring
S1101MFD2116	Camera Control Handle Spring Housing
S1101MFD2117	Camera Control Handle Docking Latch
S1101MFD2118	Camera Control Cover
S1101EFA2119	Camera Stepper Driver Board Assembly
S1101EFS2119	Camera Stepper Driver Board Schematic
S1101EFF2119	Camera Stepper Driver Board Fabrication
S1101EFL2119	Camera Stepper Driver Board Artwork
S1101EFA2120	Camera Control Power Distribution Board Assembly
S1101EFS2120	Camera Control Power Distribution Board Schematic
S1101EFF2120	Camera Control Power Distribution Board Fabrication
S1101EFL2120	Camera Control Power Distribution Board Artwork
S1101EFA2121	Camera Electronics High Voltage Power Harness Assembly
S1101MFA2122	Camera Control Air Outlet Nutplate Assembly
S1101MFD2123	Camera Control Air Outlet Nutplate
S1101EFA2124	Camera Electronics Low Voltage Power Harness Assembly
S1101EFA2125	Camera Electronics Motion Card Harness I (J5) Assembly
S1101EFA2126	Camera Electronics Motion Card Harness II (J6) Assembly
S1101EFA2127	Camera Electronics Motor Control and Communications Harness Assembly
S1101EFA2128	Camera Electronics Filter Motor Control Harness Assembly
S1101EFA2129	Modified Camera Power Board

S1101EFA2130	Modified Camera Motion Control Board
S1101MFA2200	Camera Enclosure
S1101MFA2200	Camera Enclosure Labels
S1101MFA2201	Camera Mounting Plate Assembly
S1101MFD2202	Camera Mounting Plate
S1101MFA2203	Camera Enclosure End Panel Assembly
S1101MFD2204	Camera Enclosure End Panel
S1101MFA2205	Zoom Lens Mount Assembly
S1101MFD2206	Zoom Lens Mount
S1101MFA2207	Zoom Lens Motor Cam Assembly
S1101MFD2208	Zoom Lens Motor Cam
S1101EFA2209	Zoom Lens Harness Assembly
S1101MFA2210	Camera Top Bracket Assembly
S1101MFD2211	Camera Top Bracket
S1101MFD2212	Camera Enclosure Cover
S1101MFD2213	Camera Thermal Pad
S1101EFA2214	Camera Enclosure Harness Assembly
S1101MFD2215	Sensor Board Wire Strain Relief
S1101EFA2216	Hall Effect Sensor Board Assembly
S1101EFS2216	Hall Effect Sensor Board Schematic
S1101EFF2216	Hall Effect Sensor Board Fabrication
S1101EFL2216	Hall Effect Sensor Board Artwork
S1101MFD2217	Lens Alignment Bushing
S1101MFA2218	Spur Gear Assembly
S1101MFA2219	Modified Camera Assembly
S1101MFP2220	Modified Camera
S1101EFA2221	Modified Camera Motor
S1101MFD2222	Modified Camera Gasket
S1101MFP2223	Modified Camera Enclosure Mounting Fastener
S1101MFA2224	Focus Ring Spur Gear Assembly
S1101MFA2225	Camera Enclosure Lens Cover Assembly
S1101MFD2226	Camera Enclosure Lens Cover
S1101MFA2300	Filter Enclosure Assembly
S1101MFA2301	Filter Enclosure Lower Housing Assembly
S1101MFD2302	Filter Enclosure Lower Housing
S1101MFA2303	Filter Enclosure Upper Housing Assembly
S1101MFD2304	Filter Enclosure Upper Housing
S1101MFA2305	Filter Window Gasket
S1101MFD2306	Filter Enclosure Rear Cap
S1101MFA2307	Filter Enclosure Front Cap Assembly
S1101MFD2308	Filter Enclosure Front Cap
S1101MFA2309	Filter Enclosure Window Cover Assembly
S1101MFA2309	Filter Enclosure Window Cover Assembly Labels
S1101MFD2310	Filter Enclosure Window Cover
S1101MFD2311	Filter Enclosure Front Window
S1101MFD2312	Filter Enclosure Window Clamp
2313	*not used*
S1101MFA2314	Filter Barrel with Inserts

S1101MFD2315	Filter Barrel
S1101MFA2316	Blue Filter Assembly
S1101MFA2317	Clear Filter Assembly
S1101MFA2318	Filter Assembly 3
S1101MFA2319	Filter Assembly 4
S1101MFD2320	Filter Mount
S1101MFA2321	Filter Barrel Gear Assembly
S1101MFP2322	Filter Barrel Gear
S1101MFA2323	Filter Motor Bracket Assembly
S1101MFD2324	Filter Motor Bracket
S1101MFA2325	Data Camera Mirror Assembly
S1101MFA2326	Data Camera Mirror Support Arm Assembly
S1101MFD2327	Data Camera Mirror Support Arm
S1101MFA2328	Data Camera Mirror Mount Assembly
S1101MFD2329	Data Camera Mirror Mount
S1101MFD2330	Data Camera Mirror
S1101MFA2331	Filter Motor Assembly
S1101MFP2332	Modified Cover Knob
S1101EFA2333	Filter Enclosure Harness Assembly
S1101MFD2334	Filter Clamp
S1101MFD2335	Filter Gasket
S1101MFA2336	Filter Mount Assembly
S1101MFA2337	Filter Enclosure Window Assembly
S1101MFP2338	Modified Filter Cover Fastener
S1101MFA2339	Filter Barrel Spur Gear Assembly

N. ACME Avionics Package

S1101MFA3000	ACME Avionics Package
S1101MFA3100	AVP Housing Assembly
S1101MFA3101	Housing Baseplate Assembly
S1101MFD3102	Housing Baseplate
S1101MFP3103	Housing Alignment Pin
S1101MFD3104	Housing Mounting Cleat
S1101MFD3105	Heatsink Airflow Diverter
S1101MFD3106	Air Inlet Vent Screen Frame
S1101MFD3107	Air Inlet Vent Screen
S1101MFP3108	Modified Socket Head Cap Screw, 1/4-20 x 1.00
S1101MFD3109	Spacer Washer
S1101MFA3110	Housing Front Frame Assembly
S1101MFD3111	Housing Front Frame
S1101MFA3112	Housing Left Frame Assembly
S1101MFD3113	Housing Left Frame
S1101MFA3114	Housing Right Frame Assembly
S1101MFD3115	Housing Right Frame
S1101MFA3116	Housing Rear Frame Assembly
S1101MFD3117	Housing Rear Frame
3118	*vacant*

S1101MFA3119	Housing Rear Frame Interface Bracket Assembly
S1101MFD3120	Housing Rear Frame Interface Bracket
S1101MFD3121	Housing Rear Frame Interface Panel
S1101MFA3122	Housing Top Frame Assembly
S1101MFD3123	Housing Top Frame
S1101MFD3124	Housing Left Cover
S1101MFD3125	Housing Right Cover
S1101MFD3126	Housing Front Cover
S1101MFA3127	Housing Top Cover Assembly
S1101MFD3128	Housing Top Cover
S1101MFA3129	Top Cover Airflow Restrictor Plate Assembly
S1101MFD3130	Top Cover Airflow Restrictor Plate
S1101MFA3131	Breaker Cover Assembly
S1101MFD3132	Breaker Cover
S1101MFA3133	Connector Cover Assembly
S1101MFD3134	Connector Cover
S1101MFA3200	Card Cage Assembly
S1101MFA3201	Card Cage Left Frame Assembly
S1101MFD3202	Card Cage Left Frame
S1101MFA3203	Card Cage Right Frame Assembly
S1101MFD3204	Card Cage Right Frame
S1101MFA3205	Card Cage Frame Side Assembly
S1101MFD3206	Card Cage Frame Side
S1101MFD3207	Card Cage Frame Gasket
S1101EFA3208	Modified Single Board Computer
S1101EFA3209	Modified Ethernet Board
S1101EFA3210	Modified Power Board
S1101EFA3211	Modified Serial Board
S1101EFA3212	Modified Motion Control Board
S1101EFA3213	Modified VGA Board
S1101EFA3214	VGA Interface Board Assembly
S1101EFS3214	VGA Interface Board Schematic
S1101EFF3214	VGA Interface Board Fabrication
S1101EFL3214	VGA Interface Board Artwork
S1101EFA3215	Motion Control Board Interface Board Assembly
S1101EFS3215	Motion Control Board Interface Board Schematic
S1101EFF3215	Motion Control Board Interface Board Fabrication
S1101EFL3215	Motion Control Board Interface Board Artwork
S1101EFA3216	External Battery Assembly
S1101MFA3300	Hard Drive Assembly
S1101MFA3301	Hard Drive Sleeve Assembly
S1101MFD3302	Hard Drive Sleeve
S1101MFA3303	Hard Drive Front Panel Assembly
S1101MFD3304	Hard Drive Front Panel
S1101MFA3305	Hard Drive Tray Assembly
S1101MFA3306	Hard Drive Tray Housing Assembly
S1101MFD3307	Hard Drive Tray Housing
S1101MFA3308	Hard Drive Tray Back Panel Assembly

S1101MFD3309	Hard Drive Tray Back Panel
S1101EFA3310	Hard Drive Tray Board Assembly
S1101EFS3310	Schematic
S1101EFF3310	Fabrication
S1101EFL3310	Artwork
S1101MFD3311	Hard Drive Insulating Sleeve
S1101MFA3400	Power Distribution Assembly
S1101MFA3401	Power Distribution Heatsink Assembly
S1101MFD3402	Power Distribution Heatsink
S1101EFA3403	Power Distribution Board Assembly
S1101EFS3403	Power Distribution Board Schematic
S1101EFF3403	Power Distribution Board Fabrication
S1101EFL3403	Power Distribution Board Artwork
S1101EFA3404	EMI Filter Board Assembly
S1101EFS3404	EMI Filter Board Schematic
S1101EFF3404	EMI Filter Board Fabrication
S1101EFL3404	EMI Filter Board Artwork
S1101MFD3405	Bench Connection Harness Bracket
S1101EFA3500	Avionics Power Harness Assembly
S1101EFA3501	Avionics Bench Connection/PIL1 Harness Assembly
S1101EFA3502	Card Stack Harness Assembly
S1101EFA3503	Hard Drive Power Cable Assembly
S1101EFA3504	External Interface VGA Harness Assembly
S1101EFA3505	Digital Camera Interface Harness Assembly
S1101EFA3506	Hard Drive Sata Data Cable Assembly
S1101EFA3507	External Interface USB Harness Assembly
S1101EFA3508	RS232 Motion Control Board Harness Assembly
S1101EFC3509	Avionics Package Interconnect Diagram

O. Ground Support Equipment

S1101MGA4500	ACME GSE Test Rack Assembly
S1101MBA4501	AVP Power Supply Breadboard Assembly
S1101EBA4502	Acme Avionics to Camera Package Bread Board Cable Assembly
S1101MGD4503	E-Field Mesh Screen Brazing Fixture
S1101MGA4504	E-Field Mesh Assembly Fixture
S1101MGD4505	E-Field Mesh Assembly Fixture Stop
S1101MGD4506	E-Field Mesh Assembly Fixture Mounting Plate
S1101MGA4507	Chamber Insert Assembly SLA
S1101MGD4508	E-Field Mesh Assembly Fixture Base Plate
S1101EGA4509	E-Field Power Supply Ground Support Harness Assembly
S1101EGA4510	GSE E-Field Mesh Harness Assembly
S1101EGA4511	GSE Stepper Assembly Test Harness Assembly
S1101EGA4512	LEMO FGG.0B.303.SYBD31 Test Adapter
S1101EGA4513	PAP-04-S Test Adapter
S1101EGA4514	ZHR-4 Test Adapter
S1101EGA4515	SDD15S1000G Test Adapter
S1101EGA4516	LEMO PFG.0B.303.SYBD31 Test Adapter

S1101EGA4517	Molex 6758-2000 Test Adapter
S1101EGA4518	EMI Filter Board Test Assembly
S1101EGA4519	GSE Cube Assembly Test Harness
S1101EGA4520	AVP Power Board Test Harness
S1101EGA4521	GSE Cube Communication Harness
S1101EGA4522	GSE Cube Power Harness
S1101EGA4523	GSE DNA-AO-333 to DNA-AI-225 Test Cable
S1101EGA4524	GSE DNA-AO-333 to DNA-AI-207 Test Cable
S1101EGA4525	Motion Control Interface Board Test Harness Assembly
S1101EGA4526	ACME Filter Board Test Harness Assembly
S1101EGA4527	Optics Bench Simulator Ethernet/Video Harness
S1101EGA4528	Optics Bench Simulator Power Harness
S1101EGA4529	VGA Interface Board Video Input Test Harness
S1101EGA4530	VGA Interface Board Video Output Test Harness
S1101EGA4531	Optics Bench Simulator PCTC8 Test Harness Assembly
S1101EGA4532	Optics Bench Simulator PCTC7 Test Harness Assembly
S1101EGA4533	AVP Data Camera Test Harness Assembly
S1101EBA4534	VGA Interface Breadboard
S1101EGA4535	Camera Stepper Driver and Power Distribution Board Test Kit
S1101EGA4536	ACME GSE Test Rack Schematic
S1101EGA4537	GSE Stepper Motor, Encoder, Home Switch Tester Assembly
S1101EGA4538	Upstream/Downstream Relay Board Input Signal Test Cable
S1101EGA4539	Downstream Relay Board Supply Power Test Cable
S1101EGA4540	Downstream Relay Board Output Signal Test Cable
S1101EGA4541	Upstream Relay Board Supply Power Test Cable
S1101EGA4542	Upstream Relay Board P1 Loopback Connector Assembly
S1101EGA4543	Upstream Relay Board I/O Signal Test Kit
S1101EGA4544	Fuel Mass Flow Controller Assembly Test Harness
S1101EGA4545	Inert Mass Flow Controller Assembly Test Harness
S1101EGA4546	Oxygen Mass Flow Controller Assembly Test Harness
S1101EGA4547	Upstream Relay Board Test Cable
S1101EGA4548	Standard Density D9 Femal Pigtailed Test Cable
S1101EGA4549	High Density D26 Femal Pigtailed Test Cable
S1101MGA4550	Camera Target Assembly
S1101MGD4551	Camera Target Adaptor
S1101EGA4552	UML Simulator Power Harness
S1101EGA4553	Analog Camera Assembly Test Harness
S1101MGA4554	Engineering Model Test Bed
S1101MGA4555	Manifold Standoff Assembly
S1101MGD4556	Manifold Standoff Base
S1101MGA4557	Engineering Model Nitrogen Supply System
S1101EGA4558	Optics Bench Simulator Ethernet/Video Harness to GSE Test Rack
S1101MGA4559	EMS Stand
S1101MGD4560	EMS Stand Mounting Plate
S1101MGD4561	AVP Lifting Bracket
S1101EGA4562	PCTC8 Extension Cable (PIL side Simulation)
S1101EGA4563	PCTC8 Extension Cable (EMS side Simulation)
S1101EGA4564	Optics Bench Simulator EMI Radiated Harness

S1101EGA4565	UML Simulator EMI Radiated Harness
S1101EGA4566	PCTC7 Extension Cable (PIL side Simulation)
S1101EGA4567	PCTC7 Extension Cable (EMS side Simulation)
S1101EGA4568	Optics Bench Simulator EMI Radiated Break Out Harness
S1101EGA4569	UML Simulator EMI Radiated Break Out Harness
S1101EGA4570	UML Simulator Fan Harness
S1101EGA4571	CIA Pressure Switch Jumper Connectors
S1101EGA4572	Cube Fan for Insert Testing
S1101MGD4573	ACME Main Vibe Plate
S1101MGD4574	ACME Secondary Vibe Plate 1
S1101MGD4575	ACME Secondary Vibe Plate 2
S1101MGD4576	Camera Control Package Vibe Adaptor Plate
S1101MGD4577	E-Field Power Supply Vibe Adaptor Plate
S1101MGD4578	Hall Effect Sensor Vibe Adaptor Plate
S1101MGD4579	TFP Assembly Vibration Lock
S1101MGD4580	Camera Control Package Adaptor Pin
S1101EGA4581	PCTC7 Extension Cable (VAC Outside Chamber Side)
S1101EGA4582	PCTC8 Extension Cable (VAC Outside Chamber Side)
S1101EGA4583	Radiometer Assembly Test Fixture
S1101MGD4584	Burner Test Mounting Fixture
S1101EFA4585	BRE Burner Heat Flux Test Cable For PI
S1101EFA4586	BRE Burner TC Test Cable For PI

P. Shipping Containers

S1101MGA5000	ACME CIA Shipping Container Assembly
S1101MGD5001	CIA Lower Foam
S1101MGD5002	CIA Upper Foam
S1101MGA5003	ACME AVP Shipping Container Assembly
S1101MGA5004	AVP Lower Foam
S1101MGD5005	AVP Upper Foam
S1101MGA5006	ACME Camera Shipping Container Assembly
S1101MGD5007	Camera Shipping Lower Foam
S1101MGD5008	Camera Shipping Upper Foam
S1101MGA5009	AVP Lid Foam
S1101MGD5010	Camera Vibe Lower Foam
S1101MGD5011	Camera Vibe Upper Foam
S1101MGD5012	CIA Foam Bottom Cradle
S1101MGD5013	CIA Foam Cradle Support
S1101MGD5014	CIA Foam Top Cradle
S1101MGD5015	CIA Foam Cradle Wedge
S1101MGA5016	ACME Vibe Foam A
S1101MGA5017	ACME Vibe Foam B
S1101MGA5018	ACME Vibe Foam C

Q. Trainer

S1101MGA6000	ACME Trainer
S1101MGA6001	Cube Trainer Assembly

S1101MGD6002	Cube SLA Model
S1101MGA6003	Downstream Relay Board Trainer Assembly
S1101MGD6004	Downstream Relay Board SLA Model
S1101MGA6005	E-Field Power Supply Trainer Assembly
S1101MGD6006	E-field Power Supply SLA Model
S1101MGA6007	Cold Plate Trainer Assembly
S1101MGD6008	Cold Plate SLA Model
S1101MGA6009	Analog Camera Trainer Assembly
S1101MGD6010	Analog Camera SLA Model
S1101MGA6011	Input Manifold Trainer Assembly
S1101MGD6012	Input Manifold SLA Model
S1101MGA6013	Switching Manifold Trainer Assembly
S1101MGD6014	Switching Manifold SLA Model
S1101MGD6015	Trainer Mass Flow Controller Screw
S1101MGA6016	Fuel MFC Trainer Assembly
S1101MGA6017	Inert MFC Trainer Assembly
S1101MGA6018	Oxygen MFC Trainer Assembly
S1101MGD6019	Mass Flow Controller SLA Model
S1101MGA6020	Upstream Relay Board Trainer Assembly
S1101MGD6021	Upstream Relay Board SLA Model
S1101MGA6022	TFP Array Trainer Assembly
S1101MGD6023	TFP Array SLA Model
S1101MGA6024	Stepper Driver Trainer Assembly
S1101MGD6025	Stepper Driver SLA Model
S1101MGA6026	CJC Trainer Assembly
S1101MGD6027	CJC SLA Model
S1101MGA6028	E-Field Mesh Trainer Assembly
S1101MGD6029	E-Field Mesh SLA Model
S1101MGA6030	PMT Trainer Assembly
S1101MGD6031	PMT SLA Model
S1101MGA6032	Radiometer Trainer Assembly
S1101MGD6033	Radiometer SLA Model
S1101MGA6034	Trainer Gray Water Hose Assembly
S1101MGA6035	Trainer Green Water Hose Assembly
S1101MGA6036	Trainer Oxygen Hose Assembly
S1101MGA6037	Trainer Fuel Hose Assembly
S1101MGA6038	Trainer Inert Hose Assembly
S1101MGA6039	Burner Outlet Cover
S1101MGP6040	Burner Outlet Cover Plug
S1101MGA6041	Trainer Igniter Assembly
S1101MGD6042	Trainer Igniter Mounting Bracket
S1101MGD6043	Trainer Igniter Connector Bracket
S1101MGD6044	Trainer Igniter Pivot Bushing
S1101MGA6045	Trainer Igniter Upper Arm Assembly
S1101MGD6046	Trainer Igniter Upper Arm
S1101MGD6047	Trainer Cooling Tube Bracket Clamp
S1101MGA6048	Trainer Gas Jet Burner Assembly
S1101MGA6049	Trainer 200mm TC Assembly, Igniter Side

S1101MGD6050	Trainer 200mm TC Bracket, Igniter Side
S1101MGA6051	Trainer 200mm TC Assembly, Relay Side
S1101MGD6052	Trainer 200mm TC Bracket, Relay Side
S1101MGD6053	Trainer Igniter Motor Pivot
S1101MGD6054	1/4" Spherical Burner Epoxy Fixture
S1101MGD6055	Spherical Burner Epoxy Fixture Nut
S1101MGA6056	1/4" Spherical Burner Epoxy Fixture Assembly
S1101MGA6057	Chiller Hose Flow Meter Adaptor Assembly
S1101MGA6058	Chiller Hose Adaptor Assembly
S1101MGD6059	Black Body Mounting Arm
S1101MGD6060	Black Body Mounting Arm Bracket
S1101MGA6061	Black Body Mounting Assembly
S1101MGA6062	Water Hose Pressure Gauge Assembly

6. Internet Links

These websites may be useful for understanding ACME's engineering implementation and interpreting the experimental results.

ACME	Advanced Combustion via Microgravity Experiments
CIR	Combustion Integrated Rack (ISS facility in which ACME experiments are conducted)

Photographs of the hardware and more can be seen at:

www.flickr.com/photos/space-flames
www.facebook.com/space.flames

These *Space Flames* websites are about space-based studies of combustion in general, but they are administered by Dennis Stocker, the ACME Project Scientist, and are an excellent source of photos of the hardware and crew operations.

7. Nomenclature

ACME	Advanced Combustion via Microgravity Experiments (*project conducted in the CIR*)
ADP	Acceptance Data Package
AVP	avionics package
BRE	Burning Rate Emulator (*an ACME experiment*)
CASIS	Center for the Advancement of Science in Space
CCD	Charge-Coupled Device (*such as a camera sensor*)
CIA	Chamber Insert Assembly (*inserted into the CIR combustion chamber*)
CIR	Combustion Integrated Rack (*ISS facility in which ACME experiments are conducted*)
CFI	Cool Flames Investigation (*an MDCA experiment preceding ACME in the CIR*)
CFI-G	Cool Flames Investigation with Gases (*an ACME experiment selected by CASIS and NSF*)
CLD Flame	Coflow Laminar Diffusion Flame (*an ACME experiments*)
CM	Combustion Module (*space shuttle precursor to the CIR*)
CSV	comma-separated values
DataCam	Data Camera (*developed for the ACME experiments, characterized for pyrometry, etc.*)
E-FIELD Flames	Electric-Field Effects on Laminar Diffusion Flames (*an ACME experiment*)
EMI	electromagnetic interference

EMS	Experiment Mounting Structure
ERD	Engineering Requirements Document
Fiji	Fiji Is Just ImageJ (*a version of ImageJ preloaded with many plug-in features*)
Flame Design	Flame Design (*an ACME experiment*)
FOMA	Fuel and Oxidizer Management Assembly (*CIR's gas management subsystem*)
FOV	field of view
fps	frames per second
GCIP	Gas Chromatograph Instrumentation Package (*a CIR subsystem*)
GIPOC	Glenn ISS Payload Operations Center (*also known as the TSC*)
GMT	Greenwich Mean Time
HiBMs	High Bit-depth Multi-spectral camera (*developed for the CIR facility*)
HOBJ	Halcon object (*binary uncompressed format; used for the DataCam*)
HV	high voltage
ImageJ	Image Processing and Analysis in Java (*open-source image processing software*)
IPSU	Image Processing and Storage Unit (*a CIR subsystem*)
IPSU-G	IPSU for Gigabit ethernet cameras
ISRD	Integrated Science Requirements Document (*for a set of experiments*)
ISS	International Space Station
LLL-UV	Low Light Level Ultraviolet camera ((*developed for the CIR facility*)
LSP	Laminar Soot Processes (*a shuttle experiment conducted in the Combustion Module*)
MDCA	Multi-user Droplet Combustion Apparatus (*CIR chamber insert preceding ACME*)
MFC	mass flow controller
Mist	Water-Mist Fire Suppression Experiment (*a Combustion Module experiment*)
MIUL	Materials Identification/Usage List
NSF	National Science Foundation
NTSC	National Television System Committee
OMA2	Optical Multi-channel Analyzer 2 (*open-source software for image processing and display*)
ops	operations
PaRIS	Passive Rack Isolation System
PIMS	Principal Investigator Microgravity Services
PMT	Photomultiplier Tube (*where ACME is equipped with three PMT modules*)
PSI	Physical Sciences Informatics System
PT	pressure transducer
psia	pounds per square inch
QD	quick disconnect
qty	quantity
s-Flame	Structure and Response of Spherical Diffusion Flames (*an ACME experiment*)
SAMS	Space Acceleration Measurement System
SDP	Safety Data Package
slpm	standard liters per minute (*a measure of the mass or molar flow rate of a gas*)
SOFBALL	Structure Of Flame Balls At Low Lewis-number (*a Combustion Module experiment*)
SoFIE	Solid Fuel Ignition and Extinction (*project to be conducted in the CIR after ACME*)
SRD	Science Requirements Document
TFP	Thin-Filament Pyrometry (*a technique for making gas-phase temperature measurements*)
TIFF or TIF	Tagged Image File Format (*used for the CFI intensified, HiBMs, and LLL-UV cameras*)
TM	thermistor
TS	Transport Stream (*a video format*)
TSC	Telescience Support Center (*also known as the GIPOC*)